*To my family.*

# Contents

# Chapter 1

# Introduction

## 1.1 Topic and motivation

Generative diffusion models have become the dominant paradigm for synthesising high-dimensional data: they power state-of-the-art systems for images, audio, video, and molecular structures (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Karras et al., 2022). The same construction underlies the image and video generators deployed at scale by the large technology companies, from latent diffusion (Rombach et al., 2022) onwards, and reaches beyond media: structure prediction in AlphaFold 3 places atomic coordinates with a diffusion module (Abramson et al., 2024). Their central object is the *score function*, the gradient of the log-density of progressively noised data,

$$S(x,t) \;=\; \nabla_x \log p_t(x), \tag{1.1}$$

which is all a model needs to know in order to reverse a noising process and turn pure noise back into data (Anderson, 1982). In practice the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011); the network is a black box, and the structure of the function it has to learn remains poorly understood.

A complementary research programme, initiated in the statistical-physics community, replaces the black box with *exactly solvable models*: data distributions simple enough that the score can be computed in closed form, yet rich enough to exhibit the phenomena observed in real systems, such as speciation, memorisation, and dynamical phase transitions along the generation trajectory (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024; Achilli et al., 2026). Exact scores act as a microscope: every regime, transition time, and failure mode can be traced back to explicit formulas.

This thesis works inside that programme, in a direction that is natural for data with *temporal or sequential structure*: the data are not a cloud of independent points but a *trajectory* $a_0, a_1, \dots, a_{K-1}$ of a Markov process, the prototype of a video, a time series, or any dynamic object. Each frame of the trajectory is corrupted independently by an Ornstein–Uhlenbeck (OU) channel, and the object of study is the *joint* score of the whole noisy sequence. Two classical bodies of theory then meet the modern one. First, the joint score of a Gaussian Markov chain is governed by the algebra of structured matrices: Toeplitz covariances, tridiagonal precisions, and their common eigenbasis. Second, by an identity that holds for any data distribution, computing a score is solving a Bayesian inference problem, and for chain-structured data that problem lives on a tree-shaped factor graph, the domain of *belief propagation* (BP) and of the Kalman filter, together with the Gaussian message closures of statistical physics (TAP, AMP) (Pearl, 1988; Mézard and Montanari, 2009; Kalman, 1960; Donoho et al., 2009). The thesis develops both routes in full, proves where they coincide, and measures what is lost when they are forced apart.

## 1.2 Positioning and gap

The literature this work builds on can be organised in three strands; Chapters 2–3 review them in depth.

**Statistical physics of learning machines.** Machine learning and statistical mechanics share a genealogy that runs from spin glasses through the Hopfield network and the Boltzmann machine to today's generative models, a lineage recognised by the 2024 Nobel Prize in Physics (Hopfield, 1982; Ackley et al., 1985; The Royal Swedish Academy of Sciences, 2024). Its modern branch produced closed-form analyses of diffusion generation for Gaussian mixtures, ferromagnets, and high-dimensional asymptotics: the regimes of the backward dynamics and their transitions (Biroli and Mézard, 2023; Biroli et al., 2024), the general theory of speciation (Achilli et al., 2026; Sclocchi et al., 2025b), the dynamics of memorisation and generalisation in training (Bonnaire et al., 2025; Garnier-Brun et al., 2026), and, on the training side, the demonstration that energy-based models learn through a cascade of phase transitions (Bachtis et al., 2024). These works treat the data as unstructured collections of samples; the time axis they study is the diffusion (or training) time. The internal time of a *sequence*, the fact that a video frame is correlated with its neighbours through a dynamical rule, is absent from these analyses.

**Inference on chains.** Exact posterior inference on Gauss–Markov chains is classical: the Kalman filter and smoother (Kalman, 1960; Rauch et al., 1965), hidden Markov models (Rabiner, 1989), and their unifying formulation as sum–product message passing on factor graphs (Kschischang et al., 2001; Mézard and Montanari, 2009). Continuous, non-Gaussian state spaces have a long literature of their own, and it is worth saying precisely how this work sits inside it, because the machinery is familiar enough to look like a reapplication. Kalman–RTS is exact only for linear-Gaussian models. Assumed-density filtering and expectation propagation keep tractability by projecting each message back into a parametric family, so the approximation is in the messages themselves. Gaussian-sum filters widen that family without removing the restriction. Particle filters and smoothers place no parametric restriction but substitute Monte Carlo error, which is a different thing from no error and cannot serve as a reference one measures other methods against. Grid-based filtering for continuous-state HMMs is the closest relative of what is done here, and is normally treated as an approximation to be tolerated rather than as an object whose error is itself the measurement.

Two things distinguish the present use. Deterministic quadrature is adopted precisely so that the reference carries *no* sampling error and its discretisation error can be bounded separately and reported, which is what makes it usable as ground truth rather than as another competitor. And the target is not filtering: it is the diffusion score at every noise level, obtained from the same recursion that learns the kernel from noised sequences. What the classical literature does not ask is the question generative modelling forces — how the exact posterior, and therefore the exact score, deforms as a function of the diffusion time, and what that deformation implies for anything, algorithm or network, that must compute it at every level.

**Message passing with continuous variables.** BP is exact on trees for discrete variables, where messages are finite probability vectors. For continuous variables the messages become probability densities, and practical algorithms must close them within a parametric family, almost always the Gaussian one; this is the step that connects BP to AMP and to the TAP equations of statistical physics (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016; Genovese and Piana, 2026). The classical justification of the Gaussian closure is a central-limit argument valid when every node receives many messages. On a chain each node receives exactly two: the central-limit argument fails by construction, and we are not aware of a quantitative assessment of the Gaussian message closure on chain-structured diffusion posteriors.

**Amortising inference into a network.** The three strands above concern how the score can be *computed*. A diffusion model does not compute it; it trains a network to reproduce it, and the distinction is one of kind rather than of accuracy. Sum–product on the posterior chain determines a whole distribution over the latent sequence, by making a free energy stationary over a family rich enough — because the graph is a tree — that the minimiser is exact rather than approximate. A score network is a fixed-depth feed-forward map: it spends the same computation on every input and has no mechanism to search, so it cannot represent that inference, only the answers it produces. Reversal is iterative and unbounded, but each step's score still comes from the fixed-compute map, so the network stands to the inference as an *amortisation* (LeCun et al., 2006).

Whether that amortisation is cheap or expensive is, in general, unmeasurable: it would require the very object the network was trained to stand in for. On the family of this thesis that object is available, so the question becomes an experiment — and the answer has structure. The cost is small when there is little to amortise and grows with the data, which is what one should expect of a fixed-compute approximator facing a posterior that becomes better determined as evidence accumulates. Nothing in the literature above measures it, because nothing in it has the exact target to measure against.

The gap, in one sentence: there is no end-to-end, exactly solvable account of the joint diffusion score of a structured sequence, covering its matrix structure, its message-passing computation, the price of the Gaussian closure when the sequence is not Gaussian, and the price of amortising the whole computation into a network.

## 1.3 Research questions

The thesis addresses six questions. The first four concern scores under *known* dynamics and the last two concern learning dynamics that are unknown.

**RQ1.** For a stationary Gaussian AR(1) chain observed through coordinatewise Ornstein–Uhlenbeck noise, what is the joint score, and how do its spatial couplings vary with the diffusion time?

**RQ2.** How can the same Gaussian-chain score be computed by belief propagation, and how does the resulting recursion relate to Kalman filtering and smoothing?

**RQ3.** For selected non-Gaussian innovation models, how accurate is a Gaussian-message approximation relative to a validated numerical belief-propagation reference, over the parameter ranges studied?

**RQ4.** In the Gaussian model, how rapidly does the influence of remote observations on the score decay, and what limited guidance does this provide for local approximations of the score?

**RQ5.** Under what assumptions can an *unknown* local transition law be recovered from noised sequences alone, and which numerical, optimisation and statistical checks are required before such a recovery claim is reliable?

**RQ6.** On a family where the Markov assumption holds exactly, what is supplying that assumption worth, measured against denoising-score-matching networks trained on identical data and given none of it?

These questions are deliberately confined to the models named in them. No claim is made about general Markov processes, about universal properties of learned diffusion scores, or about practical neural architectures. In particular RQ6 is a question about the value of a correct structural assumption on one family, not a comparison of learning algorithms in general: the two arms are asymmetric in supplied information by construction, which is what makes the measurement meaningful and also what bounds it.

Chapter 7 is a self-contained case study rather than a seventh question. It asks what joint observations can identify that no single-frame marginal can, on a rotating-ring model whose state space and estimand both differ from the chain's, and it is placed alongside the main line rather than on it.

## 1.4 Contributions

1. For the stationary Gaussian AR(1) chain under OU corruption, we derive the joint score in closed form, diagonalise it for all diffusion times in a single eigenbasis, and characterise the evolution of its coupling structure: a band-fill law $(Q_t)_{i,i+d} = (-1)^{d-1}(2t)^{d-1}(Q_0^d)_{i,i+d} + O(t^d)$ for small $t$, a return to isotropy at rate $e^{-2t}$, and exact bulk formulas for the posterior variance and the geometric decay rate $q(\alpha, t)$ of posterior correlations. The band-fill law corrects a prefactor error in an earlier working note of this project; the correction is verified numerically (Chapter 5).

2. We derive the belief-propagation computation of the same score on the chain's factor graph, show algebraically that Gaussian messages are closed under the updates,

identify the resulting recursion, update by update, with the Kalman filter and the Rauch–Tung–Striebel smoother, and verify that it reproduces the matrix score to within double-precision rounding across a parameter grid (Chapter 5).

3. For Laplace-innovation chains we establish which identities remain exact, locate the precise step at which closed forms stop, and measure the accuracy of Gaussian-projected BP against a numerically validated grid implementation of the functional recursion: the median relative score error is 0.24 at $t = 0.05$, falls below $10^{-2}$ by $t \approx 1$ and below $10^{-3}$ by $t \approx 2.1$, with the score direction substantially more accurate than its magnitude, in the tested regimes (Chapter 6).

4. In the Gaussian model we prove that the error of a radius-$r$ local estimator of the score decays as $q(\alpha, t)^r$, and we measure that truncating the inference (windowed message passing) is more accurate than truncating the score matrix (banding) at small diffusion times (Chapter 5).

5. As a separate, self-contained result, we characterise the per-node (AMP/TAP) closure on the same model: it returns the exact score but a different posterior variance, with an existence boundary in closed form, a breakdown time $t_c(\alpha)$, and a critical coupling $\alpha_c = \sqrt{2} - 1$. All formulas are independently re-verified; the material is placed in Appendix B because it branches off the main line (statement of results in Section 10.1).

## 1.5 Methodology

The methodology is deliberate: closed forms first, numerics as audit. Analytical results lead, and every approximation is measured against a reference computed by an independent method rather than assumed small. Networks appear in exactly one role — as the baseline RQ6 measures the structured estimator against — and never as an instrument for establishing anything about the model itself. Every claim is either proved by explicit algebra (with the longer passages isolated in numbered *toolboxes*) or verified by deterministic numerical checks against ground truth computed by an independent method, in the spirit of the exactly solvable programme of Biroli and Mézard (2023). Where an approximation is introduced (the Gaussian closure of BP messages, local truncations), its error is *measured* against an exact reference (matrix algebra, or dense grid quadrature validated on the solvable case). All numbers in the thesis are regenerated by deterministic audit scripts, documented in Appendix C; we regard this reproducibility as a strength of the method, not as a contribution in itself.

## 1.6 Scope and limitations

The exact results concern one-dimensional state per frame, linear AR(1) dynamics, and stationary initial conditions. The non-Gaussian results are numerical, valid within the validated resolution regime of the reference solver and the tested parameter ranges. The locality results are proved for the Gaussian bulk only.

The learning results (RQ5, RQ6) carry their own limits, and they are tighter than the analytical ones. The estimator is given the Markov factorisation, the autoregressive form, homogeneity across sites, and an innovation family flexible enough to represent the truth closely; nothing here speaks to a setting where those do not hold. The last of these is a representation rather than an exact containment: a $C$-component Gaussian mixture does not contain a Laplace density for any finite $C$, so what maximum likelihood approaches is the best approximation within the fitted family, and the recovery statements of Chapter 9 are about specific summaries — $\alpha$, the innovation variance, its excess kurtosis — rather than about the transition density as a whole. The comparison in RQ6 is therefore a measurement of what a correct structural assumption buys on one family, and not evidence that likelihood learning through belief propagation is a better learning algorithm than score matching — a distinction the thesis maintains wherever the numbers appear. Any implication for trained score networks in practice remains a hypothesis suggested by a controlled model, not an established property of practical systems.

Two lines of experiment are reported as unresolved rather than as findings: the innovation-mixture capacity sweep, in which most shape coordinates had not settled within the iteration budget used, and the reverse-generation study built on those same fits. Both are recorded in Appendix C as methodological negative results, and neither supports a conclusion in Chapter 10.

## 1.7 Structure of the thesis

Chapters 2–3 contain the background and literature review; Chapters 4–9 contain the research. Throughout, each concept is developed once, in the chapter where it naturally belongs, and then used without repetition.

Chapter 2 develops the statistical mechanics this thesis uses (phase space and Liouville's theorem, the Boltzmann–Gibbs distribution, free energy, the Ising model) and then follows the discipline into machine learning: spin glasses, the Hopfield network, the Boltzmann machine and its restricted variant, and the statistical mechanics of learning. Chapter 2 supplies the probabilistic dynamics: stochastic processes and stationarity, Markov chains, the AR(1) running example, sampling by dynamics, the Itô calculus, the Ornstein–Uhlenbeck corruption channel, the Fokker–Planck equation, and Anderson's time-reversal theorem. Chapter 2 reviews modern generative modelling: energy-based models in LeCun's formulation, score matching, the DDPM construction, the score-SDE synthesis, flow models, the score–posterior identity, the statistical-physics analyses of diffusion, and the cascade of phase transitions in energy-based-model training. Chapter 3 builds the inference machinery: the multivariate Gaussian and its precision-matrix reading, the spectral theorem, Markov random fields, factor graphs, the sum–product algorithm, Gaussian BP and the Kalman filter, the TAP/AMP closures, and the architectural mirror of locality in convolutional networks and U-Nets.

Chapter 4 records the preliminary work: the toy models, the first supervision models, and the corrections that produced the final research questions. Chapter 5 solves the Gaussian chain exactly (RQ1, RQ2, and the Gaussian half of RQ4), with the spectral form of the precision matrix as the organising object and belief propagation identified with the Kalman smoother. Chapter 6 leaves Gaussianity in the most controlled way possible, Laplace innovations, and measures what the Gaussian message family costs when the data are no longer Gaussian (RQ3). Chapter 7 extends the state at each site from a scalar to a point on a rotating ring, a nonlinear dynamic object, and shows that a Cartesian surrogate is exactly solvable while the model as drawn is solvable numerically to a measured accuracy, both admitting the same structural description of the joint score. Chapters 8–9 leave the innovation law itself unknown: an EM procedure built on the exact chain inference of Chapter 5 estimates it from noised data alone, and the structured estimator is compared against trained neural denoisers, both pointwise and generatively, along axes of model capacity and of misspecification of the Markov assumption itself. Chapter 10 states findings per research question, limitations, and the questions that follow. The appendices collect the Gaussian identities with proofs, the AMP/TAP analysis with its exact phase boundary, the reproducibility protocol, the background derivations supporting Chapter 2, the graphical and rotating-ring derivations, and the robustness of the headline aggregation.

# Chapter 2

# From Stationary Action to Generative Diffusion

This chapter builds, in one arc, the background the thesis needs: a variational principle, the statistical mechanics that grew out of it, the stochastic processes that turn it into an algorithm, and the diffusion models that are the subject of the work. It is deliberately a narrative rather than a development. Every derivation it states is carried out in full in Appendices D, E and F, which are unbounded in length; the body's job is the thread connecting them.

The thread is one idea, restated four times. A physical trajectory is a stationary point of the action (§2.1). An equilibrium distribution is a stationary point of the free energy (Appendix D). A diffusion model is trained by making a variational bound stationary (Appendix F). And the transition kernel of Chapter 8 is estimated by making a likelihood stationary. The objects change; the shape of the argument does not, and that continuity is why this thesis is written in the physics tradition rather than only borrowing its vocabulary.

## 2.1 The principle of stationary action

Mechanics can be built from a single variational statement, and it is worth starting there rather than with Hamilton's equations, because the same statement — *a physical trajectory is a stationary point of a functional* — returns three more times in this thesis: as the free-energy principle of §D.4, as the variational bound that trains a diffusion model (Chapter 2), and as the maximum-likelihood problem solved by expectation–maximisation in Chapter 8. The objects change; the shape of the argument does not.

Let a system be described by generalised coordinates $q(t)$ and let the *Lagrangian* $L(q, \dot{q}, t)$ be the difference between kinetic and potential energy, $L = T - U$. For a path $q(t)$ joining fixed endpoints $q(t_1) = q_1$ and $q(t_2) = q_2$, define the *action*

$$S[q] \;=\; \int_{t_1}^{t_2} L\Big(q(t), \dot{q}(t), t\Big)\, \mathrm{d}t. \tag{2.1}$$

The action assigns one number to a whole path. Hamilton's principle states that the path actually taken is a *stationary point* of $S$: for any smooth variation $\delta q(t)$ vanishing at the endpoints, $\delta S = 0$.

*Remark* 2.1 (Stationary, not minimal). It is usually called the principle of *least* action, and that is slightly wrong. The physical path is stationary, not necessarily minimal — it can be a saddle. The same caution applies everywhere the pattern recurs: expectation–maximisation converges to a stationary point of the likelihood, not to a global maximum (Chapter 8), and it is a recurring source of false confidence to read "the algorithm converged" as "the algorithm found the best answer".

Carrying out the variation and integrating by parts, with the boundary term vanishing because $\delta q$ is fixed at the endpoints, gives the Euler–Lagrange equations

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} \;-\; \frac{\partial L}{\partial q_i} \;=\; 0, \qquad i = 1, \dots, n, \tag{2.2}$$

which for $L = \sum_i \frac{1}{2} m_i \dot{q}_i^2 - U(q)$ return Newton's second law (Appendix D.1). The variational principle is therefore not a reformulation for its own sake: it derives the dynamics from a scalar functional, and that is the move which generalises. A Legendre transform then carries (2.2) to Hamilton's equations (D.2), whose gain is not the equations but the arena — phase space, where a state is a *point* whose evolution preserves volume, which is exactly the structure statistical mechanics needs in order to speak about probability densities over states.

## 2.2 From the Ising model to neural networks

Everything so far holds for arbitrary $H$; the physics, and the difficulty, enters when the energy couples many degrees of freedom. The canonical example is the *Ising model* (Ising, 1925): binary spins $\sigma_i \in \{-1, +1\}$ on the nodes of a graph $G = (V, \mathcal{E})$, with

$$H(\sigma) \;=\; -\sum_{(i,j)\in\mathcal{E}} J_{ij}\,\sigma_i\sigma_j \;-\; \sum_{i\in V} h_i\,\sigma_i, \tag{2.3}$$

and the Gibbs distribution $p(\sigma) \propto e^{-\beta H(\sigma)}$. Three features of it shape the rest of the thesis.

*The partition function is the hard part.* $Z = \sum_\sigma e^{-\beta H(\sigma)}$ runs over $2^{|V|}$ configurations and is #P-hard on general graphs; all of approximate inference exists because of that wall. On *trees*, however, $Z$ and every marginal are computable exactly in linear time by the transfer-matrix recursion — the same recursion Chapter 3 presents as belief propagation. That tractable island is precisely where the research model of this thesis lives.

*Collective order emerges.* Beyond a critical $\beta$ the two-dimensional model magnetises spontaneously and the symmetric state splits into two ergodic components. The dynamical counterpart of that symmetry breaking — a generation trajectory committing to one mode of a multimodal distribution — is the *speciation* phenomenon of diffusion models (Biroli et al., 2024; Raya and Ambrogioni, 2023).

*Disorder creates complexity.* Draw the $J_{ij}$ at random with both signs (Edwards and Anderson, 1975; Sherrington and Kirkpatrick, 1975) and transparency collapses: a triangle with two positive and one negative bond cannot satisfy all three, and the large-$N$

consequence of this *frustration* is a free-energy landscape shattered into exponentially many valleys. Controlling it took a decade of replica and cavity theory (Mézard et al., 1987) and the Thouless–Anderson–Palmer equations (Thouless et al., 1977), and earned Parisi the 2021 Nobel Prize. Two exports matter here: a language of rugged landscapes and metastable states, which machine learning speaks whenever it discusses loss surfaces; and a technology, since the cavity and TAP equations are recursions on *local* quantities solving a *global* inference problem, and are the direct ancestors of the message-passing algorithms (Mézard and Montanari, 2009; Zdeborová and Krzakala, 2016; Krzakala and Zdeborová, 2024) that Chapter 3 develops. That connection is historical descent, not analogy.

**Memory as a phase of matter.** Hopfield (1982) asked whether simple binary units could *store* and *retrieve* patterns collectively, the way a magnet stores a magnetisation. His construction is (2.3) in disguise: wire $N$ neurons to $P$ patterns $\xi^\mu$ by the Hebb rule (Hebb, 1949),

$$J_{ij} \;=\; \frac{1}{N}\sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu, \qquad H(\sigma) \;=\; -\frac{1}{2}\sum_{i\neq j} J_{ij}\,\sigma_i\sigma_j, \tag{2.4}$$

and iterate the simplest possible dynamics, aligning each neuron with its local field:

$$\sigma_i \;\leftarrow\; \operatorname{sign}\Big(\textstyle\sum_{j\neq i} J_{ij}\,\sigma_j\Big). \tag{2.5}$$

This descends $H$ at every flip (Appendix D.5), so on a finite configuration space every initial condition falls into an attractor, and retrieval *is* that fall. How many memories fit is a genuine phase-transition question: for random patterns with $P = \alpha N$, retrieval survives only below $\alpha_c \approx 0.138$ (Amit et al., 1985), beyond which the memories are crushed into a glassy phase. Memory here is a phase of matter in the technical sense — entered and left through sharp transitions. The 2024 Nobel Prize in Physics went to Hopfield and Hinton (The Royal Swedish Academy of Sciences, 2024): formal recognition that this lineage is physics rather than merely inspired by it.

**Learning enters.** The Hopfield network stores what its designer wires into it; the decisive step was to make the couplings the unknowns. Run the dynamics stochastically at temperature $1/\beta$ and its stationary law is exactly the Gibbs measure of (2.3). A *Boltzmann machine* (Ackley et al., 1985) is that object run backwards — find the $J$ making the Gibbs measure reproduce a dataset — and gradient ascent on the log-likelihood gives a rule of striking symmetry,

$$\frac{\partial}{\partial J_{ij}} \mathbb{E}_{\text{data}}\Big[\log p(\sigma)\Big] \;=\; \beta\left( \underbrace{\langle \sigma_i \sigma_j \rangle_{\text{data}}}_{\text{positive phase}} \;-\; \underbrace{\langle \sigma_i \sigma_j \rangle_{\text{model}}}_{\text{negative phase}} \right), \tag{2.6}$$

which stops when model and data correlations agree. This is the maximum-entropy principle of Toolbox D.3 turned into an algorithm, and it contains the field's central obstruction in miniature: the positive phase is a cheap average over data, while the negative phase is an average over the model's Gibbs measure — a partition-function computation, approachable only by sampling (Section E.2). Restricting the couplings to a bipartite visible/hidden graph (Smolensky, 1986) makes each layer conditionally independent given the other and made training practical (Hinton, 2002; Hinton et al., 2006). The lineage then split: the supervised branch scaled through backpropagation and convolutional networks (Rumelhart et al., 1986; Krizhevsky et al., 2012; LeCun et al., 2015) and left equilibrium physics behind, while the *generative* branch — the one this thesis lives in — kept the Gibbsian core, in which a model is an energy, learning is moment matching, and sampling is dynamics.

## 2.3 The statistical mechanics of learning

Physics also built the theory of these machines' behaviour. Gardner (1988) showed that learning itself is an equilibrium problem: compute, by replica methods, the volume of coupling space compatible with a set of examples, and capacity and generalisation come out as thermodynamics in *weight* space. That programme matured into the modern statistical physics of inference (Zdeborová and Krzakala, 2016), whose central lesson recurs throughout this thesis: high-dimensional inference problems have phases, and algorithms fail or succeed at sharp boundaries.

Two strands frame the research questions directly. The first is algorithmic: Mézard (2017) derived the TAP equations of the Hopfield model, exhibited the equivalence between Hopfield networks and restricted Boltzmann machines, and showed retrieval can be organised as message passing — the same machinery, transplanted from memories to diffusion scores, that the research chapters run. The second is dynamical: training undergoes phase transitions. Gradient descent in linear networks learns the singular modes of the data covariance in sequence (Saxe et al., 2014); an RBM's weight matrix aligns with the data's principal components early in training (Decelle et al., 2017), and hidden units acquire compositional receptive fields through condensation transitions (Tubiana and Monasson, 2017). The sharpest result in this line, Bachtis et al. (2024), examined in Section 2.13, shows RBM training to be an entire *cascade* of second-order transitions, one per learned feature. Learning, like memory, is a sequence of symmetry breakings.

The order of a transition — which derivative of the free energy first misbehaves — matters for reading these results, and is set out in Appendix D.6. The cascade above is second-order, which is why it is detected through susceptibilities rather than jumps. Both notions require the thermodynamic limit: in the finite models of Chapters 5 to 7, a correlation length that grows until it reaches the system size and then stops is a finite-size ceiling, not a divergence.

## 2.4 What this thesis takes from physics

A *complex system* is, operationally, a system whose macroscopic behaviour is not readable off its microscopic rules: interactions generate collective phenomena (order, phases, transitions, slow dynamics) that must be derived, not postulated. Statistical mechanics is the toolbox for that derivation, and after this chapter's story its objects map one-to-one onto the objects of modern generative modelling:

| **Statistical mechanics** | **Generative modelling** |
|---|---|
| microstate $x$ | data point / latent configuration |
| energy $H(x)$ | negative log-density $-\log p(x)$ (up to $Z$) |
| Boltzmann distribution $e^{-\beta H}/Z$ | model distribution |
| partition function $Z$ | normalising constant / evidence |
| free energy $F$ | negative ELBO (up to constants) |
| Hebbian couplings, attractors | stored memories, retrieval |
| moment matching (2.6) | maximum-likelihood training |
| Langevin dynamics | sampling / reverse diffusion |
| phase transition | feature condensation, speciation |
| cavity / TAP equations | belief propagation / AMP |

The next chapter supplies the probabilistic dynamics, Markov chains and their continuous-time limits, that turn the static ensembles of this chapter into processes that move probability around, which is what a diffusion model ultimately is.

## 2.5 Stochastic processes and stationarity

A *stochastic process* is a collection of random variables indexed by time, $(X_k)_{k\geq 0}$ in discrete time or $(X_t)_{t\geq 0}$ in continuous time, defined on a common probability space; equivalently, a probability law on the space of trajectories. Two notions of time homogeneity are used constantly in this thesis and are worth stating precisely (Brockwell and Davis, 1991).

**Definition 2.2** (Strict and weak stationarity)**.** A process $(X_k)$ is *strictly stationary* if all its finite-dimensional laws are invariant under time shifts: for every $n$, every set of indices $k_1, \dots, k_n$, and every shift $h$,

$$(X_{k_1}, \dots, X_{k_n}) \stackrel{d}{=} (X_{k_1+h}, \dots, X_{k_n+h}).$$

It is *weakly (second-order) stationary* if its mean is constant and its autocovariance depends only on the lag, $\mathrm{Cov}(X_k, X_{k+d}) = \gamma(d)$.

Strict stationarity implies weak stationarity whenever second moments exist; for *Gaussian* processes the two notions coincide, because a Gaussian law is determined by its first two moments. Stationarity has an immediate matrix consequence that the research chapters exploit: the covariance matrix of $K$ consecutive observations of a weakly stationary process, $(\Sigma)_{ij} = \gamma(i-j)$, is constant along diagonals, a symmetric *Toeplitz* matrix. Whenever a covariance in this thesis is Toeplitz, stationarity is the reason.

## 2.6 Markov chains: kernels, stationarity, reversibility

**Definition 2.3** (Markov chain)**.** A sequence of random variables $(X_k)_{k\geq 0}$ with values in a state space $\mathcal{X}$ is a *Markov chain* if for every $k$ and every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{P}\Big(X_{k+1} \in A \,\Big|\, X_k, X_{k-1}, \ldots, X_0\Big) \;=\; \mathbb{P}\Big(X_{k+1} \in A \,\Big|\, X_k\Big). \tag{2.7}$$

The conditional law is encoded by a *transition kernel* $M(x' \mid x)$: a density (or matrix, for discrete $\mathcal{X}$) with $\int M(x'|x)\,\mathrm{d}x' = 1$ for every $x$.

The Markov property (2.7) is exactly a statement about *factorisation*: the joint law of the first $K$ states is

$$p(x_0, x_1, \ldots, x_{K-1}) \;=\; p_0(x_0) \prod_{k=0}^{K-2} M(x_{k+1} \mid x_k). \tag{2.8}$$

This factorisation is the most heavily used equation of the thesis. It says that the joint distribution of a Markov trajectory is a product of *local* terms, each touching two consecutive variables: the structure that makes the chain a tree in the factor-graph sense of Chapter 3, that makes its precision matrix tridiagonal in the Gaussian case of Chapter 5, and that makes belief propagation exact.

Marginal distributions evolve by integrating the kernel: $p_{k+1}(x') = \int M(x'|x)\, p_k(x)\,\mathrm{d}x$, compactly $p_{k+1} = Mp_k$. A distribution $\pi$ is *stationary* (invariant) for $M$ if $\pi = M\pi$;

a chain started in $\pi$ stays in $\pi$ forever, and the trajectory is then a strictly stationary process in the sense of Definition 2.2. The kernel satisfies *detailed balance* with respect to $\pi$ if

$$\pi(x)\, M(x' \,|\, x) \;=\; \pi(x')\, M(x \,|\, x') \qquad \text{for all } x, x' : \tag{2.9}$$

the equilibrium probability flux from $x$ to $x'$ equals the reverse flux, so the chain, watched in equilibrium, is statistically indistinguishable run forwards or backwards. Detailed balance implies stationarity (integrate (2.9) over $x$), and its power is practical: it is a local, pointwise condition, checkable without computing any integral.

Stationarity alone does not guarantee that the chain *finds* $\pi$; for that one needs the chain to be able to reach everywhere (*irreducibility*) without getting trapped in cycles (*aperiodicity*). Under these conditions the fundamental convergence theorem holds (Robert and Casella, 2004): the marginal law of $X_k$ converges to $\pi$ from any initial condition, and time averages converge to expectations under $\pi$ (the ergodic theorem),

$$\frac{1}{N} \sum_{k=1}^{N} f(X_k) \;\xrightarrow[N\to\infty]{\text{a.s.}}\; \mathbb{E}_\pi[f]. \tag{2.10}$$

Equation (2.10) is the rigorous form of the ergodic postulate invoked in Section D.3, with the physical dynamics replaced by an algorithmic one.

## 2.7 The Ornstein–Uhlenbeck process: the corruption channel

The noising process used by diffusion models, and by every result of this thesis, is the *Ornstein–Uhlenbeck process* (Uhlenbeck and Ornstein, 1930): Brownian motion pulled back to the origin by a linear spring,

$$\mathrm{d}X_t \;=\; -X_t\,\mathrm{d}t \;+\; \sqrt{2}\,\mathrm{d}W_t, \qquad X_0 = a. \tag{2.11}$$

(The drift and diffusion constants are a normalisation choice, the variance-preserving convention of the diffusion literature (Song et al., 2021); any other choice rescales time.) The OU process is the unique Gaussian, Markov, stationary process in continuous time, and it is exactly solvable.

**Toolbox 2.1 — Exact solution of the OU SDE by integrating factor**

Multiply (2.11) by $e^t$ and consider $Y_t = e^t X_t$. By Itô's lemma with $\phi(x,t) = e^t x$ (here $\partial^2\phi/\partial x^2 = 0$, so no correction term arises):

$$\mathrm{d}Y_t = e^t X_t\,\mathrm{d}t + e^t\Big( - X_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t\Big) = \sqrt{2}\,e^t\,\mathrm{d}W_t.$$

The drift cancels exactly; that is what the integrating factor is for. Integrate from 0 to $t$ and undo the substitution:

$$X_t \;=\; a\,e^{-t} \;+\; \sqrt{2}\int_0^t e^{-(t-s)}\,\mathrm{d}W_s.$$

The integral is Gaussian (limit of Gaussian sums), with mean zero and variance given by the Itô isometry:

$$\mathrm{Var}(X_t) = 2\int_0^t e^{-2(t-s)}\,\mathrm{d}s = 2\,e^{-2t}\,\frac{e^{2t}-1}{2} = 1 - e^{-2t}.$$

Hence the *transition kernel* of the OU channel is

$$p(x_t \,|\, X_0 = a) \;=\; \mathcal{N}\Big(x_t;\; a\,e^{-t},\; \Delta_t\Big), \qquad \boxed{\Delta_t \;=\; 1 - e^{-2t}}.$$

□

The boxed kernel is used on every page of the research chapters, so we fix its reading now. Corrupting a data point $a$ for a time $t$ produces

$$x \;=\; a\,e^{-t} \;+\; \sqrt{\Delta_t}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1)\,: \tag{2.12}$$

the signal is *contracted* by $e^{-t}$ while isotropic noise of variance $\Delta_t$ fills in. At $t = 0$: $x = a$ (no corruption); as $t \to \infty$: $x \sim \mathcal{N}(0, 1)$ regardless of $a$ (all information destroyed, the equilibrium of the spring). The signal-to-noise ratio $e^{-2t}/\Delta_t$ sweeps continuously from $\infty$ to 0, and it diverges as $1/2t$ for small $t$, a factor that reappears throughout the thesis as an error amplifier. The OU channel interpolates between the data distribution and a standard Gaussian, which is precisely what a diffusion model needs.

Two structural remarks, both load-bearing later. First, the OU kernel is Gaussian *in $a$ as well as in $x$*: viewed as a function of the clean variable, $\mathcal{N}(x; ae^{-t}, \Delta_t) \propto \exp[-(e^{-2t}/2\Delta_t)(a - xe^t)^2]$, a Gaussian likelihood factor of precision $e^{-2t}/\Delta_t$ centred at the deconvolved observation $e^t x$. This is the fact that makes the posteriors of Chapter 5 jointly Gaussian and the messages of the BP computation closable. Second, if a random vector $a \sim \mathcal{N}(\mu, \Sigma_0)$ is pushed through (2.12) coordinate-wise with *independent* noises, the output is Gaussian with

$$\mu_t = e^{-t}\mu, \qquad \Sigma_t \;=\; e^{-2t}\,\Sigma_0 \;+\; \Delta_t\, I, \tag{2.13}$$

since means and covariances transform linearly and the added noise is white. Equation (2.13), shrink the covariance and add a multiple of the identity, is the entire forward process of this thesis in one line.

## 2.8 Time reversal: Anderson's theorem

One last piece of SDE theory converts everything above into a generative method. Suppose data $X_0 \sim p_0$ are corrupted by a forward SDE (E.4) up to time $T$, producing marginals $p_t$. Can the corruption be undone? Can we find an SDE which, run from $X_T \sim p_T$ backwards, reproduces the same marginals? The answer (Anderson, 1982) is yes, and the reverse drift needs exactly one unknown object:

$$\mathrm{d}X_t \;=\; \Big[f(X_t, t) \;-\; g(t)^2\,\nabla_x \log p_t(X_t)\Big]\,\mathrm{d}t \;+\; g(t)\,\mathrm{d}\bar{W}_t, \tag{2.14}$$

where time runs backwards from $T$ to 0 and $\bar{W}$ is a reverse-time Wiener process. The correction to the drift is the *score* $\nabla_x \log p_t$ of the noised marginal: the object defined in (1.1) and computed exactly, for Markov-chain data, in the research chapters. A heuristic that makes (2.14) plausible follows from the Fokker–Planck equation: rewriting its diffusion term as a transport term,

$$\frac{g^2}{2}\frac{\partial^2 p_t}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{g^2}{2}\, p_t\, \frac{\partial \log p_t}{\partial x}\right),$$

one sees that the density evolution of (E.4) is *identical* to that of the noise-free ODE with velocity $f - \frac{g^2}{2}\nabla \log p_t$ (the *probability-flow ODE* of Song et al., 2021); running that transport backwards, and re-injecting noise consistently, yields (2.14). The rigorous statement and proof are in Anderson (1982).

## 2.9 From algorithmic chains to data chains

A closing change of perspective, to set up the research problem. In this chapter Markov chains began as *algorithms*: their stationary law was the target, their trajectory a means to an end. From Chapter 5 onwards the chain (E.1) plays the opposite role: it is the *data-generating process*, its trajectory $(a_0, \ldots, a_{K-1})$ is one data point (a video of $K$ frames), and the object of interest is the probability law of the whole trajectory after each frame has been independently corrupted by the OU channel (2.12). The factorisation (2.8) then stops being a computational convenience and becomes the structure to be discovered: the research programme of this thesis asks how much of (2.8) survives noising, and how an inference algorithm, or a neural network, can exploit what survives. In one sentence: the OU channel destroys data at a known rate, the Fokker–Planck equation describes the destruction at the level of densities, and Anderson's theorem says the destruction is invertible by anyone who knows the score, which is why the rest of this thesis is about computing that score exactly when the data have Markov structure.

## 2.10 The diffusion idea

Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021) solve the generative problem with a two-process design inspired by non-equilibrium thermodynamics: a fixed *forward process* gradually corrupts data into pure noise, and a learned *reverse process* undoes the corruption step by step. A recent book-length survey of the family is Lai et al. (2025). The forward process used throughout this thesis is the OU channel of Section 2.7: at diffusion time $t$,

$$x_t \;=\; a\,e^{-t} + \sqrt{\Delta_t}\,\varepsilon, \qquad a \sim p_0, \quad \varepsilon \sim \mathcal{N}(0, I), \qquad \Delta_t = 1 - e^{-2t}, \tag{2.15}$$

so that the noised marginal is the convolution

$$p_t(x) \;=\; \int p_0(a)\,\mathcal{N}\Big(x;\, ae^{-t},\, \Delta_t I\Big)\,\mathrm{d}a. \tag{2.16}$$

As $t$ grows, $p_t$ interpolates from the data distribution to $\mathcal{N}(0, I)$. By Anderson's theorem (Section 2.8), the corruption is undone by the reverse SDE (2.14), whose only unknown ingredient is the score $\nabla_x \log p_t(x)$ of the noised marginals. Generative diffusion is therefore, at bottom, *score estimation*: whoever knows $\nabla \log p_t$ for all $t$ can sample from $p_0$. Note what this design buys relative to Section F.1: no energy is ever normalised, no equilibrium is ever awaited. The Markov-chain convergence problem of MCMC is replaced by a *fixed-horizon* integration of an SDE whose drift is learned. The partition-function problem has not been solved; it has been side-stepped, by working only with objects (scores) from which $Z$ has already been differentiated away.

### 2.10.1 The thermodynamic reading, and the direction of the arrow

The appeal to non-equilibrium thermodynamics in Sohl-Dickstein et al. (2015) is structural rather than decorative. Their forward process is a chain of small perturbations converting a structured distribution into a simple one, that is an annealing schedule, and the objective they derive is a variational bound of the same algebraic form as the work identities of non-equilibrium statistical mechanics: the log-likelihood is bounded by an average of log-ratios along the forward path, exactly as free-energy differences are bounded in Jarzynski-type equalities and estimated in annealed importance sampling. That analogy is what supplies the training objective, and it is the reason the field speaks of temperature, annealing and entropy at all.

The direction of the forward process is fixed by the second law, which for the Fokker–Planck equation (E.6) takes an exact form. Let $p_t$ evolve with a gradient drift and stationary law $p_\infty \propto e^{-V}$, and consider the free energy $\mathcal{F}[p_t] = D_{\mathrm{KL}}(p_t \,\|\, p_\infty)$, which is the mean energy minus the entropy up to a constant. Writing the dynamics as $\partial_t p = \nabla \cdot \big(p \nabla \log(p/p_\infty)\big)$ and integrating by parts,

$$\frac{\mathrm{d}\mathcal{F}}{\mathrm{d}t} = \int \partial_t p \, \log \frac{p}{p_\infty} = -\int p \left\| \nabla \log \frac{p}{p_\infty} \right\|^2 \;\leq\; 0, \tag{2.17}$$

with equality only at $p_t = p_\infty$. This is the H-theorem: the free energy decreases monotonically and the relative entropy is consumed at a rate given by a relative Fisher information. The forward channel therefore destroys information about the data monotonically and irreversibly, and (2.17) identifies the rate. The reverse direction is not forbidden, but it is not free either: by Anderson's theorem it requires the score of the intermediate laws to be supplied from outside. The entire modelling effort of a diffusion model is the estimation of that one object, and Chapters 5 to 7 are concerned with what that object looks like when the data are trajectories.

## 2.11 The score as a posterior expectation

The middle line of Vincent's derivation deserves its own section, because every result of this thesis passes through it. Rearranging $\nabla \log p_t(x) = \mathbb{E}[-(x - ae^{-t})/\Delta_t \mid x]$:

$$\boxed{\nabla_x \log p_t(x) \;=\; \frac{e^{-t}\,\mathbb{E}[\,a\,|\,x\,] - x}{\Delta_t} \qquad \Longleftrightarrow \qquad \mathbb{E}[\,a\,|\,x\,] = e^t\Big(x + \Delta_t\,\nabla_x \log p_t(x)\Big).} \tag{2.18}$$

We refer to (2.18) throughout as the *score–posterior identity*; in the empirical-Bayes literature it is known as Tweedie's formula (Robbins, 1956; Miyasawa, 1961; Efron, 2011). The score of the noisy marginal and the Bayes-optimal denoiser $\mathbb{E}[a|x]$ determine each other exactly, with no assumption whatsoever on $p_0$. Three readings will recur:

1. *Geometric*: the score points from $x$ towards (the contracted image of) the posterior mean of the clean data; denoising direction and score direction coincide.

2. *Statistical*: estimating the score is solving a Bayesian inference problem, namely computing a posterior expectation. The score is not *like* an inference target; it *is* one.

3. *Algorithmic*: for sequence data with a Markov prior, that inference problem lives on a tree-shaped graph, so the score is computable by message passing, the programme made precise in Chapter 3 and executed in Chapter 5.

For *vector-valued* data $a \in \mathbb{R}^K$ noised coordinate-wise, (2.18) holds componentwise with the *full* posterior:

$$\Big(\nabla_x \log p_t(x)\Big)_k \;=\; \frac{e^{-t}\,\mathbb{E}[\,a_k\,|\,x_0,\ldots,x_{K-1}\,] - x_k}{\Delta_t}. \tag{2.19}$$

The conditioning is on *all* coordinates: even though the noise is independent across frames, the posterior mean of frame $k$, and hence the $k$-th score component, depends on every observation, because the prior couples the frames. Equation (2.19) is the precise reason the *joint* score differs from the per-frame marginal score, a distinction that is easy to blur and whose importance for structured data was impressed on this project in supervision (Chapter 4); Chapter 5 quantifies exactly how much coupling the conditioning induces, as a function of $t$.

## 2.12 The statistical physics of generative diffusion

A line of work initiated by Biroli, Mézard, and collaborators analyses diffusion models with the tools of Chapter 2, in the regime where the score is *exact*: either computed from a solvable $p_0$ or from the empirical measure of $n$ training points. Because this is the literature our research extends, we review it in some detail.

**High dimensions and the cost of asymmetry.** Biroli and Mézard (2023) analyse generative diffusion in the high-dimensional limit for two controlled data models, a Gaussian model and the Curie–Weiss ferromagnet. In the latter, the reverse dynamics undergoes the symmetry-breaking mechanism of Section 2.2: at a critical diffusion time the trajectory commits to one of the two low-temperature states. Their sharpest quantitative point concerns data requirements: to reconstruct the *relative weights* of two asymmetric modes, the number of training samples must be much larger than the data dimension, and the paper characterises the scaling laws in dimension and sample size for efficient generation.

**The three regimes of the backward dynamics.** Biroli et al. (2024) study the backward process when the score has been trained optimally on $n$ samples in dimension $d$, and identify three dynamical regimes separated by two transitions. Starting from pure noise, the trajectory first behaves as a free (Brownian-like) process; at the *speciation* time it commits to the gross structure of the data (a class, a mode), through a mechanism analogous to spontaneous symmetry breaking; at the later *collapse* time the trajectory is attracted to an individual memorised training point, through a mechanism analogous to condensation in a glass. The speciation time is computable from a spectral analysis of the data correlation matrix, and the collapse time from an excess-entropy estimate; the dependence of the collapse time on $n$ and $d$ quantifies the curse of dimensionality for diffusion models. Related transition phenomena along the generation trajectory were exhibited by Raya and Ambrogioni (2023).

**Speciation with general class structure.** Achilli et al. (2026) develop the general theory of the speciation transition: where earlier analyses required classes separated by their first moments (Gaussian mixtures with distinct means), they characterise speciation times for arbitrary class structures through a free-entropy difference criterion, recovering the known Gaussian-mixture results and extending them to classes that differ only through higher-order or collective features, with successive speciation events for multiple classes. A hierarchical counterpart is Sclocchi et al. (2025b): in a hierarchical generative model of data, the backward process started from time $t$ exhibits a phase transition at a threshold time at which the probability of preserving the high-level class drops abruptly, while low-level features evolve smoothly; beyond the transition a regenerated sample can retain low-level elements of the original image while its class has changed. Sclocchi et al. (2025a) turn that transition into an instrument: forward–backward experiments around the threshold expose the latent hierarchy of real text and image data, since the correlated blocks that change together grow as the transition is approached. Favero et al. (2025a) follow the same hierarchy through training rather than through the schedule, finding that grammatical rules are acquired level by level, each needing more data than the one below, so compositional generalisation and creativity improve in stages. The statistical-physics treatment of generative diffusion in solvable models is developed at book length in the doctoral thesis of Achilli (2026).

**Memorisation and generalisation.** When the score is learned from finitely many samples rather than computed exactly, the question becomes when generation generalises and when it memorises. Bonnaire et al. (2025) identify two timescales in the training dynamics: an early time $\tau_{\text{gen}}$ at which the model begins to generate high-quality samples, and a later time $\tau_{\text{mem}}$ beyond which memorisation of training data emerges; $\tau_{\text{mem}}$ grows linearly with the training-set size $n$ while $\tau_{\text{gen}}$ stays constant, opening a widening window in which overparameterised models generalise, a form of implicit dynamical regularisation. Garnier-Brun et al. (2026) refine the picture by identifying a phase of *biased* generalisation in between: the test loss still decreases while the model increasingly favours samples anomalously close to training points, so the standard early-stopping criterion does not certify genuine novelty. Ghio et al. (2024) map generative samplers (flows, diffusions, autoregressive networks) onto statistical-physics sampling problems and delimit where each is efficient, importing the algorithmic-hardness picture of spin glasses (Zdeborová and Krzakala, 2016).

Three further readings of the same puzzle are worth separating, because they locate the explanation in three different objects. Favero et al. (2025b) place it in the training dynamics, as Bonnaire et al. (2025) do, and show that the safe window widens with overparameterisation rather than closing — bigger models are not straightforwardly more prone to memorising. Kadkhodaie et al. (2024) place it instead in the learned *representation*: two denoisers trained on disjoint halves of a dataset converge to nearly the same function, and the inductive bias they share is a geometry-adaptive harmonic basis, which is a property of the architecture and not of the samples. Achilli et al. (2024) place it in the geometry of the empirical score itself, reading memorisation off the eigenvalue spectrum of its Jacobian as a progressive collapse of dimensions, so that memorisation is directional rather than global. A solvable two-layer model in which the whole trajectory can be tracked asymptotically is given by Cui et al. (2025).

The common obstacle is stated most clearly by putting these side by side: each compares a trained network against a target that is itself unavailable in closed form, so the network's error and the difficulty of its target cannot be separated. This is exactly the separation that Chapter 5 onwards buys by choosing a data model whose exact score is computable.

**The exactly solvable programme, and what it holds fixed.** The common method of these works is to choose $p_0$ rich enough to be interesting and simple enough that $\nabla \log p_t$ is available in closed form, then study the exact generation dynamics. What none of them vary is the *internal structure of a single data point*: their coordinates are exchangeable or i.i.d. conditional on a mode. There is no ordering and no dynamics *inside* the sample. That axis is precisely where this thesis works.

## 2.13 Learning as a cascade of phase transitions

The works above put phase transitions on the *sampling* side of generative modelling. A complementary result, Bachtis et al. (2024), which this section examines in detail, puts them on the *training* side, and completes the argument of Section 2.3 that learning itself is a sequence of symmetry breakings. The object of study is the restricted Boltzmann machine of Section 2.2, the minimal trainable EBM, with energy $E(v,h) = -v^{\mathsf{T}}Wh - b^{\mathsf{T}}v - c^{\mathsf{T}}h$ and the moment-matching gradient (2.6). The central idea is to track training as a sequence of symmetry breakings in the student's singular modes, and the result is that an RBM learning a Hopfield teacher with two correlated patterns undergoes not one transition but a *cascade*: each singular direction condenses at its own critical time, in order of the strength of the data covariance along it, and each condensation is a second-order transition of Curie–Weiss type. The mapping onto Curie–Weiss, the two-rate argument that produces the ordering, and the confirmation on MNIST, CelebA and human-genome data are in Appendix D.7.

**What this means for the thesis.** Phase transitions now appear on both faces of generative diffusion: in *sampling*, as speciation events along the reverse trajectory (Biroli et al., 2024; Achilli et al., 2026), and in *training*, as the cascade just described. Both are hierarchies of symmetry breakings ordered by the data's spectrum. Our research adds the axis that both literatures hold fixed: the structure of the inference problem *inside a single data point*, and how the diffusion time $t$ deforms it. The chain we solve is the minimal model in which that question is non-trivial and answerable exactly.

## 2.14 The gap, restated

Real sequential data (video, audio, trajectories) have exactly the structure the solvable programme leaves out: an internal (frame) time with a Markovian dependence along it. For such data the score (2.19) is a smoothing posterior on a chain, an object with seventy years of inference theory behind it. Yet the questions a diffusion modeller asks about it are not answered by the classical literature, which fixes the noise level, nor by the diffusion literature, which ignores internal structure: how does the coupling structure of $\nabla \log p_t$ deform with $t$? Can it be computed locally? What survives a Gaussian parametrisation of the messages when the data are not Gaussian? What limited guidance does that give for score architectures? Answering these questions, in the fully solvable setting of the AR(1) chain and its non-Gaussian perturbations, is the contribution of Chapters 5 and 6, with the inference tools prepared in Chapter 3 and the path that led to the precise formulation recorded in Chapter 4.

# Chapter 3

# Graphical Models and Message Passing

This chapter introduces the representation the thesis computes in: a distribution written as a product of local factors, and the recursion that returns its marginals exactly when those factors form a tree. It is short by design. The multivariate Gaussian in both parametrisations, the spectral and matrix-inversion tools, the Gaussian special case of the recursion, and its relation to TAP and AMP are all in Appendix G; what is kept here is the structure, because it is the structure and not the algebra that decides what is computable.

## 3.1 Markov random fields

The Gaussian story above assigned structure to matrices. Graphical models assign it to distributions, for arbitrary (not necessarily Gaussian) variables.

**Definition 3.1** (Markov random field)**.** Let $G = (V, \mathcal{E})$ be an undirected graph with one random variable $x_i$ per node. The collection $x = (x_i)_{i \in V}$ is a *Markov random field* (MRF) if it satisfies the local Markov property: for every node $i$,

$$p\Big(x_i \,\Big|\, x_{V \setminus \{i\}}\Big) \;=\; p\Big(x_i \,\Big|\, x_{\partial i}\Big), \tag{3.1}$$

where $\partial i$ is the set of neighbours of $i$ in $G$.

The fundamental structure theorem connects this conditional-independence definition to the energy picture of Appendix F: by the Hammersley–Clifford theorem (proved in Besag, 1974), a strictly positive density is an MRF on $G$ if and only if it factorises over the *cliques* of $G$,

$$p(x) \;=\; \frac{1}{Z} \prod_{C \in \mathrm{cliques}(G)} \psi_C(x_C) \;=\; \frac{1}{Z} \exp\Big[ -\sum_C E_C(x_C) \Big]. \tag{3.2}$$

Conditional independence, graphical locality, and additive local energies are one and the same thing. The Ising model (2.3) is the textbook example. The decisive example for us is one-dimensional: the chain factorisation (2.8) makes a Markov chain an MRF on a line, with pairwise nearest-neighbour energies. For *Gaussian* MRFs the correspondence becomes matrix-concrete: the energy is the quadratic form $\frac{1}{2}a^\top J a - h^\top a$, its pairwise terms are the entries $J_{ij}$, and the graph of the MRF is exactly the sparsity pattern of the precision matrix. The dual reading of zeros of Section G.1 and the Hammersley–Clifford theorem are the same statement, once in linear algebra and once in probability. A Gaussian Markov chain therefore has a tridiagonal precision, a fact Chapter 5 derives explicitly and exploits throughout.

One more structural remark that the research chapters rely on: if each variable of an MRF is observed through its own independent noisy channel, the likelihood factors touch one variable each, so no clique is enlarged. *The posterior of an MRF under per-node observations is an MRF on the same graph.* Observation changes the potentials, never the graph.

## 3.2 Factor graphs

MRFs record *which* variables interact but not *how* the joint factorises: a triangle of pairwise terms and a single triple term have the same graph. The representation that makes the factorisation itself graphical, and on which message passing is most naturally defined, is the *factor graph* (Kschischang et al., 2001; Koller and Friedman, 2009).

**Definition 3.2** (Factor graph)**.** Given a factorisation $p(x) = \frac{1}{Z}\prod_a \psi_a(x_{\partial a})$, its factor graph is the bipartite graph with a *variable node* for each $x_i$, a *factor node* for each $\psi_a$, and an edge $(i, a)$ whenever $x_i \in x_{\partial a}$.

A connected factor graph with one fewer edge than nodes is a *tree*: it contains no cycles, so any edge separates it into two disjoint components. Trees are the tractable case of inference: on them, marginals and partition functions are computable exactly by the recursion of the next section, whatever the alphabet of the variables. The factor graph of a Markov chain observed through per-node noise is a tree of a particularly simple shape (a spine of variables and transition factors, with one evidence leaf per variable), as Chapter 5 verifies by counting.

## 3.3 The sum–product algorithm

On a factor graph, sum–product belief propagation (BP) passes messages along every edge (Pearl, 1988; Kschischang et al., 2001; Mézard and Montanari, 2009). Variable-to-factor messages multiply the incoming factor messages from all *other* neighbours, and factor-to-variable messages integrate the factor against the incoming variable messages:

$$\nu_{i\to a}(x_i) \propto \prod_{b\in\partial i\setminus a} \hat{\nu}_{b\to i}(x_i), \qquad \hat{\nu}_{a\to i}(x_i) \propto \int \psi_a(x_{\partial a}) \prod_{j\in\partial a\setminus i} \nu_{j\to a}(x_j)\, \mathrm{d}x_{\partial a\setminus i}, \tag{3.3}$$

with the belief at node $i$ proportional to $\prod_{a\in\partial i} \hat{\nu}_{a\to i}(x_i)$. The "exclude the recipient" rule is the heart of the algorithm: the message $i \to a$ summarises everything node $i$ knows *except* what it learned from $a$, so that no piece of evidence is ever counted twice along a path.

**Theorem 3.3** (Tree exactness of sum–product)**.** *On a tree-shaped factor graph, the equations* (3.3) *have a unique solution, reached after one inward and one outward pass, and the resulting beliefs are the exact marginals of $p$ (Mézard and Montanari, 2009, Theorem 14.1).*

The reason is the separation property of trees: each edge splits the graph into two disjoint halves, and the message crossing it summarises exactly the half behind it. On graphs with cycles the same updates, iterated, define *loopy* BP, which is extraordinarily useful but approximate (Weiss and Freeman, 2001; Malioutov et al., 2006); none of that is needed in this thesis, because our graphs are trees.

**The representation problem.** For *discrete* variables each message is a finite probability vector and each update in (3.3) is finite arithmetic. For *continuous* variables each message is a probability *density*, an infinite-dimensional object, and the update integrals must be carried out in function space. Tree exactness (a property of the graph) says nothing about representability of the messages (a property of the model's function class); keeping the two apart is essential, and the distinction is where all the difficulty of the non-Gaussian research chapter lives. There are three ways to make continuous BP computable: restrict to models whose messages close in a finite-dimensional family (the Gaussian route, next section); represent messages numerically on a grid (the route our non-Gaussian reference solver takes, Chapter 6); or project each message onto a tractable family and accept the error (the route whose cost this thesis measures).

## 3.4 Locality in learned architectures: convolutions, receptive fields, U-Nets

The message-passing story has an architectural mirror, and it supplies the vocabulary for the locality questions of the research chapters. A convolutional network processes a signal by applying the same local filter at every position, stacking such layers so that depth enlarges the *receptive field*: the window of input coordinates that can influence a given output coordinate (LeCun et al., 2015). Convolutional architectures powered the breakthrough of deep learning in vision (Krizhevsky et al., 2012), and their central inductive bias is precisely locality plus translation invariance: the value of an output pixel is computed from a bounded neighbourhood, identically across the image. The standard denoising backbone of diffusion models, the U-Net (Ronneberger et al., 2015; Ho et al., 2020), is a convolutional encoder–decoder with skip connections, whose receptive field grows along the contracting path and whose skip connections reinject fine-scale local information along the expanding path.

Why local architectures should be *sufficient* for score estimation is a theoretical question, and message passing has begun to answer it. Mei (2024) shows that for generative hierarchical models (tree-structured graphical models over image-like data), the denoising function computed by belief propagation can be implemented efficiently by a U-Net, with the network's layers mirroring the sweeps of the message-passing schedule; convolutional layers are efficient implementations of the local BP updates. This gives the U-Net an inference-algorithm reading: the architecture is good at denoising because denoising, on structured data, *is* message passing, and message passing is local.

The correspondence is not confined to convolutional architectures. Garnier-Brun et al. (2024) train transformers on data generated by a hierarchical model in which the exact posterior is again computable by belief propagation, and find that the trained attention maps come to reproduce the message-passing schedule — the network discovers the inference algorithm rather than approximating the function some other way. Read together with Mei (2024), the claim is architecture-independent: what a denoiser must implement is determined by the graphical structure of the data, and the natural depth is set by that structure. This is precisely why the chain matters here. A chain is a degenerate tree, of depth equal to its length and width two, so the depth these results would predict is not the logarithmic depth of a balanced hierarchy but a depth linear in the correlation length — a sharply different architectural prediction, on a model where it can be checked.

The present thesis works one step below this correspondence, on a model small enough that the locality of the score is not an architectural hypothesis but a computable quantity. For the Gaussian chain, the exact score at frame $k$ depends on the observation at frame $k \pm d$ with a weight that decays geometrically in $d$, at a rate $q(\alpha, t)$ computed in closed form in Chapter 5; the error of the best radius-$r$ local approximation decays as $q^r$, and the comparison between truncating the inference and truncating the score matrix quantifies, in one solvable case, how a local computation should be organised. These results describe the function a network would have to learn rather than the dynamics of learning it; the trained baselines of Chapter 9 supply the other half of that comparison.

# Chapter 4

# The Object of Study, and the Minimal Model

Before any of the analysis, two choices have to be justified: that the quantity worth studying is the *joint* score of a whole trajectory rather than a per-frame score, and that the stationary Gaussian AR(1) chain under coordinatewise OU noise is the right model in which to study it. This short chapter settles both. The exploratory models that led here, and the sequence of formulations they ruled out, are recorded in the companion compendium rather than in this document.

## 4.1 Why the joint score

Standard diffusion models generate *static* objects; the data of interest here are *dynamic*, a data point being a sequence of frames indexed by an internal time $u = 0, 1, \ldots, K-1$ — a video, a rotating shape, a stochastic trajectory. Two times therefore appear from the outset, and keeping them apart is a permanent discipline of this thesis: the *internal* time $u$ of the sequence, and the *diffusion* time $t$ of the noising channel, which acts independently on every frame.

It is tempting to treat such a trajectory as $K$ separate samples, computing each frame's noisy marginal score independently. For the stationary models of this thesis that is not merely lossy — it is empty. Under the variance-preserving channel every noisy frame is marginally $\mathcal{N}(0,1)$ at every diffusion time, so the per-frame marginal score is exactly $-x_k$: independent of $t$, independent of the dynamics, and identical for every model considered anywhere in this thesis. A per-frame treatment of a stationary sequence model cannot distinguish a fast chain from a slow one, or a Gaussian innovation from a Laplace one, because it returns the same function in all cases.

A sequence is *one* sample from a joint law, and the quantity that drives the reverse dynamics is the joint score of the whole noisy trajectory, the gradient of $\log P_t(x_0,\ldots,x_{K-1})$, whose $k$-th component is a conditional expectation over the *full* posterior of the clean frames, as in (2.19). All information about temporal structure lives in that object and in no marginal of it. Chapter 5 exhibits the contrast in closed form at $K=2$.

This also settles the shape of the structural question. One might hope that consecutive frame-blocks of the score are related by a fixed map, a "propagator",

$$S_{:,u}(x,t) \;\approx\; \mathcal{P}_t\Big(S_{:,u-1}(x,t)\Big), \tag{4.1}$$

so that a learning system could hold one block and one map instead of $K$ blocks. What the joint view shows is that the structure worth studying is not a map between per-frame scores at all, but the coupling pattern of the joint score — a statement about a matrix in the Gaussian case, and about a posterior in general. That is the object the rest of this thesis analyses.

## 4.2 The minimal model

Among the candidate dynamic objects — rotations, coupled phases, bounded trajectories with reflections, curved data manifolds — the stationary Gaussian AR(1) chain under coordinatewise OU noise is the minimal one in which the joint score is nontrivial yet exactly computable: one coupling parameter $\alpha$ for the internal dynamics against one noise parameter $t$ for the corruption, linear algebra available at every step, and every

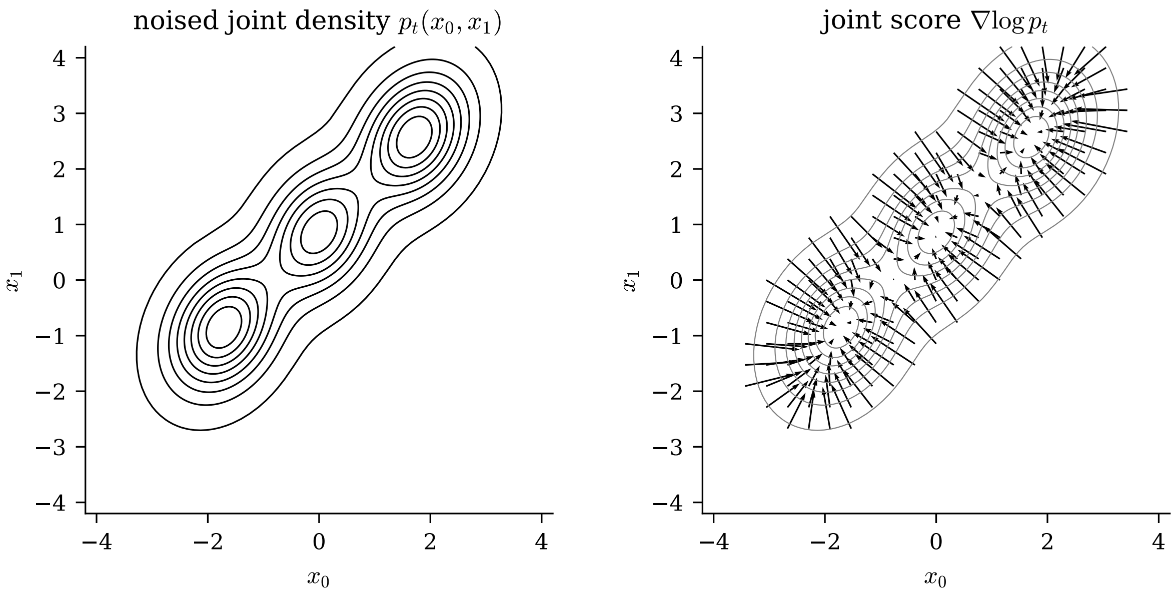


**Figure 4.1:** The coupling made visible, in the smallest case where it exists: $x_1 = x_0 + c + \eta$ with a trimodal Gaussian-mixture prior on $x_0$, drift $c = 1$, innovation standard deviation $\sigma = 0.45$, after diffusion time $t = 0.15$. Left: the noised joint density $p_t(x_0, x_1)$. Right: the joint score field $\nabla \log p_t$ (arrows) over the same density (contours). Both panels are computed from the closed-form mixture expressions on a grid. The arrows point towards the modes and tilt along the diagonal, and that tilt is the entire effect of the coupling between frames — it is exactly what the per-frame score $-x_k$ discards.

structural question (coupling range, locality, message passing) posable in closed form.

The requirement doing the work here is *exact computability.* The whole programme of this thesis is to measure what approximations cost — Gaussian message closure in Chapter 6, an estimated transition kernel in Chapters 8 and 9, a finite receptive field in Chapter 5 — and every one of those measurements needs a reference that is known rather than approximated. A model in which the answer is available in closed form is not a simplification of the question; it is what makes the question answerable at all.

The nonlinear models are set aside at that point deliberately, as targets for after the solvable case is understood. Chapter 7 takes up the first of them.

# Chapter 5

# The Gaussian Chain: Exact Score, Belief Propagation, and Locality

This chapter answers RQ1, RQ2, and the Gaussian half of RQ4. For the stationary AR(1) chain under the OU channel we compute the joint score exactly by two independent routes (linear algebra and Bayesian smoothing), characterise how its coupling structure evolves with diffusion time, derive the belief-propagation computation of the same score and identify it with the Kalman filter and smoother, and quantify how rapidly the influence of remote observations decays. Everything in this chapter is exact unless explicitly marked as a numerical check; each boxed formula is covered by a named check of the audit suite (Appendix C).

## 5.1 The joint law and the object of study

A data point is a trajectory $a = (a_0, \ldots, a_{K-1})$ of the stationary AR(1) chain (E.1) with $\sigma_\eta^2 = 1 - \alpha^2$. Each frame passes independently through the OU channel (2.12) for a time $t$,

$$x_k \;=\; e^{-t} a_k + \sqrt{\Delta_t}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0,1) \text{ i.i.d.}, \tag{5.1}$$

and the object of study is the joint score of the noisy sequence,

$$S(x,t) \;=\; \nabla_x \log P_t(x) \;\in\; \mathbb{R}^K, \qquad P_t(x) = \int_{\mathbb{R}^K} P_0(a) \prod_{k=0}^{K-1} \mathcal{N}\Big(x_k;\, e^{-t}a_k,\, \Delta_t\Big)\, \mathrm{d}a. \tag{5.2}$$

The gradient is taken with respect to all $K$ noisy frames simultaneously, under the law of the whole noisy trajectory (Section 2.11).

## 5.2 The clean law: dense covariance, sparse precision

The chain factorisation (2.8) gives the clean joint density

$$P_0(a) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{K-2} \mathcal{N}\Big(a_{k+1};\, \alpha a_k,\, \sigma_\eta^2\Big) = \mathcal{N}(a;\, 0,\, \Sigma_0). \tag{5.3}$$

**Proposition 5.1** (Stationary covariance). $(\Sigma_0)_{ij} = \alpha^{|i-j|}$*: a dense, symmetric Toeplitz matrix with geometrically decaying off-diagonals (Toolbox E.1).*

**Proposition 5.2** (Tridiagonal precision). $Q_0 := \Sigma_0^{-1}$ *is exactly tridiagonal:*

$$(Q_0)_{kk} = \begin{cases} \dfrac{1}{\sigma_\eta^2}, & k \in \{0, K-1\}, \\[2ex] \dfrac{1+\alpha^2}{\sigma_\eta^2}, & 0 < k < K-1, \end{cases} \qquad (Q_0)_{k,k\pm1} = -\frac{\alpha}{\sigma_\eta^2}, \tag{5.4}$$

*and every entry at distance $\geq 2$ vanishes identically.*

**Toolbox 5.1 — Deriving the tridiagonal precision from the log-density**

For a centred Gaussian, the precision is the Hessian of the negative log-density. From (5.3),

$$-2\log P_0(a) = \frac{a_0^2}{\sigma_0^2} + \sum_{k=0}^{K-2} \frac{(a_{k+1} - \alpha a_k)^2}{\sigma_\eta^2} + \text{const}, \qquad \sigma_0^2 = 1.$$

Read off the quadratic form term by term. The only term coupling $a_k$ and $a_{k+1}$ is the transition $(a_{k+1} - \alpha a_k)^2/\sigma_\eta^2$, with cross coefficient $-2\alpha/\sigma_\eta^2$, hence precision entry $-\alpha/\sigma_\eta^2$; no term couples variables at distance $\geq 2$, so those entries are exactly zero. An interior $a_k$ appears in its left transition with coefficient $1/\sigma_\eta^2$ and in its right transition with $\alpha^2/\sigma_\eta^2$, summing to $(1+\alpha^2)/\sigma_\eta^2$. At the left boundary the prior term $1/\sigma_0^2 = (1-\alpha^2)/\sigma_\eta^2$ plus the right transition $\alpha^2/\sigma_\eta^2$ give $1/\sigma_\eta^2$; the right boundary is symmetric. $\square$

*Remark* 5.3 (Tridiagonality is Markovianity, not Gaussianity). A zero in the precision matrix is a conditional independence given all other variables; by the Hammersley–Clifford theorem (Section 3.1), factorisation over the cliques of a graph and conditional independence with respect to that graph are the same property. Gaussianity entered the derivation only to identify the precision with the Hessian of $-\log P_0$; the support of that Hessian, nearest neighbours only, comes from the chain factorisation alone. Any Markov chain with smooth transitions has this sparsity pattern; the Gaussian case makes the entries constants.

The covariance is dense (every frame correlates with every other) while the precision is tridiagonal: correlation is global, conditional dependence is local. The score is governed by the second object.

## 5.3 The score, two ways

### 5.3.1 Linear algebra

**Proposition 5.4** (Noised law). $P_t(x) = \mathcal{N}(x;\, 0,\, \Sigma_t)$ *with*

$$\boxed{\Sigma_t \;=\; e^{-2t}\,\Sigma_0 \;+\; \Delta_t\, I_K} \tag{5.5}$$

This is (2.13). The diagonal is $e^{-2t} + \Delta_t = 1$ at every $t$ (variance preservation); only the off-diagonal entries, $\alpha^{|i-j|}e^{-2t}$, depend on the diffusion time, because the noise of different frames is independent. Since $\log P_t(x) = -\frac{1}{2}x^\top \Sigma_t^{-1} x + \text{const}$,

$$\boxed{S(x,t) \;=\; -\,Q_t\, x, \qquad Q_t := \Sigma_t^{-1}.} \tag{5.6}$$

The joint score is linear in $x$, the hallmark of Gaussianity, and its entire structure is the noisy precision matrix $Q_t$: which frames couple, how strongly, and at which $t$, are questions about this one matrix, answered in Sections 5.4 and 5.7.

### 5.3.2 Bayesian inference

The score–posterior identity (2.19) expresses the same score through the posterior mean of the clean chain,

$$S_k(x,t) = \frac{e^{-t}\,\mathbb{E}[a_k \,|\, x] - x_k}{\Delta_t}, \tag{5.7}$$

and the posterior is Gaussian with parameters used throughout:

**Proposition 5.5** (Posterior of the clean chain)**.** $P(a \,|\, x) = \mathcal{N}(a;\, m,\, J^{-1})$ *with*

$$\boxed{J \;=\; \frac{e^{-2t}}{\Delta_t} I \;+\; Q_0, \qquad J\,m \;=\; h := \frac{e^{-t}}{\Delta_t}\,x.} \tag{5.8}$$

**Toolbox 5.2 — Posterior precision and field, by completing the square**

Bayes' rule with the likelihood (5.1):

$$\begin{aligned} -2\log P(a|x) &= a^{\mathsf{T}} Q_0\, a + \sum_k \frac{(x_k - e^{-t} a_k)^2}{\Delta_t} + \text{const} \\ &= a^{\mathsf{T}}\Big(Q_0 + \tfrac{e^{-2t}}{\Delta_t} I\Big) a - 2\,\tfrac{e^{-t}}{\Delta_t}\, x^{\mathsf{T}} a + \text{const}. \end{aligned}$$

Matching against the generic form $a^{\mathsf{T}} J a - 2h^{\mathsf{T}} a$ gives (5.8). Two structural observations. (i) The likelihood is diagonal in $a$: observing each frame through its own channel adds the channel precision $e^{-2t}/\Delta_t$ to every diagonal entry of the prior precision and touches nothing else. (ii) Consequently $J$ is tridiagonal at every $t$: the *posterior* of the clean chain remains a nearest-neighbour Markov field; what fills up with $t$ is the precision $Q_t$ of the noisy marginal law. Keeping $J$ (tridiagonal, posterior) and $Q_t$ (dense, noisy marginal) distinct prevents much confusion. □

Substituting $m = J^{-1}h$ into (5.7) must return $-Q_t x$; the operator identity behind this is Woodbury's (Appendix A), and the audit suite checks the two routes agree to $2.7\times10^{-15}$ across sweeps of $(K, \alpha, t)$. The agreement is an end-to-end check of two computations built from different formulas.

### 5.3.3 The smallest example, $K = 2$

For $K = 2$, with $r := \alpha\, e^{-2t}$,

$$\Sigma_t = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} \qquad \Longrightarrow \qquad S_0(x,t) = -\frac{x_0 - r\,x_1}{1 - r^2}, \qquad S_1(x,t) = -\frac{x_1 - r\,x_0}{1 - r^2}. \tag{5.9}$$

By contrast, each $x_k$ is marginally $\mathcal{N}(0,1)$ at every $t$ (variance preservation), so the per-frame marginal score is $-x_k$: independent of $t$ and of the other frame, it carries no information about the dynamics. The joint score contains the cross term $-rx_1/(1-r^2)$: the denoiser of frame 0 uses evidence from frame 1 with a weight that equals $\alpha$ at $t = 0$ and decays as $e^{-2t}$. This is the coupling structure whose full lifecycle the next section derives, in miniature.

## 5.4 The spectral form and the lifecycle of the precision

How does $Q_t$ interpolate between the tridiagonal $Q_0$ at $t = 0$ and the identity at $t = \infty$? The analysis rests on the spectral theorem and its affine-family corollary (Section G.2): affine images of a symmetric matrix share its eigenbasis, $(cA + dI)\,u_i = (c\,\omega_i + d)\,u_i$. Applying this to the forward map (5.5), which is affine in $\Sigma_0$, gives the representation from which everything in this section follows.

**Theorem 5.6** (Spectral form). *Let $\Sigma_0 = U\operatorname{diag}(\omega_i)U^\top$. Then for every $t \geq 0$,*

$$\boxed{\Sigma_t = U\operatorname{diag}\big(\lambda_i(t)\big)U^\top, \qquad Q_t = U\operatorname{diag}\Big(\frac{1}{\lambda_i(t)}\Big)U^\top, \qquad \lambda_i(t) = e^{-2t}\omega_i + \Delta_t.} \tag{5.10}$$

Diffusion never rotates the problem: the eigenvectors of $\Sigma_t$ and $Q_t$ are those of $\Sigma_0$, frozen for all $t$, while each eigenvalue flows along the affine path $\omega_i \mapsto e^{-2t}\omega_i + \Delta_t$, monotonically towards 1. Because $\Sigma_0$ is symmetric Toeplitz, its eigenvectors are asymptotically the Fourier modes of the sequence (Section G.2): in frequency space the denoising problem is $K$ independent scalar problems at every diffusion time. Two computational corollaries are recorded for later use: factoring $\Delta_t$ out of (5.5) gives the SNR form

$$Q_t = \tfrac{1}{\Delta_t}\Big(I + \gamma_t\,\Sigma_0\Big)^{-1}, \qquad \gamma_t := e^{-2t}/\Delta_t, \tag{5.11}$$

and multiplying (5.5) by $Q_0$ and inverting gives the resolvent form

$$Q_t = e^{2t}\,Q_0\Big(I + c_t Q_0\Big)^{-1} = e^{2t}\sum_{m\geq 0}(-c_t)^m\,Q_0^{m+1}, \qquad c_t := e^{2t} - 1, \tag{5.12}$$

whose Neumann series converges for $\|c_t Q_0\| < 1$, i.e. at small $t$.

### 5.4.1 Losing tridiagonality: when, how, and how much

Sparsity is a property of a matrix in a basis. In the eigenbasis the problem is diagonal at all times; in the frame basis, the operationally meaningful one, the zeros of $Q_0$ are a cancellation that diffusion destroys. The destruction is quantifiable, and it answers three questions: when tridiagonality is lost, how, and how much structure survives at each $t$.

**When: immediately.** By (5.10) every entry of $Q_t$ is analytic in $t$. The distance-$d$ entries, identically zero at $t = 0$ for $d \geq 2$, are non-zero for every $t > 0$: no exact zero survives any positive diffusion time.

**How: one diagonal per order in $t$.** The loss is nevertheless graded, and the resolvent form gives the exact rate at which each diagonal switches on.

**Theorem 5.7** (Band-fill leading order). *For small $t$ and frame distance $d \geq 1$,*

$$\boxed{(Q_t)_{i,i+d} \;=\; (-1)^{d-1}\,(2t)^{d-1}\left(Q_0^d\right)_{i,i+d} \;+\; O(t^d).} \tag{5.13}$$

**Toolbox 5.3 — Band fill from the Neumann series**

In the resolvent form (5.12), substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$:

$$Q_t = \Big(1 + O(t)\Big)\Big[Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \cdots\Big].$$

$Q_0$ is tridiagonal, so $Q_0^m$ has bandwidth $m$: the entry at distance $d$ receives its first contribution from the single term $m = d$, with prefactor $(-c_t)^{d-1} = (-1)^{d-1}(2t)^{d-1}(1+O(t))$; every later term is $O(t^d)$. Each additional frame of coupling range costs one power of $t$: the band fills outward, one diagonal per order. □

*Remark* 5.8 (Correction of an earlier formula). An earlier working note of this project stated (5.13) with an extraneous factor $1/(d-1)!$. The factor does not follow from the expansion (the leading distance-$d$ term is a single Neumann term, with no combinatorial sum behind it), and the numerical audit rejects it: the no-factorial coefficient matches the measured $(Q_t)_{i,i+d}/t^{d-1}$ to better than 0.9% for every $d \in \{1, \ldots, 5\}$ (at $\alpha = 0.9$, $K = 25$), while the factorial version fails by 86% already at $d = 3$. The error had survived because slope fits in $\log t$ are blind to constant prefactors; every prefactor in this thesis has since been audited, not only every exponent.

**How much: the off-tridiagonal mass.** Define the off-tridiagonal Frobenius mass of $Q_t$: the norm of everything outside the band that Markov structure would allow. At small $t$ the band-fill law makes it rise from zero, dominated by the $d = 2$ diagonal, linearly in $t$. At large $t$, expanding the inverse of $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ gives

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}), \tag{5.14}$$

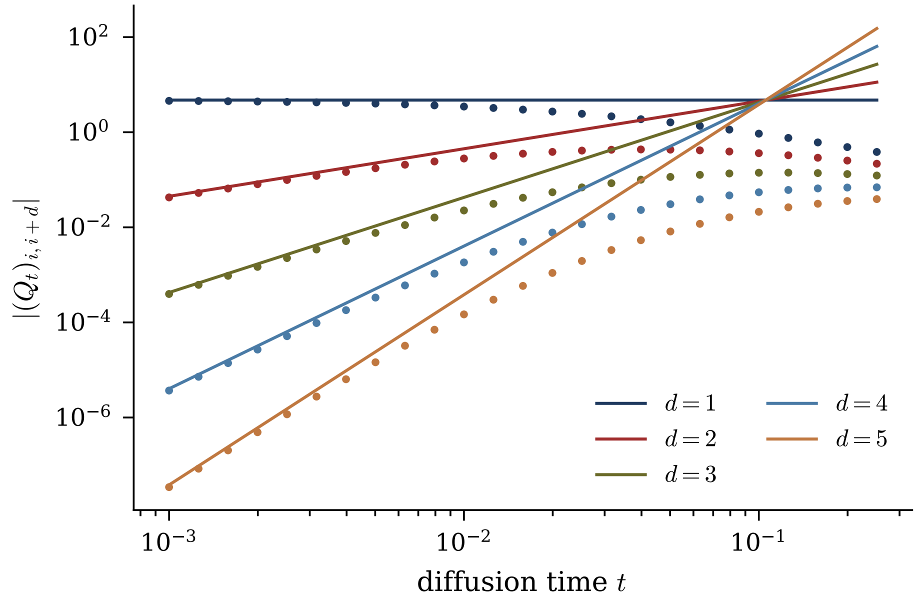


**Figure 5.1:** Band-fill law. Measured $|(Q_t)_{i,i+d}|$ against $t$ (dots) and the leading order (5.13) (solid lines, no fitted parameters), for $\alpha = 0.9$, $K = 25$, $d = 1, \dots, 5$. The slopes are $d-1$; agreement at small $t$ is at the percent level for every $d$.

so $\|Q_t - I\|_F \simeq e^{-2t}\|\Sigma_0 - I\|_F$: all structure, off-tridiagonal and tridiagonal alike, decays at the rate $e^{-2t}$, and the score approaches the isotropic $-x$. In between, the off-tridiagonal mass peaks (Figure 5.2): the noisy sequence is most non-Markov at intermediate diffusion times. The lifecycle in one line: tridiagonal at $t = 0$; densifying at rate $t^{d-1}$ per diagonal $d$; maximally coupled at intermediate $t$; isotropic as $t \to \infty$ at rate $e^{-2t}$.

## 5.5 Belief propagation on the chain

The matrix route inverts a $K \times K$ matrix, a global computation over the whole sequence. The score–posterior identity says the same object is a posterior expectation, and the posterior has a structure that supports local computation. This section develops that computation once, in full.

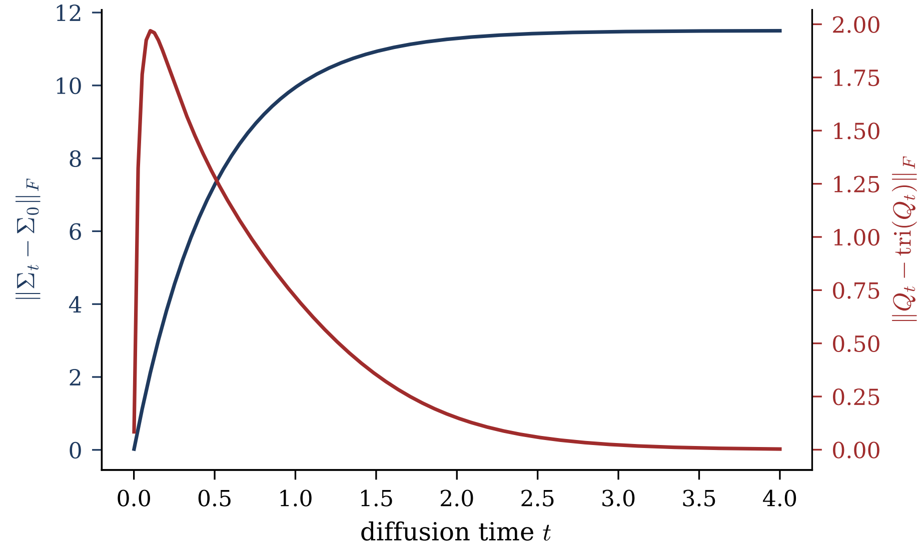


**Figure 5.2:** Structure across diffusion time ($\alpha = 0.8$, $K = 40$). $\|\Sigma_t - \Sigma_0\|_F$ (navy) grows monotonically as the covariance is driven to the identity; the off-tridiagonal Frobenius mass of $Q_t$ (red) rises from zero, peaks at intermediate $t$, and decays to zero: the band fills, then empties.

### 5.5.1 The factor graph of the posterior

For any Markov-chain prior and per-frame OU evidence, Bayes' rule gives

$$p(a\,|\,x) \;\propto\; \psi_{\mathrm{pr}}(a_0) \prod_{k=0}^{K-2} \psi_{\mathrm{tr}}^{(k)}(a_k, a_{k+1}) \prod_{k=0}^{K-1} \psi_{\mathrm{ob}}^{(k)}(a_k), \qquad \psi_{\mathrm{ob}}^{(k)}(a_k) = \mathcal{N}\Big(x_k;\, e^{-t}a_k,\, \Delta_t\Big), \tag{5.15}$$

a hidden Markov model (Rabiner, 1989): a latent chain observed through independent per-frame noise. By (2.19), the $k$-th score component requires exactly the smoothing marginal $\mathbb{E}[a_k|x]$ of (5.15).

The factor graph of this posterior (Section 3.2) has $K$ variable nodes, $2K$ factor nodes (one prior, $K-1$ transitions, $K$ observations; the data $x_k$ are constants clamped inside their factors) and $3K-1$ edges. A connected graph with one fewer edge than nodes is a tree: here, a spine of variables and transition factors with one evidence leaf per variable.

**Proposition 5.9.** *For any Markov-chain prior and any per-frame observation model, the posterior $p(a\,|\,x)$ factorises over a tree-shaped factor graph, and all $K$ smoothing marginals* *are computed exactly by one forward and one backward sweep of sum–product messages.*

Proposition 5.9 counts sweeps, not work: it says nothing about how the messages are represented, and for continuous variables each message is a function (Section 3.3). Keeping tree-exactness of the algorithm (a graph property) separate from representability of the messages (a property of the model's function class) is the conceptual hinge of this thesis; the Gaussian case closes the gap, and Chapter 6 measures what happens when it no longer does.

### 5.5.2 The sum–product recursion, with the bookkeeping explicit

Specialising the sum–product rules (3.3) to this tree, the schedule collapses into one forward and one backward sweep (Theorem 3.3), and the bookkeeping deserves care.

**Proposition 5.10** (Chain sweeps, Convention A)**.** *Define messages*

$$\mu_{\to 0}(a_0) := \psi_{\mathrm{pr}}(a_0), \quad \mu_{\to k+1}(a_{k+1}) := \int \psi_{\mathrm{tr}}^{(k)}(a_k, a_{k+1})\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\to k}(a_k)\, \mathrm{d}a_k, \tag{5.16}$$

$$\mu_{\leftarrow K-1}(a_{K-1}) := 1, \qquad \mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\mathrm{tr}}^{(k-1)}(a_{k-1}, a_k)\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\leftarrow k}(a_k)\, \mathrm{d}a_k. \tag{5.17}$$

*Then for every $k$,*

$$p(a_k \,|\, x) \;\propto\; \mu_{\to k}(a_k)\; \psi_{\mathrm{ob}}^{(k)}(a_k)\; \mu_{\leftarrow k}(a_k)\;: \tag{5.18}$$

*the forward message into $a_k$ carries the evidence $x_0, \ldots, x_{k-1}$ only, the backward message carries $x_{k+1}, \ldots, x_{K-1}$ only, and the local evidence enters once, at combination time.*

The proof is direct marginalisation: integrating (5.15) over all $a_j$, $j \neq k$, splits at $a_k$ by the Markov property into independent left and right halves, and peeling the innermost variable of each half yields the recursions.

*Remark* 5.11 (The double-counting trap). An alternative convention absorbs the local evidence into the forward message, $\mu^{\mathrm{B}}_{\to k} = \mu_{\to k} \cdot \psi_{\mathrm{ob}}^{(k)}$; the product (5.18) then counts $x_k$ twice and must be corrected by dividing one factor back out. An early version of this project's notes fell into this trap, caught by the numerical audit: under Convention A the BP score matches the matrix score to $10^{-14}$, under the naive alternative the disagreement is of order one. Every message in this thesis is therefore stated with its evidence content explicit.

## 5.6 The Gaussian closure and the Kalman smoother

For the Gaussian chain, the two closure lemmas of Toolbox G.1 are the only facts needed: the recursions (5.16)–(5.17) perform exactly one product and one affine marginalisation per step. The local evidence, read as a function of $a_k$, is itself Gaussian, $\psi_{\mathrm{ob}}^{(k)}(a_k) \propto \mathcal{N}(a_k;\, e^t x_k,\, \Delta_t e^{2t})$: a pseudo-observation at the deconvolved location $e^t x_k$ with precision $e^{-2t}/\Delta_t$, the pinning strength that appears throughout this chapter.

**Theorem 5.12** (Gaussian closure of BP on the chain)**.** *For the Gaussian AR(1) chain under the OU channel, every BP message and every belief is exactly Gaussian, at every node, for every $(K, \alpha, t)$. Each message is carried by two numbers, each update is a few arithmetic operations, and the two sweeps compute the exact posterior marginals, hence the exact joint score. Here, and only here, the sweep count of Proposition 5.9 translates into an actual computation of fixed, small cost per frame.*

*Derivation.* Induction along each sweep. The initial messages are Gaussian (the prior; the flat message is the zero-precision limit of the family). Each step of (5.16)–(5.17) is one application of Lemma 1 of Toolbox G.1 (absorb the local evidence) followed by one of Lemma 2 (push through the transition); the combination (5.18) is Lemma 1 twice. Exactness follows from tree exactness of sum–product (Theorem 3.3). □

*Remark* 5.13 (Closure is algebra, not a central limit theorem). The closure just proved holds at node degree two exactly as it would at any degree: it is algebra, not an aggregation effect, as discussed in Section G.3. The central-limit argument matters only for the per-node closures of statistical physics (Section G.4), and on the chain those fail in a way that can be made exact, with a breakdown time $t_c(\alpha)$ and a critical coupling $\alpha_c = \sqrt{2} - 1$ (Appendix B).

Executing the Gaussian sweeps in mean–variance arithmetic reproduces, update by update, the classical algorithms of linear estimation.

**Toolbox 5.4 — The Kalman filter, derived from the BP sweep**

Write the forward message in Convention A as $\mu_{\to k} = \mathcal{N}(a_k;\, m_{k|k-1},\, P_{k|k-1})$, the predicted state given past observations. One BP step does two things.

*Update (absorb the evidence; Lemma 1).* Multiplying by the pseudo-observation $\mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$ gives the filtered Gaussian $\mathcal{N}(a_k; m_{k|k}, P_{k|k})$ with

$$\frac{1}{P_{k|k}} = \frac{1}{P_{k|k-1}} + \frac{e^{-2t}}{\Delta_t},$$
$$m_{k|k} = P_{k|k}\left(\frac{m_{k|k-1}}{P_{k|k-1}} + \frac{e^{-t}x_k}{\Delta_t}\right) = m_{k|k-1} + G_k\Big(e^t x_k - m_{k|k-1}\Big),$$

where $G_k = P_{k|k}\, e^{-2t}/\Delta_t \in (0,1)$ is the Kalman gain: the filtered mean is the prediction corrected by a gain-weighted innovation (Kalman, 1960).

*Predict (push through the dynamics; Lemma 2).*

$$m_{k+1|k} = \alpha\, m_{k|k}, \qquad P_{k+1|k} = \alpha^2 P_{k|k} + \sigma_\eta^2.$$

The forward BP sweep is the Kalman filter, message by message.

*Smoothing (combine both directions).* The belief (5.18) multiplies the filtered forward Gaussian by the backward message (Lemma 1 once more), producing the posterior marginal given all observations, $\mathcal{N}(a_k; m_{k|K-1}, P_{k|K-1})$; carrying out the same algebra for the backward recursion yields the fixed-interval smoother of Rauch et al. (1965). The two-sweep BP schedule on the chain is the Kalman filter followed by the RTS smoother, exactly as the generic correspondence of Section G.3 predicts. □

Combining with the score–posterior identity answers RQ2: the $k$-th component of the joint diffusion score is an affine function of the Kalman-smoothed estimate of frame $k$,

$$S_k(x,t) \;=\; \frac{e^{-t}\, m_{k|K-1} - x_k}{\Delta_t}. \tag{5.19}$$

For Markov sequences, score computation is a smoothing problem, and the classical theory of linear smoothing applies directly.

**Numerical agreement of the two routes.** The audit suite runs BP against the matrix route over 80 configurations ($K \in \{2,3,5,8,16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$, random $x$): maximum disagreement $1.3\times10^{-15}$ on posterior means, $3.9\times10^{-15}$ on variances, $1.2\times10^{-14}$ on scores, i.e. double-precision rounding accumulated along the recursion. An independent implementation, written from scratch against the recursion specification alone, reproduces the same agreement (Appendix C). One qualitative detail from a worked $K = 3$ instance: at $x = (1.2, -0.4, 0.8)$ the score component $S_1$ is positive although $x_1 < 0$, because the neighbours pull the posterior mean of the middle frame positive. No per-frame marginal score can produce this; it is inter-frame evidence sharing in numbers.

## 5.7 The bulk in closed form, and the price of locality

For a long chain the interior (bulk) is translation invariant and can be solved exactly. By Proposition 5.5 the posterior precision in the bulk is tridiagonal Toeplitz with

$$\boxed{J_d \;=\; \frac{e^{-2t}}{\Delta_t} + \frac{1+\alpha^2}{\sigma_\eta^2}, \qquad \beta \;=\; -\frac{\alpha}{\sigma_\eta^2}, \qquad \sigma_\eta^2 = 1-\alpha^2.} \tag{5.20}$$

The diagonal $J_d$ is the sum of the evidence precision $e^{-2t}/\Delta_t$ (divergent as $1/2t$ at small $t$, vanishing at large $t$) and the constant prior precision; the coupling $\beta$ is pure prior. The dimensionless ratio $|\beta|/J_d \in (0, \frac{1}{2})$ controls everything below.

**Theorem 5.14** (Exact bulk variance and geometric covariance)**.** *In the bulk of a long* *chain, the posterior variance and covariance are*

$$\boxed{V \;=\; \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \qquad \Big(J^{-1}\Big)_{i,\,i+d} \;=\; q^{\,d}\, V, \qquad q \;=\; \frac{J_d - \sqrt{J_d^2 - 4\beta^2}}{2\,|\beta|} \in (0,1).} \tag{5.21}$$

**Toolbox 5.5 — Transfer-matrix inversion of a tridiagonal Toeplitz matrix**

Seek the inverse row in geometric form, $(J^{-1})_{i,i+d} = Cq^{\,d}$ for $d \geq 0$, symmetric in $\pm d$. The defining relation $\sum_j J_{ij}(J^{-1})_{j,i+d} = \delta_{0d}$ reads, for $d \geq 1$,

$$\beta\, Cq^{\,d-1} + J_d\, Cq^{\,d} + \beta\, Cq^{\,d+1} = 0 \quad \Longleftrightarrow \quad \beta q^2 + J_d\, q + \beta = 0,$$

a quadratic whose root inside the unit disc is $q = (J_d - \sqrt{J_d^2 - 4\beta^2})/(2|\beta|)$ (for $\alpha > 0$ all inverse entries are positive; for $\alpha < 0$ they alternate in sign with the same modulus). The $d = 0$ relation fixes the prefactor: $J_d C + 2\beta C q = 1$, and since $2\beta q = -(J_d - \sqrt{J_d^2 - 4\beta^2})$, $C = 1/\sqrt{J_d^2 - 4\beta^2} = V$. Existence of the real square root requires $J_d > 2|\beta|$, which always holds here:

$$J_d - 2|\beta| = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2 - 2|\alpha|}{\sigma_\eta^2} = \frac{e^{-2t}}{\Delta_t} + \frac{(1 - |\alpha|)^2}{\sigma_\eta^2} \;>\; 0$$

for every $|\alpha| < 1$ and $t > 0$. This positivity margin is also what guarantees the belief-propagation fixed point exists for every coupling and diffusion time (Appendix B). $\square$

The number $q(\alpha, t)$ measures how far evidence travels along the chain: the posterior correlation length is $\xi = 1/\log(1/q)$. Its limits are

$$\begin{aligned} t \to 0 :& \quad q \approx \frac{|\beta|}{J_d} \to 0 \quad \text{(each frame pinned by its own observation);} \\ t \to \infty :& \quad q \to |\alpha| \qquad\quad \text{(posterior} \to \text{prior),} \end{aligned} \tag{5.22}$$

the second because $q \to \big((1+\alpha^2)-(1-\alpha^2)\big)/(2|\alpha|) = |\alpha|$ when the evidence term vanishes. As diffusion time runs, the posterior correlation length climbs monotonically from 0 to the prior's own $1/\log(1/|\alpha|)$.

### 5.7.1 The locality law

RQ4 asks how rapidly the influence of remote observations decays. Define the local estimator of radius $r$: the posterior mean of $a_k$ given only the observations within distance $r$, $m_k^{(r)} = \mathbb{E}[a_k \mid x_{k-r}, \ldots, x_{k+r}]$. Two facts make the comparison clean. Any contiguous window of a stationary AR(1) chain is itself a stationary AR(1) chain, so $m_k^{(r)}$ is exact on its window; the only approximation is the truncation of range. And both estimators are linear in $x$, so the error $m_k - m_k^{(r)} = e^\top x$ has a closed-form RMS over the data law, $\mathrm{RMS}_r = \sqrt{e^\top \Sigma_t e}$, a deterministic number with no sampling involved.

**Proposition 5.15** (Geometric locality: influence coefficients)**.** *In the bulk, the influence of the observation at distance $d$ on the exact posterior mean is $(J^{-1})_{k,k\pm d} \propto q(\alpha, t)^d$, exactly, by Theorem 5.14. Truncating the evidence at radius $r$ therefore discards a leading term of order $q^{r+1}$.*

What Proposition 5.15 controls is the size of the *discarded* evidence, and that part is exact. It does not by itself prove that the radius-$r$ estimator's error inherits the rate: $m_k^{(r)}$ is the posterior mean of a *re-normalised* model on the window, so truncation changes the boundary conditions as well as removing the leading coefficient, and bounding the difference between those two effects is not done here.

The rate is instead confirmed numerically, and closely. Fitted slopes of $\log \mathrm{RMS}_r$ over $r = 2, \ldots, 13$ at the centre of a $K = 121$ chain match $\log q$ within 0.2% at $t \in \{0.1, 0.5, 2.0\}$ ($\alpha = 0.8$); see Figure 5.3, where the dashed lines carry the predicted slope with no fitting. The claim this thesis makes is therefore: an exact geometric law for the influence coefficients, and a measured geometric law with the same rate for the windowed estimator's error, over the stated range of $r$. A two-sided bound establishing the second from the first is left open.

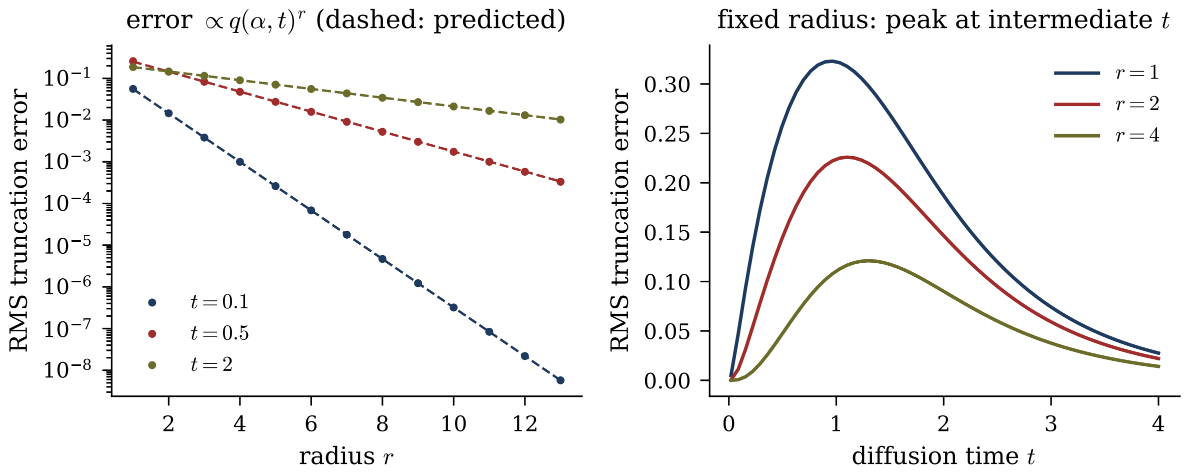


**Figure 5.3:** Locality of the joint score ($\alpha = 0.8$, $K = 121$, deterministic RMS over the data law). Left: exact truncation error of the radius-$r$ local estimator (dots) at three diffusion times; the dashed lines have the predicted slopes $q(\alpha, t)^r$ from (5.21), with no fitting. Right: the same error at fixed radius as a function of $t$: it vanishes at both ends and peaks at intermediate $t$, where the band of $Q_t$ is fullest.

### 5.7.2 Truncating the matrix versus truncating the inference

A truncated score can be built in two inequivalent ways. *Truncate the answer*: band the matrix, $\widehat{S} = -Q_t^{(b)} x$ with entries beyond $|i - j| \le b$ zeroed. *Truncate the algorithm*: let messages travel at most $r$ hops, i.e. run exact inference on the window $x_{k-r}, \ldots, x_{k+r}$ (the local estimator above). Both are linear in $x$, so their RMS errors are again deterministic. At equal locality budget ($b = r = 1$, $K = 40$, $\alpha = 0.8$):

| relative RMS score error | $t = 0.05$ | $t = 0.3$ | $t = 1$ | $t = 3$ |
|---|---|---|---|---|
| banded matrix ($b = 1$) | 0.210 | 0.302 | 0.135 | 0.0035 |
| truncated inference ($r = 1$) | 0.123 | 0.235 | 0.130 | 0.0035 |

Two observations. The cost of locality is non-monotone in $t$ and peaks at intermediate times, consistent with Section 5.4.1. And at small $t$ truncating the algorithm beats truncating the answer (12.3% against 21.0% at $t = 0.05$): the windowed inference solves a correctly normalised problem on its window, while zeroing entries of an inverse produces a matrix that is the inverse of nothing and misstates the diagonal. The two coincide at large $t$, where the off-diagonal content vanishes. Either error is measured to decay geometrically in the allowed range at the rate $q(\alpha, t)$ of Proposition 5.15.

## 5.8 Status of the results

Exact, by derivation: the score formulas (5.6) and (5.7), the spectral form (5.10), the band-fill law (5.13) as a small-$t$ asymptotic, the large-$t$ law (5.14), tree exactness and Gaussian closure of BP, the Kalman identification, and the bulk formulas (5.20)–(5.21), including the $q^d$ decay of the influence coefficients. Numerical, within stated grids and tolerances: the agreement of the two score routes, the audited band-fill coefficients, the locality slopes of the *windowed estimator* (whose rate is measured rather than derived, see Proposition 5.15), and the truncation table. Interpretation, limited to this model: within the Gaussian benchmark, a local score computation with receptive field $r \gtrsim \xi \log(1/\varepsilon)$, $\xi = 1/\log(1/q)$, achieves relative error $\varepsilon$, and the requirement is most demanding at intermediate diffusion times; whether trained score networks on real sequence data behave accordingly is not tested here.

# Chapter 6

# Beyond Gaussianity: Laplace Innovations and the Gaussian-Message Approximation

Every exact result of Chapter 5 rests on the closure of the Gaussian family under the two operations the model performs, multiplying densities and marginalising. This chapter removes that property in the most controlled way available and answers RQ3. We keep the AR(1) recursion, the OU channel, and the factor-graph structure, and change only the innovation law from Gaussian to Laplace: heavy-tailed and kinked at the origin, yet log-concave and analytically tractable, it is a minimal step outside the Gaussian family, so every deviation we find is attributable to non-Gaussianity alone. The chapter first establishes what remains exact (the score–posterior identity, the tree structure, one boundary integral, and a differential characterisation of the $K = 1$ marginal), then locates precisely where closed forms stop, and finally measures, against a validated numerical reference, what a Gaussian message family costs.

## 6.1 What changes, and what survives

The Laplace AR(1) chain replaces the Gaussian innovations of (E.1) with Laplace ones, keeping the initial law and the second-order structure fixed:

$$a_0 \sim \mathcal{N}(0,1), \qquad a_k = \alpha\, a_{k-1} + \varepsilon_k, \qquad \varepsilon_k \sim \mathrm{Laplace}(0,b) \text{ i.i.d.}, \qquad k = 1,\dots,K-1, \tag{6.1}$$

observed through the same per-frame OU channel. The innovation scale is *not* free: $b = \sqrt{(1-\alpha^2)/2}$, so that $\mathrm{Var}(\varepsilon_k) = 1-\alpha^2$ and the chain has unit marginal variance and $\mathrm{Cov}(a_j, a_k) = \alpha^{|j-k|}$, exactly as in the Gaussian case. This is what makes the comparison in this chapter a comparison: the Gaussian and Laplace chains agree on every second moment and differ *only* beyond them, the innovation's excess kurtosis (0 against $+3$) being the single non-Gaussianity knob. The same convention — Gaussian initial state, variance-matched innovations — is used for every non-Gaussian family in this thesis, and carries forward unchanged to the estimated chain of (8.1). Three structural facts survive unchanged, because none of them used Gaussianity of the prior. The score–posterior identity (2.19) holds for any $p_0$. The posterior factorises over the same tree-shaped factor graph (Proposition 5.9), so the sum–product recursion (5.16)–(5.17) still computes exact smoothing marginals in one forward and one backward sweep. And the OU likelihood factors are still Gaussian in $a_k$. What is lost is representability: the messages are no longer Gaussian, no finite-dimensional family we know of is closed under the updates, and the cost of an update is no longer bounded. The score also ceases to be linear in $x$, which is the first thing to make precise.

## 6.2 One frame: the score becomes nonlinear

For $K = 1$ there is no chain; the model isolates the ingredient "non-Gaussian prior meets Gaussian channel". Let $a \sim \mathrm{Laplace}(0,b)$, $p_0(a) = e^{-|a|/b}/2b$, and write $\mu := e^{-t}$. The

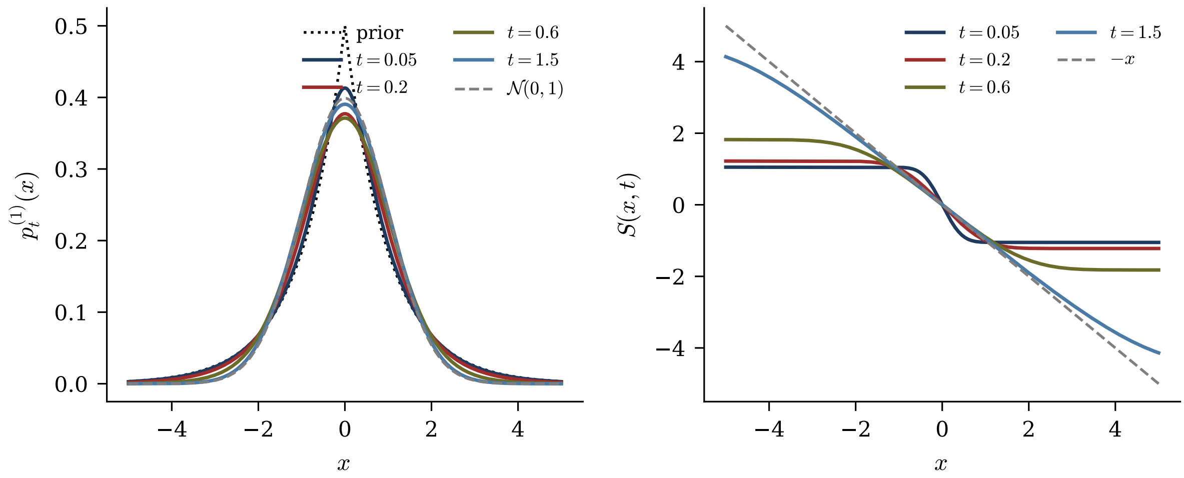


**Figure 6.1:** The Laplace $K = 1$ problem ($b = 1$). Left: noised marginal $p_t^{(1)}$ at increasing diffusion times, from the kinked Laplace prior (dotted) to the Gaussian attractor (dashed). Right: the score at the same times; the dashed line is the attractor $-x$. The nonlinearity lives near $x = 0$ and flattens as $t$ grows; the limits are the exact statements (6.3). Curves computed by deterministic quadrature.

noised marginal is the convolution

$$p_t^{(1)}(x) = \int p_0(a)\, \mathcal{N}\Big(x;\, \mu a,\, \Delta_t\Big)\, \mathrm{d}a. \tag{6.2}$$

As $t \to 0$ the Gaussian kernel concentrates and $p_t^{(1)} \to p_0$; as $t \to \infty$, $p_t^{(1)} \to \mathcal{N}(0, 1)$, the OU attractor. By the score–posterior identity, $S(x, t) = (\mu\, \mathbb{E}[a|x] - x)/\Delta_t$; for a Laplace prior the conditional mean is a nonlinear function of $x$, and the score inherits the nonlinearity. Its limits are exact: for every fixed $x \neq 0$,

$$t \to 0: \quad S(x, t) \;\longrightarrow\; \partial_x \log p_0(x) = -\frac{\operatorname{sign}(x)}{b}, \qquad\qquad t \to \infty: \quad S(x, t) \;\longrightarrow\; -x. \tag{6.3}$$

At small $t$ the score converges pointwise to the sign-shaped score of the prior, with the nonlinearity concentrated in the kink near $x = 0$; at large $t$ it flattens into the linear attractor score $-x$ (Figure 6.1). The audit verifies the posterior-expectation route against central finite differences of $\log p_t^{(1)}$ computed by quadrature: maximum disagreement $1.19 \times 10^{-7}$ over a grid of $(b, t, x)$, at the truncation floor of the finite difference.

### 6.2.1 A differential characterisation of the noised marginal

The channel writes $x$ as a sum of independent variables, so (6.2) is a convolution, which factorises in Fourier space:

$$\widehat{p}_t^{(1)}(k) = \frac{1}{1+b^2\mu^2k^2}\, e^{-\Delta_t k^2/2}, \tag{6.4}$$

the product of the (rescaled) Laplace and Gaussian characteristic functions.

**Theorem 6.1** (Screened Poisson equation)**.** *The Laplace noised marginal solves the linear, constant-coefficient ODE*

$$\boxed{\left(1 - b^2\mu^2\,\partial_x^2\right) p_t^{(1)}(x) \;=\; G_{\Delta_t}(x),} \qquad G_{\Delta_t}(x) = \frac{e^{-x^2/2\Delta_t}}{\sqrt{2\pi\Delta_t}}. \tag{6.5}$$

**Toolbox 6.1 — From the characteristic function to the screened Poisson equation**

Multiply (6.4) by $(1+b^2\mu^2k^2)$:

$$\left(1+b^2\mu^2k^2\right)\widehat{p}_t^{(1)}(k) = e^{-\Delta_t k^2/2},$$

whose right-hand side is the characteristic function of $G_{\Delta_t}$. Multiplication by $k^2$ in Fourier space is $-\partial_x^2$ in real space, so the polynomial $(1+b^2\mu^2k^2)$ becomes the operator $(1-b^2\mu^2\partial_x^2)$, and (6.5) follows by inverse transformation. The audit checks (6.4) against direct quadrature of the characteristic function to $2.9\times10^{-6}$, the floor of oscillatory quadrature. □

Equation (6.5) states that the noised marginal is the solution of a screened Poisson equation with a Gaussian source, with screening length $b\mu = be^{-t}$: the memory of the prior's tails shrinks as $e^{-t}$. The deviation from Gaussianity is governed by a single local differential operator whose strength dies with diffusion time.

One further exact consequence identifies what a Gaussian approximation must get wrong. Differentiating the score–posterior identity once more,

$$H_t(x) := -\partial_x^2 \log p_t^{(1)}(x) = \frac{1}{\Delta_t}\left(1 - \frac{\mu^2}{\Delta_t}\,\mathrm{Var}[a\,|\,x]\right). \tag{6.6}$$

For a Gaussian prior the posterior variance is independent of $x$ and $H_t$ is constant. For the Laplace prior $\mathrm{Var}[a|x]$ depends on $x$ and $H_t$ is a nontrivial function: the breakdown of a constant curvature is the second-order form of the breakdown of score linearity. A Gaussian message is exactly the object that cannot represent an $x$-dependent curvature, so (6.6) states what the Gaussian closure of Section 6.5 will miss, before it is measured.

## 6.3 The chain: where closed forms stop

**Theorem 6.2** (Joint score from chain messages)**.** *For any chain prior, in particular* (6.1)*, and the OU likelihood, the joint score is, componentwise,*

$$S_k(x,t) = \frac{\mu\,\mathbb{E}[a_k|x] - x_k}{\Delta_t}, \qquad \mathbb{E}[a_k|x] = \frac{\displaystyle\int a_k\; \mu_{\to k}(a_k)\; \psi_{\mathrm{ob}}^{(k)}(a_k)\; \mu_{\leftarrow k}(a_k)\; \mathrm{d}a_k}{\displaystyle\int \mu_{\to k}(a_k)\; \psi_{\mathrm{ob}}^{(k)}(a_k)\; \mu_{\leftarrow k}(a_k)\; \mathrm{d}a_k}, \tag{6.7}$$

*with the messages of Proposition 5.10.*

This is a complexity statement: the $K$-dimensional posterior integral is replaced by two one-dimensional message recursions plus one one-dimensional integral per frame. Computational locality survives non-Gaussianity; what does not survive is the collapse of those one-dimensional integrals into closed-form arithmetic. Exactly one such integral has a closed form.

**Proposition 6.3** (One-step boundary integral)**.** *For the Laplace transition* $M(a'|a) = \frac{1}{2b}e^{-|a'-\alpha a|/b}$ *and OU evidence* $\mathcal{N}(x';\mu a',\Delta_t)$*,*

$$\int_{\mathbb{R}} M(a'|a)\;\mathcal{N}(x';\mu a',\Delta_t)\;\mathrm{d}a' \;=\; p_t^{(1)}\!\left(x' - \mu\alpha\,a\right) : \tag{6.8}$$

*the first backward message of the chain is the $K = 1$ noised marginal evaluated at a shifted* *argument.*

The proof is the innovation substitution $r := a' - \alpha a$, which turns the transition into the bare Laplace density at $r$ and the evidence into $\mathcal{N}(x' - \mu\alpha a; \mu r, \Delta_t)$; the integral is then the definition (6.2). The shift $\mu\alpha a$ is what frame $a$ predicts for the next noisy observation, and the whole $K = 1$ analysis, screened Poisson equation included, recycles as the chain's boundary building block.

*Remark* 6.4 (Why iteration destroys the closed form). Equation (6.8) works because the integrand contains exactly one Laplace kink times exactly one Gaussian. Every subsequent BP step violates this: the incoming message already carries the propagated structure of several frames, so the next integrand contains two or more absolute-value kinks. On each sign-quadrant the integral reduces to half-line Gaussians, but globally it is a sum of error functions whose arguments depend on the conditioning variable non-affinely, and the clean form does not survive one iteration. In the Gaussian chain every step is Gaussian-times-Gaussian and closure propagates forever; this is the precise structural location of the difference between the two models, and the reason the Gaussian projection below is an approximation with an error to be measured, not an identity.

A change of coordinates shows the obstruction is intrinsic, not an artefact of the frame basis. In innovation coordinates $r_0 = a_0$, $r_k = a_k - \alpha a_{k-1}$ (unit Jacobian), the prior factorises into independent variables — $r_0$ Gaussian and $r_{k\geq 1}$ i.i.d. Laplace — but the likelihood becomes $p(x|r) \propto \exp(-\frac{\mu^2}{2\Delta_t} r^\mathsf{T} L^\mathsf{T} L r + \frac{\mu}{\Delta_t} x^\mathsf{T} L r)$ with $L$ the unit lower-triangular map $a = Lr$, and $L^\mathsf{T} L$ is dense. The coupling does not disappear; it migrates from the prior to the likelihood. There is no basis in which both are diagonal at once.

## 6.4 A validated numerical reference: grid BP

Since functional messages admit no closed form, the exact recursion is approximated numerically: messages are represented by their values on $M$ equispaced points in $[-A, A]$ and the update integrals are carried out by trapezoidal quadrature, each update touching every pair of grid points. This *grid BP* is a numerical approximation of the functional-message recursion, and it can serve as a reference only inside a validated regime, which we establish on the exactly solvable Gaussian chain. There the trapezoid rule on analytic, rapidly decaying integrands converges spectrally, so failure is abrupt rather than gradual: at $M = 101$, $A = 8$ the score error sits at the $10^{-15}$ floor down to $t = 0.02$ and fails only when the likelihood width $\sqrt{2t}$ approaches the grid step. The working configuration for all non-Gaussian experiments is $M = 401$, $A = 8$, with a reference solution at $M = 801$ and the resolution constraint $\sqrt{2t} \gtrsim 3\,\mathrm{d}x$ checked automatically; grid self-convergence error remains $10^4$ to $10^7$ times smaller than every effect measured below. Within this regime we treat grid BP as the reference score; outside it no claim is made.

All score errors are amplified posterior-mean errors: for any two solvers with means $\widehat{m}$ and $m_{\mathrm{ref}}$, the score–posterior identity gives, exactly,

$$\widehat{s} - s_{\mathrm{ref}} = \frac{e^{-t}}{\Delta_t}\Big(\widehat{m} - m_{\mathrm{ref}}\Big), \qquad \frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t} \ \ (t \downarrow 0). \tag{6.9}$$

The numerical residual of this identity is logged in every experiment and stays at $10^{-15}$ to $10^{-12}$. Equation (6.9) explains in advance why small diffusion times are the difficult regime: whatever an approximation does to the posterior mean is amplified by $1/2t$ in the score.

## 6.5 The Gaussian-message approximation, measured

The object under test is Gaussian-projected BP: the same sweeps, with every message forced into the Gaussian family by moment matching. This is the algorithm one would deploy when quadrature is unaffordable, and the chain analogue of the per-node closures of statistical physics (Appendix B).

On the Laplace chain (chain correlation $\alpha = 0.85$, $K = 40$ frames, 60 trials per $t$, reference $M = 801$), the median relative score error of the Gaussian closure is

| $t$ | 0.05 | 0.32 | 0.98 | 2.07 | 3.0 |
|---|---|---|---|---|---|
| median relative score error | 0.24 | $4.6 \times 10^{-2}$ | $6.8 \times 10^{-3}$ | $2.5 \times 10^{-4}$ | $7.3 \times 10^{-6}$ |
| $\mathbb{P}(\text{err} > 0.1)$ | 1.00 | 0.08 | 0.00 | 0.00 | 0.00 |

(Figure 6.2). Two thresholds are worth separating, because they are different statistics and sit an order of magnitude apart in $t$. The probability of a trial exceeding 10% relative error first reaches zero at $t \approx 0.98$; the *median* error first falls below $10^{-3}$ only at $t \approx 2.1$. In between — around $t \approx 1$, where the failure probability has already vanished — the typical error is still $\sim 7 \times 10^{-3}$.

The closure error is of order tens of percent at small diffusion time and decays as diffusion noise Gaussianises the tilted posteriors. Inspection of the worst trials shows the mechanism: the reference belief is sharply non-Gaussian, occasionally bimodal; the single Gaussian commits to a compromise mean; and (6.9) amplifies the shift. The score *direction* is much more robust than its magnitude: the median cosine similarity between projected and reference scores is 0.976 at $t = 0.05$ (where the magnitude error is 24%), exceeds 0.99 for all $t \geq 0.11$, and the worst single trial of the sweep stays above 0.879. These are numerical observations for the stated parameter ranges, not theorems.

### 6.5.1 Varying the innovation law

Sweeping six innovation families with identical covariance $\alpha^{|i-j|}$ (Laplace, Student-$t$ with $\nu = 5, 8$, and symmetric Gaussian mixtures of increasing separation) isolates the shape of the innovation law as the only moving part. Three observations, all within the tested grids. First, the noise level dominates: every family's error decays to about $10^{-3}$ or below by $t = 2.4$; all substantial closure error lives at $t \lesssim 0.4$. Second, bimodality is the worst case: at $t = 0.05$, $\alpha = 0.5$, the median error is 0.94 for a strongly bimodal mixture, 0.52 for Laplace, 0.32 for Student-$t_{\nu=5}$, and 0.07 for a mildly bimodal mixture; a single Gaussian cannot represent two posterior modes and commits to the gap between them. Third, stronger chain correlation helps rather than hurts: for Laplace at $t = 0.05$ the error is 0.43, 0.27, 0.13 at $\alpha = 0.5$, 0.85, 0.95; informative neighbours act like additional

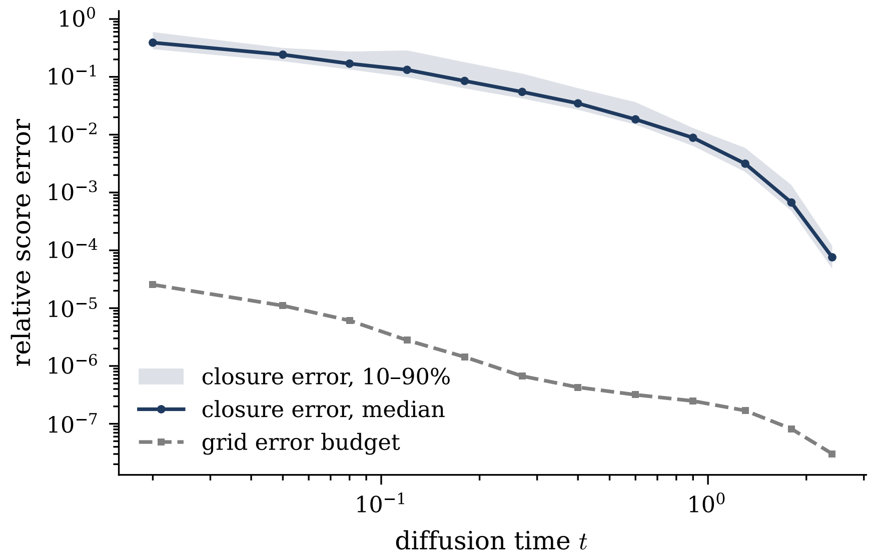


**Figure 6.2:** Gaussian-closure score error on the Laplace chain ($\alpha = 0.85$, $K = 32$, 60 trials per $t$; median and 10–90% band) against the grid error budget measured on identical data. The closure error exceeds the reference error by $10^4$ to $10^7$ at every $t$, so the measurement is trustworthy where it is used.

Gaussian observations and Gaussianise the local tilted posteriors.

## 6.6 What can and cannot be concluded

Exact: the surviving structure of Section 6.1, the limits (6.3), the factorisation (6.4), the screened Poisson equation (6.5), the curvature identity (6.6), the message form (6.7), the boundary integral (6.8), the breakdown argument of Remark 6.4, and the error identity (6.9). Numerical, within validated regimes: the quadrature audits of Section 6.2, the grid-BP calibration, and the closure-error measurements, which are specific to the stated innovation laws, correlation levels, chain lengths, and diffusion times. Not concluded: closed forms for general (non-boundary) messages at $K \geq 2$; any locality law for the non-Gaussian score beyond the structural delimitation above; and any statement about learned scores. An earlier conjecture of this project, that the Laplace $K = 2$ posterior Hessian is approximately rank-two, proved regime-dependent under numerical tests and is not asserted.

# Chapter 7

# The rotating ring: a two-dimensional dynamic object

Chapters 5 and 6 studied a chain of scalar frames, where the joint score is either exactly Gaussian or a controlled departure from it. Chapter 4 set aside a second family of dynamic objects — planar trajectories near a circle — because they are nonlinear in the coordinates in which the diffusion acts. This chapter takes up the first of them. It shows that a Cartesian surrogate of the model is exactly solvable, that the model as drawn is solvable numerically to a measured accuracy, and that both admit the same structural description of the joint score.

The question the chapter answers is narrower than the models it uses.

> How can the Markov structure of the clean rotating trajectory be exploited to compute and interpret the joint diffusion score?

Four measurable subquestions make this operational: how far the joint score differs from a collection of one-frame marginal scores; how strongly the observation at frame $j$ influences the score block at frame $k$; how that influence decays with the temporal lag $|j - k|$; and how many neighbouring frames are needed to reproduce the score block to a stated accuracy. Every answer below is a statement about one model at one parameter set, in the sense fixed in Section 1.3.

# 7.1 Model and notation

## 7.1.1 The clean process

The clean state is a radius and a phase, evolving in the internal time $u$ as two independent diffusions,

$$\mathrm{d}R_u = -\kappa(R_u - 1)\,\mathrm{d}u + \sqrt{2D_r}\,\mathrm{d}B_u^{(r)}, \tag{7.1}$$

$$\mathrm{d}\Theta_u = \omega\,\mathrm{d}u + \sqrt{2D_\theta}\,\mathrm{d}B_u^{(\theta)}, \tag{7.2}$$

observed through the polar embedding

$$a_u = R_u\Big(\cos\Theta_u,\ \sin\Theta_u\Big) \in \mathbb{R}^2. \tag{7.3}$$

We use the simplest constant-angular-drift reading of the model; Section 7.7 records the alternatives.

A point of $\mathbb{R}^2$ is described by its distance from the origin and the angle it makes with the first axis, and at each point the plane carries the orthonormal moving frame

$$e_r = (\cos\theta,\ \sin\theta), \qquad e_\theta = (-\sin\theta,\ \cos\theta), \tag{7.4}$$

in which $e_\theta$ is $e_r$ turned by a quarter revolution. Moving along $e_r$ changes the radius at fixed angle, moving along $e_\theta$ changes the angle at fixed radius, and the area element is $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$. The decomposition matters because the two directions are treated very differently by (7.1) and (7.2): the radius is pulled back towards one, the phase drifts and diffuses freely. Measuring the consequence of that asymmetry for the score is one purpose of the chapter.

### 7.1.2 The radial drift is a confining force

Equation (7.1) is the overdamped Langevin equation (E.7) for the quadratic potential

$$V(r) = \tfrac{\kappa}{2}\,(r-1)^2. \tag{7.5}$$

A potential stores energy, and the force it generates is minus its slope,

$$F(r) = -V'(r) = -\kappa\,(r-1). \tag{7.6}$$

The sign carries the content: a system driven by $-V'$ moves downhill in $V$, since $\frac{\mathrm{d}}{\mathrm{d}u}V(r_u) = V'(r)\,\dot{r} = -V'(r)^2 \leq 0$. Reading the two cases of (7.6), if $r > 1$ then $F < 0$ and the force points inward, and if $r < 1$ then $F > 0$ and it points outward. The radius is driven towards one from either side at a rate proportional to the displacement, with $\kappa$ setting the stiffness. By (E.8) the stationary radial law is $\mathcal{N}(1, D_r/\kappa)$.

### 7.1.3 One trajectory is one data point

The trajectory is sampled at internal times $u_k = kh$ for $k = 0, \ldots, K-1$, and the complete datum is the stacked vector

$$a = (a_0, a_1, \ldots, a_{K-1}) \in \mathbb{R}^{2K}. \tag{7.7}$$

The $K$ frames are correlated coordinates of one data point, not $K$ samples. Relative to Chapter 5 the only change of setting is that each frame is now two-dimensional, so the score has $K$ blocks of size two rather than $K$ scalar components; the forward channel is unchanged,

$$x = e^{-t}a + \sqrt{\Delta_t}\,\xi, \qquad \xi \sim \mathcal{N}(0, I_{2K}), \tag{7.8}$$

which is the channel of Section 2.7 acting on all $2K$ coordinates at once. Figure 7.1 shows the model and the two clocks: internal time orders the frames along the path, diffusion time corrupts them simultaneously.

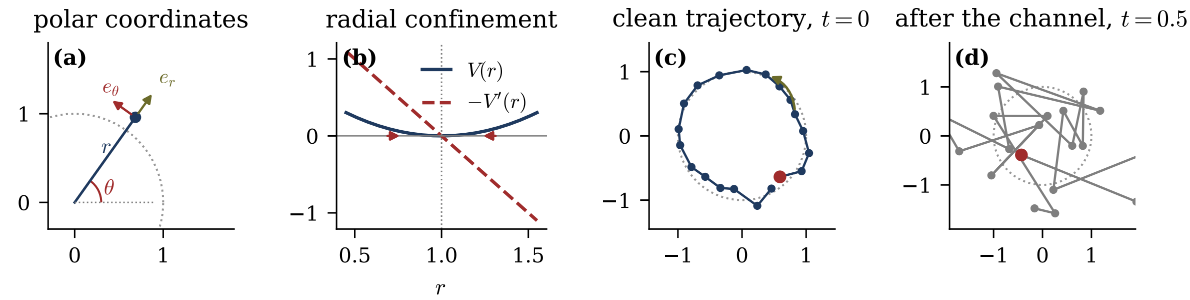


**Figure 7.1:** The rotating-ring model, at the parameters used throughout this chapter ($K = 20$, $\kappa = 2$, $D_r = 0.015$, $D_\theta = 0.005$, $\omega = 1$, $h = 2\pi/20$). (a) Polar coordinates and the moving frame (7.4). (b) The potential (7.5) and the force (7.6) it generates; the force changes sign at $r = 1$, which is therefore the stable radius. (c) One clean trajectory, a noisy coherently turning path near the unit circle; the arrow marks the sense of rotation. (d) The same trajectory after the channel (7.8) at $t = 0.5$, from a single standardised noise realisation.

## 7.2 The gauge, and the exact score

### 7.2.1 The surrogate, and what it keeps

The surrogate retains a ring, a coherent rotation, and dependence along the trajectory, and drops the rest:

$$z_0 \sim p_\lambda(z) \propto \exp\Big[-\frac{(\|z\|-1)^2}{2\lambda}\Big], \tag{7.9}$$

$$z_{k+1} = R_\psi z_k + \sigma\eta_{k+1}, \qquad \eta_{k+1} \overset{\text{iid}}{\sim} \mathcal{N}(0, I_2), \tag{7.10}$$

with $R_\psi$ the planar rotation through the one-frame angle $\psi = \omega h$.

*Remark* 7.1 (The surrogate is not a discretisation of the polar model). In (7.10) the ring acts only on $z_0$. Later frames form a rotating Cartesian random walk and their radial variance grows without bound, whereas in (7.1) a confining force acts at every internal time. The surrogate is a solvable companion whose role is to exhibit the mechanism in closed form, not to approximate the model of Section 7.1.

### 7.2.2 The known rotation is an orthogonal change of frame

The rotation is the object to be removed, so we write it out:

$$R_\psi = \begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}. \tag{7.11}$$

Its columns are the standard basis vectors, each turned through $\psi$. They are unit vectors, since $\cos^2\psi + \sin^2\psi = 1$, and orthogonal, since $\cos\psi(-\sin\psi) + \sin\psi\cos\psi = 0$. In matrix form,

$$R_\psi^\top R_\psi = I_2, \qquad \det R_\psi = 1, \qquad R_\psi^\top = R_{-\psi}, \qquad R_\psi R_\varphi = R_{\psi+\varphi}, \tag{7.12}$$

and orthogonality gives $\|R_\psi z\|^2 = z^\top R_\psi^\top R_\psi z = \|z\|^2$: lengths and inner products are preserved.

**Lemma 7.2** (Rotations preserve the standard isotropic Gaussian)**.** *If $\eta \sim \mathcal{N}(0, I_2)$ and $R$ is orthogonal, then $R\eta \sim \mathcal{N}(0, I_2)$.*

*Proof.* Appendix H. □

The isotropy is essential. A rotation would not preserve $\mathcal{N}(0, \Sigma)$ for general $\Sigma$, and the reduction below is exact only because both the innovation in (7.10) and the channel (7.8) are isotropic.

**Lemma 7.3** (Gauge reduction)**.** *Define the de-rotated frame $y_k = R_{-k\psi} z_k$ and stack the per-frame rotations into $U_\psi = \operatorname{diag}(I_2, R_{-\psi}, \dots, R_{-(K-1)\psi})$. Then*

$$y_{k+1} = y_k + \sigma\tilde{\eta}_{k+1}, \qquad \tilde{\eta}_{k+1} \overset{iid}{\sim} \mathcal{N}(0, I_2), \tag{7.13}$$

*$U_\psi$ is orthogonal, and the noised density and score satisfy*

$$P_t(z \mid \psi) = P_t^{(0)}(U_\psi z), \qquad S_\psi(z, t) = U_\psi^\top S^{(0)}(U_\psi z, t), \tag{7.14}$$

*where the superscript zero denotes the quantities of the non-rotating model.*

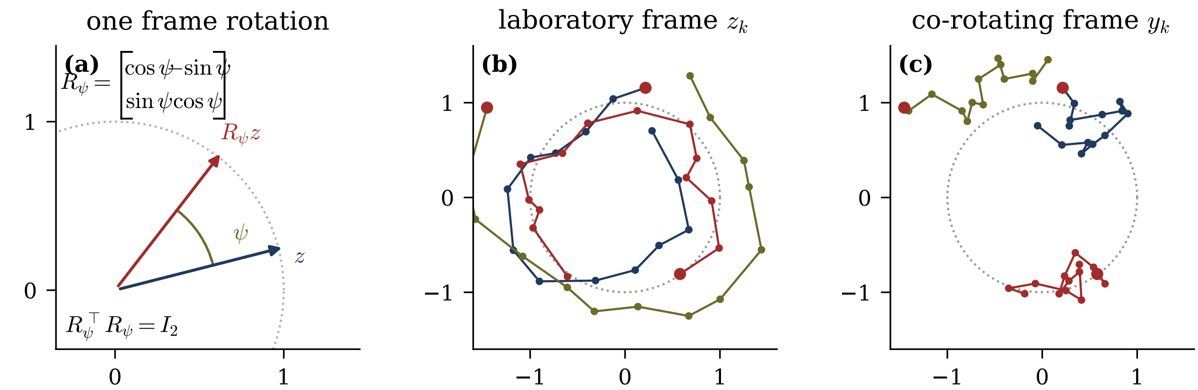


**Figure 7.2:** The rotation and the gauge. (a) $R_\psi$ turns every vector through the same angle without changing its length, which is the statement $R_\psi^\top R_\psi = I_2$. (b) Three surrogate trajectories in the laboratory frame ($K = 14$, $\sigma = 0.16$, $\psi = 2\pi/14$; the marked point is $k = 0$), coherently circulating. (c) The same trajectories after $U_\psi$: a random walk anchored on the ring. Because $U_\psi$ is orthogonal the reduction is exact at every diffusion time; the numerical round trip agrees to machine precision (Table 7.1).

*Proof.* Appendix H. □

The term *gauge* is used in the sense standard in physics, of a change of description that leaves the content untouched and may be chosen to simplify the equations. Equation (7.14) says that a known rotation adds no statistical difficulty: de-rotate the data, solve the non-rotating problem, rotate the score back. Since $U_\psi^\top = U_\psi^{-1}$, the operation is exactly invertible and destroys no information. It would not hold for an *unknown* $\psi$, which is a genuine latent variable (Section 7.7). Figure 7.2 shows the two frames.

### 7.2.3 The clean law after the gauge

Iterating (7.13) and cancelling the intermediate terms gives the anchored representation (H.1), whose derivation, together with the second-difference identity it satisfies, is in Appendix H. The identity is verified to machine zero (Table 7.1).

### 7.2.4 The noised law and the exact score

Conditioning on the anchor, (H.1) gives $\mathbb{E}[y_i \mid A = \alpha] = \alpha$ and

$$\operatorname{Cov}(y_i, y_j \mid A) = \sigma^2\, \mathbb{E}\Big[\Big(\textstyle\sum_{r\le i} \tilde{\eta}_r\Big)\Big(\textstyle\sum_{s\le j} \tilde{\eta}_s\Big)^{\mathsf{T}}\Big] = \sigma^2 \min(i,j)\, I_2, \tag{7.15}$$

since $\mathbb{E}[\tilde{\eta}_r \tilde{\eta}_s^{\mathsf{T}}] = \delta_{rs} I_2$ leaves only the $\min(i,j)$ indices common to both sums. Compactly, $y \mid A = \alpha \sim \mathcal{N}(\mathbf{1} \otimes \alpha,\ G \otimes I_2)$ with $G_{ij} = \sigma^2 \min(i,j)$, where $\mathbf{1} \in \mathbb{R}^K$ is the all-ones vector. The two Kronecker products separate the temporal index from the spatial one: $\mathbf{1} \otimes \alpha$ repeats the same two-dimensional anchor in every frame, and $(G \otimes I_2)_{(i,\alpha'),(j,\beta')} = G_{ij}\, \delta_{\alpha'\beta'}$, so $G$ carries all the temporal correlation while $I_2$ states that the two spatial coordinates are uncorrelated and share the same temporal covariance. Applying the channel and using $I_{2K} = I_K \otimes I_2$ to combine the covariances,

$$x \mid A = \alpha \ \sim\ \mathcal{N}\Big(e^{-t}\, \mathbf{1} \otimes \alpha,\ C_t \otimes I_2\Big), \qquad C_t = e^{-2t} G + \Delta_t\, I_K, \tag{7.16}$$

in which $C_t$ does not depend on $\alpha$: the noised law is a two-parameter location mixture of one fixed Gaussian.

**Proposition 7.4** (Exact surrogate score and response)**.** *Let $q = C_t^{-1}\mathbf{1}$, $\beta = e^{-2t}\mathbf{1}^{\mathsf{T}} C_t^{-1}\mathbf{1}$ and $h = e^{-t}\sum_j q_j x_j \in \mathbb{R}^2$. Then the anchor posterior is*

$$p(\alpha \mid x) \propto \exp\Big[-\frac{(\|\alpha\| - 1)^2}{2\lambda} - \frac{\beta}{2}\,\|\alpha\|^2 + h^{\mathsf{T}}\alpha\Big], \tag{7.17}$$

*and the score and response blocks are*

$$S_k(x,t) = -\Big(C_t^{-1} x\Big)_k + e^{-t} q_k\, \mathbb{E}[A \mid x], \qquad J_{kj}(x,t) = -\Big(C_t^{-1}\Big)_{kj} I_2 + e^{-2t}\, q_k q_j \operatorname{Cov}(A \mid x). \tag{7.18}$$

*Proof.* Appendix H. □

The $2K$-dimensional integral over clean trajectories has collapsed to a two-dimensional posterior over the shared anchor, which is the whole simplification and a direct conse-

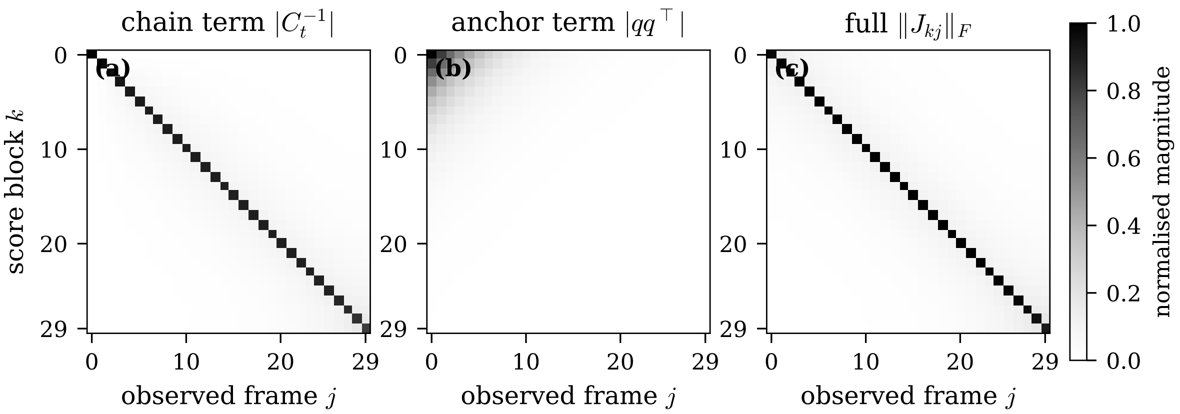


**Figure 7.3:** Exact decomposition of the surrogate response (7.18) at $t = 0.20$, $K = 30$, $\lambda = 0.05$, $\sigma = 0.15$, averaged over 250 noisy trajectories, each panel normalised by its own maximum. (a) The chain term $|C_t^{-1}|$ is concentrated near the diagonal. (b) The anchor coefficient $|qq^\mathsf{T}|$ is rank one and spread over all lags. (c) The mean norm of the complete $2 \times 2$ blocks. The analytic Jacobian agrees with central finite differences to $9.2 \times 10^{-6}$ (Table 7.1).

quence of (H.1). Its two moments are cheap: the prior and the $\|\alpha\|^2$ term are rotationally symmetric, so the posterior depends on the direction of $\alpha$ only through $h^\mathsf{T}\alpha$, and $\mathbb{E}[A \,|\, x] = \mu(\|h\|)\,\widehat{h}$ with $\mathrm{Cov}(A \,|\, x) = v_\perp I_2 + (v_\parallel - v_\perp)\widehat{h}\widehat{h}^\mathsf{T}$, three scalar functions of one scalar argument, tabulated once per diffusion time by one-dimensional quadrature.

The structural content of (7.18) is that the temporal coefficient of the second term is the outer product $qq^\mathsf{T}$, which is of rank one. The nonlocal part of the response is therefore not a chain of local couplings but a single shared latent variable seen by every frame. Figure 7.3 separates the two terms.

**Proposition 7.5** (What the surrogate establishes)**.** *Clean Markov locality does not survive marginalisation through the OU channel as locality of the noised score: by* (7.18) *the response has nonzero blocks at every lag for every* $t > 0$. *The nonlocal part is nonetheless structured, decomposing exactly into a near-diagonal chain term and a rank-one correction generated by the shared ring anchor.*

**Table 7.1:** Validation of the rotating-ring computations. The response identity and the analytic surrogate Jacobian are compared with central finite differences of the score; the grid check compares two polar resolutions.

| check | discrepancy | tolerance |
|---|---|---|
| posterior response identity vs finite differences | $6.63 \times 10^{-9}$ | $10^{-7}$ |
| analytic surrogate Jacobian (7.18) vs finite differences | $9.18 \times 10^{-6}$ | $10^{-4}$ |
| $41 \times 128$ against $51 \times 192$ polar response grid | $3.04 \times 10^{-11}$ | $10^{-7}$ |
| rotation gauge round trip (Lemma 7.3) | $2.22 \times 10^{-16}$ | $10^{-12}$ |
| clean score on an exact co-rotating ring path | 0 (exact) | $10^{-12}$ |

## 7.3 Numerical validation

Every identity of this chapter is checked against an independently computed reference, in the protocol of Appendix C.

Beyond Table 7.1 the checks include row normalisation of the transition kernels, global rotation equivariance of the score, stability under halving the finite-difference step, and the exact vanishing of the clean surrogate score on a perfect co-rotating ring path. One caveat on grid convergence: the difference between the $41 \times 128$ and $51 \times 192$ grids is small because the baseline likelihoods and kernels are already smooth at those resolutions, and a deliberately coarse $11 \times 24$ grid does show visible deviations. The production grid is converged for these parameters rather than trivially grid-independent.

## 7.4 Marginal blindness

Everything above concerns how far influence *reaches* in the noised score. This section establishes something stronger and of a different kind: there is a parameter of this model about which the per-frame marginals carry not less information, but none.

**Theorem 7.6** (Marginal blindness)**.** *For the rotating ring with uniform initial phase, every single-frame marginal is independent of the rotation $\psi$, at every noise level. Consequently a model built from per-frame marginals carries exactly zero Fisher information about the rotation.*

*Proof.* Unrolling the recursion, $z_u = R_\psi^u z_0 + \sigma \sum_{k \le u} R_\psi^{u-k} \eta_k$. Each $\eta_k$ is isotropic Gaussian, and a rotation of an isotropic Gaussian has the same law, so $R_\psi^{u-k}\eta_k \stackrel{d}{=} \eta_k$ independently across $k$. The initial phase $\theta_0$ is uniform on the circle, so $z_0$ has a rotationally invariant law and $R_\psi^u z_0 \stackrel{d}{=} z_0$. The two terms are independent, hence

$$z_u \stackrel{d}{=} z_0 + \sigma\sqrt{u}\,\xi, \qquad \xi \sim \mathcal{N}(0, I_2), \tag{7.19}$$

in which $\psi$ does not appear. The channel acts coordinatewise and independently of $\psi$, so the noised frame is likewise $\psi$-free. Therefore $\partial_\psi \log P_t(x_u) \equiv 0$ for every frame and every noise level, and a sum of per-frame log-densities cannot contain information no term contains. □

*Remark* 7.7 (What the theorem does not say). The marginals are not uninformative in general. They determine the width and centre of the confining well and the walk noise perfectly well, and by (7.19) the per-frame law is a visible, non-Gaussian ring whose radius grows as $\sqrt{u}$. They are blind to the *rotation* specifically. Nor is the joint easy: the joint carries the information, and extracting it is what the estimator does.

The sharpness is what matters. A per-frame model is not less informative about the dynamics, nor informative only at low noise. It carries zero, exactly, at every noise level. Radii are not evidence.

## 7.5 Three estimation targets, and what was measured

The theorem furnishes an unusually clean experiment. This model carries a parameter the marginals can see — the confining potential $V(r) = (r - r_*)^2/2\lambda$ — and a parameter they provably cannot, so running both through the same estimator isolates the effect of the structure rather than of the method. Three targets, each in two arms differing only in which likelihood scores a trajectory:

| | target | free parameters | visible to marginals? |
|---|---|---|---|
| (a) | the potential $(\lambda, r_*)$ | two | **yes** |
| (b) | a single rotation $\psi$ | one | **no** |
| (c) | a law over rotations $p(\psi)$ | a distribution | **no** |

For (c) the estimator is expectation–maximisation over a grid of angles, with responsibilities $r_\mu(\psi) \propto p(\psi) P_t(x^\mu \mid \psi)$ and the update $p(\psi) \leftarrow M^{-1} \sum_\mu r_\mu(\psi)$. The same sufficiency that makes the expected-transition statistic a one-shot quantity on the chain (Chapter 8) appears here: the $\psi$-conditional log-likelihood does not depend on $p(\psi)$, so the matrix of log-densities is computed once and every iteration reweights it.

The blind arm is not a baseline chosen to lose. It is the theorem's own control: its likelihood is constant in $\psi$, so its responsibilities equal $p(\psi)$, its update is the identity, and $p(\psi)$ cannot leave its uniform initialisation. If it moves, the implementation is wrong rather than the theorem.

**The blind arm returns exactly the null.** Over sixteen seeds, seven budgets and twelve noise levels — 1344 cells — the per-frame arm's estimate moves 0.0000 in total variation from its uniform initialisation, and over the 480 cells of target (b) its profile likelihood in $\psi$ has range $0.00 \times 10^0$ nats. This is a machine-precision zero rather than a small number, which is what a theorem predicting exactly no information requires.

It is worth being precise about what that zero does and does not evidence. The per-frame likelihood is implemented by a function that takes no $\psi$ argument, so its independence of $\psi$ is a property of the code path, not a measurement: the zero confirms that the estimator is wired to the marginal it is supposed to use, and would catch a leak, but it cannot corroborate the theorem against an independently constructed alternative.

The independent check is a paired one. Trajectories are simulated separately at $\psi \in \{0, \pi/6, \pi/3, 1.1\}$, sharing no code with the estimator, and two statistics are computed on the *samples*. As a negative control, the energy distance between one-frame clouds at different $\psi$ is compared against the same distance between two independent draws at *equal* $\psi$, which is the sampling floor: the floor is $1.5 \times 10^{-2}$ and the cross-$\psi$ distances are 2.6

to $4.7 \times 10^{-3}$ — strictly *below* the noise of having drawn twice, both on clean trajectories and after OU noising at $t = 0.5$. As a positive control, the lag-one cross-covariance of those same samples must equal $s^2 R_\psi$, so its antisymmetric part is $\sin\psi$; measured, it is $-0.001$, $0.503$, $0.865$, $0.893$ against $\sin\psi = 0$, $0.500$, $0.866$, $0.891$.

The pairing is the point. A null result on its own cannot distinguish a true invariance from a statistic too blunt to detect anything; here the same samples, at the same size, move by 0.9 in the joint statistic and by less than their own resampling noise in the marginal one. Both controls are in `tests/test_ring_blindness_controls.py`.

**The joint arm recovers the rotation, and loses it to the channel.** Estimating a single angle, the error is $2.1 \pm 0.7^\circ$ pooled over the whole schedule. Recovering the two-species law $p^*(\psi) = \frac{1}{2}\delta_{+\omega} + \frac{1}{2}\delta_{-\omega}$ against a truth of $\omega = 30^\circ$ and a null of $90^\circ$:

| $t$ | 0.05 | 0.32 | 0.68 | 0.98 | 1.43 | 3.00 |
|---|---|---|---|---|---|---|
| median $\lvert\hat{\omega} - \omega\rvert$ | $0.15^\circ$ | $0.24^\circ$ | $0.90^\circ$ | $2.86^\circ$ | $7.09^\circ$ | $58.8^\circ$ |

Information about the rotation survives until $t \approx 1$ and is gone by $t = 3$, where the estimate has returned to the null. That is the channel doing what a channel does, and it is legible here in a way it is not on the chain, because the null is known exactly rather than estimated.

*Remark* 7.8 (A normalisation that fails silently). The radial density of the confining well is naturally written unnormalised, and its normaliser depends on both $\lambda$ and $r_*$. While only $\psi$ is estimated the term is constant and cancels in every responsibility — which is why it can be absent for a long time without anything appearing wrong. The moment $\lambda$ is estimated it stops being constant: a wider well contains more mass, the likelihood grows without bound in $\lambda$, and the estimate pins to the top of whatever grid it is given. The failure produces a smooth, confident, wrong answer rather than an error. It is recorded here because the same shape of mistake — a quantity that is constant in the case one has been testing, and is not constant in the case one moves to — is the most common way a correct derivation becomes an incorrect measurement.

## 7.6 What this chapter establishes

Three things, in increasing order of how much they constrain what a method can do.

First, the rotation is a *gauge*. Because $U_\psi$ is orthogonal it commutes with the isotropic channel, so at known $\psi$ the model reduces to $\psi = 0$ and the conditional density is closed form at any trajectory length — a quadratic form plus one one-dimensional radial integral. There is a ground truth here, computed rather than estimated, and it is what everything else is measured against.

Second, the clean law is local but the noised score is not. For $t > 0$, $S_k(x, t)$ is not a function of $x_{k-1}, x_k, x_{k+1}$ alone: integrating out uncertain latent states creates posterior correlations at every lag, and the anchor term of (7.18) is rank one but touches every pair of frames. Clean Markovity buys a local *generative* law, not a local score.

Third, and this is the result that does not have an analogue on the chain, there is a parameter of the model about which per-frame marginals carry *exactly zero* information (Theorem 7.6), and it is nonetheless recoverable from the joint. The blind arm of §7.5 does not merely do worse; it cannot move off its initialisation, and over 1,824 cells it does not, to machine precision. That is a sharper statement than a comparison of error rates, because the null is known exactly rather than estimated.

## 7.7 Limitations

1. **One parameter set for the geometry.** The ring width, walk noise and trajectory length are held at a single baseline. Only the noise level and the data budget are swept, and nothing is claimed about scaling in $K$.

2. **Radius positivity.** The radial process is formally supported on all of $\mathbb{R}$. At the baseline the probability of a negative radius is negligible, but a strict model would reflect at the origin or use a positive diffusion.

3. **Known rotation in the gauge argument.** Lemma 7.3 assumes $\psi$ known; §7.5 is what happens when it is not, and the two-species case is the smallest version of

**Table 7.2:** Status of the results of this chapter.

| result | status |
|---|---|
| Gauge reduction and score transformation (Lemma 7.3) | exact |
| Exact surrogate score and rank-one response (Proposition 7.4) | exact |
| Marginal blindness (Theorem 7.6) | exact |
| Recovery of $p(\psi)$ from the joint, and its decay with $t$ | numerical, sixteen seeds |
| The blind arm returning the null to machine precision | numerical, 1,824 cells; a consistency check on the e |
| Paired sample-level controls: one-frame law invariant to $\psi$ below the resampling floor, lag-one joint tracking $\sin\psi$ | numerical, independent of the estimator |
| Recovery of the confining potential $(\lambda, r_*)$ | numerical, in progress |
| Behaviour in $K$, and inference of the phase rather than its use | open |

the speciation phenomenon discussed in Section 2.12.

4. **Grid approximation.** The radial integral is numerically converged at these parameters but is a quadrature, not a symbolic closed form. The validation of §7.3 bounds the error; it does not remove it.

5. **The potential experiment is incomplete.** The $(\lambda, r_*)$ sweep runs 300 gradient steps per cell and had not finished at the time of writing. Its preliminary reading — that both arms recover the well, which is the expected contrast with the rotation — is stated as preliminary and nothing else depends on it.

The immediate extensions are to vary $K$ systematically, to infer the angular phase rather than use an oracle basis, to add an unknown rotation class, and to test whether a time-dependent local architecture with receptive field $L_\epsilon(t)$ can match the full score.

# Chapter 8

# Learning the Innovation Law: EM for Non-Gaussian Chains

Chapters 5 and 6 treated the transition law of the clean chain as given: Gaussian in the first case, a fixed non-Gaussian family in the second. Belief propagation then computed the diffusion score exactly, in the Gaussian case, or against a validated numerical reference, in the Laplace case. Neither chapter asked where the transition law itself comes from. This chapter does. The clean data generating process is the same first-order Markov chain of (6.1), but the innovation law is now a member of an estimated family, and the question is how much the Markov structure this thesis has exploited throughout still helps once part of the model must be learned rather than assumed, and what breaks first when the assumptions themselves are wrong.

> How much does known Markov structure reduce the statistical and computational burden of learning a diffusion score, and what is lost when the transition law or the continuous messages must be approximated?

The chapter answers the *how* — the estimator, what is exact in it and what is numerical, and what the implementation was checked against. Chapter 9 answers the *how much*.

## 8.1 Model: an estimated mixture innovation

The clean chain is the one of (6.1), in the index convention of this chapter,

$$a_1 \sim \mathcal{N}(0,1), \qquad a_i = \alpha\, a_{i-1} + \varepsilon_i, \qquad i = 2, \ldots, n, \tag{8.1}$$

with innovations $\varepsilon_i$ i.i.d., mean zero, variance $q = 1-\alpha^2$ — the same Gaussian initial state and the same variance-matching that Chapter 6 used — but now drawn from a member of a *parametric family* rather than a fixed law. Forward noising is the coordinatewise Ornstein–Uhlenbeck channel of Section 2.7, $x = e^{-t}a + \sqrt{\Delta_t}\,\xi$, unchanged from every earlier chapter.

The estimated family is a $C$-component Gaussian mixture on the innovation,

$$K_\theta(a' \mid a) = \sum_{c=1}^{C} \pi_c \mathcal{N}\Big(a' - \alpha a;\ \nu_c,\ s_c^2\Big), \qquad \theta = \{\alpha, (\pi_c, \nu_c, s_c)_{c=1}^{C}\}, \tag{8.2}$$

so both the autoregressive coefficient $\alpha$ and the shape of the innovation density are free parameters, estimated jointly by maximum marginal likelihood from noised data alone. At $C = 1$ this collapses to a single Gaussian, recovering the model of Chapter 5; larger $C$ can represent skew, heavy tails, and multimodality to the resolution of the grid. The initial law $a_1 \sim \mathcal{N}(0,1)$ is *held fixed* at estimation time — it is the value used to generate every family compared in Chapter 9, so there is no misspecification in it, but neither is there direct evidence it could itself be recovered from noised data; that question is not addressed here.

Two assumptions are load-bearing and, consistent with the practice of this thesis, are stated rather than tested by the estimator itself: the clean process is exactly first-order Markov, and the forward noise is exactly coordinatewise. Section 9.5 in the next chapter measures what happens when the first of these is false by a controlled amount; the second is never relaxed anywhere in this thesis.

*Remark* 8.1 (Covariance-stationary, not strictly stationary). Choosing $q = 1 - \alpha^2$ makes the chain covariance-stationary — $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \alpha^{|i-j|}$ hold exactly for *every* innovation law, by the same second-moment recursion as the Gaussian case. It does not make the chain strictly stationary: for a non-Gaussian innovation the invariant marginal law of the chain is not $\mathcal{N}(0,1)$, so fixing $a_1 \sim \mathcal{N}(0,1)$ starts every family off its own invariant distribution. Nothing compared in this or the next chapter depends on the distinction, because every family compared shares the same $a_1$ and the same covariance structure. The claim available here is covariance-stationarity, and it is the one used.

## 8.2 What remains exact

Two facts from Section 6.4 carry over unchanged and are worth restating as the load-bearing structure of the whole estimator.

**Proposition 8.2** (The posterior factor graph is still the prior's chain)**.** *Coordinatewise forward noising contributes exactly one* unary *factor per site,* $x_i \mapsto \phi_i(a_i) \propto \exp\Big(-(x_i - e^{-t}a_i)^2/2\Delta_t\Big)$*. It reweights node potentials and creates no new edges among the latent* $a_i$*, for any innovation law, mixture or otherwise. The posterior over the clean chain given the noised observation therefore factorises on exactly the same chain graph as the prior.*

**Proposition 8.3** (The E-step is exact)**.** *Since the posterior graph is a tree, one forward and one backward sum-product sweep return the exact single-site and pairwise marginals of the posterior, at the level of functional messages, for* any *innovation law. Consequently the E-step of an EM procedure for* $\theta$ *requires no variational approximation and no sampling: the sufficient statistics EM needs are exact conditional expectations under the true posterior, mixture label included.*

Both are consequences of the chain being a tree (Theorem 3.3) and hold independently of whether the innovation law admits a closed-form message update. What is *not* exact is the representation of a functional message on a computer, which is where Section 8.4 and the grid-BP machinery of Section 6.4 enter.

## 8.3 The EM algorithm

Write $\Xi(\theta)$ for the vector of exact posterior sufficient statistics that BP supplies given the current parameter $\theta$ — the conditional expectations of the pairwise sums, mixture-label indicators, and second moments that the mixture kernel's complete-data log-likelihood needs. The gradient of the *marginal* log-likelihood $\mathcal{L}(\theta)$ is given by Fisher's identity,

$$\nabla_\theta \mathcal{L}(\theta) = \Big\langle \Xi(\theta),\ \nabla_\theta \log K_\theta \Big\rangle. \tag{8.3}$$

The practical content of (8.3) is that belief propagation is never differentiated through: BP supplies the conditional expectation $\Xi(\theta)$ at the current parameter, and the gradient with respect to $\theta$ is then an ordinary derivative of the kernel's log-density. This is what makes the E-step of Proposition 8.3 directly usable inside an optimiser.

The M-step for the mixture kernel is *not* an exact maximisation. Given $\Xi(\theta^{(k)})$, updating $(\pi_c, \nu_c, s_c)_{c=1}^{C}$ exactly would require solving a mixture-likelihood problem with its own latent label, so the implementation alternates closed-form *conditional* maximisers — the mixture block and $\alpha$ — until the per-edge gain in $Q$ falls below a tolerance or a cap of sixteen sweeps is reached, recording on the returned kernel which of the two occurred. The algorithm is therefore generalised EM, or ECM (Meng and Rubin, 1993), rather than exact EM: a sweep ascends $Q$ without maximising it.

The cap of sixteen is set by measurement, not convention. Varying only this budget at $C = 8$, $n = 256$, and a fixed outer count of 40, the fitted innovation excess kurtosis moves by 28% between four sweeps and sixteen ($1.68 \to 2.14$, against a true value of 3.0) while $\alpha$ moves in the fourth decimal — the same rate asymmetry between shape and correlation that Section 8.5 finds in the outer loop, appearing one level down. The effect is confined to short outer runs: by 60 outer iterations the two loops substitute for one another and the inner budget stops mattering. Sixteen removes the interaction rather than requiring a reader to establish which regime a given cell is in.

Monotone ascent of $\mathcal{L}$ is guaranteed by the standard EM argument applied to the surrogate $Q(\theta \mid \theta^{(k)}) = \langle \Xi(\theta^{(k)}), \log K_\theta \rangle$ — ascending $Q$ is all monotonicity requires —

and is checked directly: across an executable audit of the implementation, the minimum increase in $Q$ over twelve consecutive steps was $+2.61$ for a Gaussian kernel, $+4.34$ for Laplace, and $+4.18$ for an eight-component mixture, with zero decreases recorded in any logged run. The returned kernel's evidence equals `trace.log_evidence[-1]` by construction, with a regression test asserting it, so the reported convergence trace always describes the model actually returned.

## 8.4 The numerical representation: grid BP, again

Messages remain functions on $\mathbb{R}$, so they are represented exactly as in Section 6.4: values on $M$ equispaced points in $[-A, A]$, updates by trapezoidal quadrature. Two error sources are tracked separately here as well. Truncation error — the probability mass the finite domain $[-A, A]$ misses — is diagnosed by the maximum boundary mass over the sites of a chain. That is a statistic of a sampled trajectory rather than a bound, so it is recorded over 256 chains per configuration: the worst chain falls from $2.3 \times 10^{-3}$ at $A = 4$ to $1.2 \times 10^{-6}$ at $A = 8$, the 90th percentile from $2.7 \times 10^{-4}$ to $1.5 \times 10^{-8}$, and neither depends on the grid resolution $M$. The gap between the two rows matters, because a figure read off a single trajectory sits near the median and so understates the tail by two orders of magnitude. Quadrature error, from the finite spacing $h = 2A/(M - 1)$ — the grid holds $M$ points including both endpoints — on an otherwise exact interior update, falls as $O(h^2)$: a factor of four per halving of $h$, matching the trapezoidal rate on a smooth integrand. At the working configuration used for every experiment in this and the next chapter, $M = 401$, $A = 8$, both sources sit far below any effect subsequently measured: the discretised recursion reproduces the closed-form Gaussian posterior mean to $9.2 \times 10^{-15}$, and agrees with an independently coded brute-force enumeration — sharing no code with the BP recursion — to $1.6 \times 10^{-14}$ on posterior means and $1.8 \times 10^{-15}$ on the log-evidence.

*Remark* 8.4 (The boundary-mass diagnostic is empirical, not a bound)*.* The worst-boundary-mass figures above were originally computed from a single sampled chain and reported

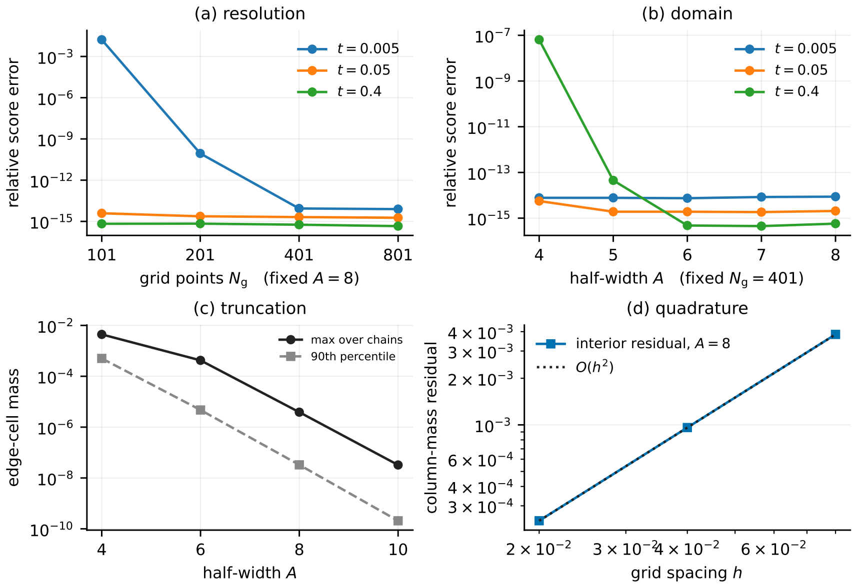


**Figure 8.1:** Grid and domain control on a Laplace chain ($\alpha = 0.85$, $n = 32$ sites), four panels. (a) Relative score error against a high-resolution reference as grid resolution $M$ grows at fixed half-width $A = 8$; (b) the same as $A$ grows at fixed $M = 401$ — both saturate at the floating-point floor well inside the working configuration. (c) Forward-message boundary mass against $A$, the truncation diagnostic, drawn as *both* the maximum and the 90th percentile over 256 chains, which differ by roughly two orders of magnitude for the reason given in Remark 8.4. (d) Interior column-mass residual against spacing at $A = 8$, the quadrature diagnostic, with an $O(h^2)$ reference. Neither diagnostic alone certifies convergence. Source: `outputs/frozen/exp_01/` and `outputs/frozen/exp_18/`.

as if they bounded the truncation error for every run. They do not: reseeding the same diagnostic over 256 independent chains per configuration shows the single-chain value sits near the *median* over chains, while the maximum observed is up to 152× higher (range 2.9× to 8369× across configurations). The diagnostic should be read as an empirical estimate of typical truncation loss, not a certified bound. What it does *not* affect is the separation of truncation from quadrature error reported above, which is a pure function of $(A, M)$ and the prior and involves no sampled chain at all — that comparison reproduces to 0.000 relative difference under reseeding and is exactly the result Section 6.4 already relies on.

One further identity is worth restating from Section 6.4 because it governs every result in the next chapter exactly as it governed Chapter 6: whatever an approximation

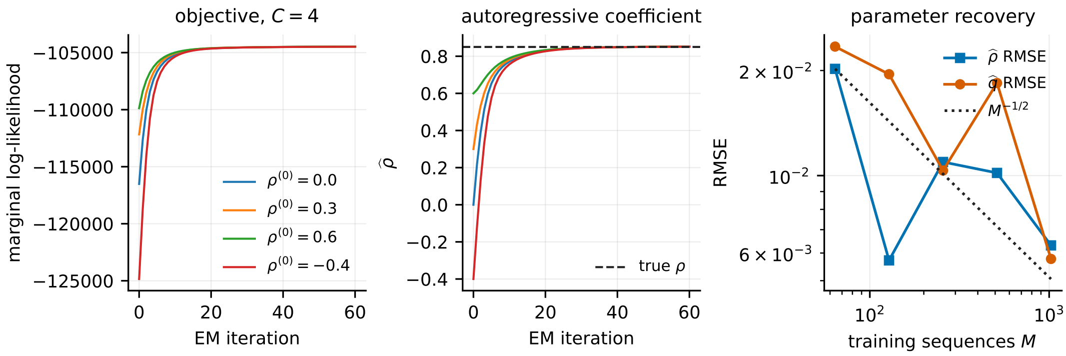


**Figure 8.2:** Optimisation diagnostics for the EM procedure of Section 8.3. Left: marginal log-likelihood by iteration from four initialisations of $\alpha$; the largest decrease across iterations is zero in every run. Centre: the estimated $\alpha$ over the same runs, converging to $[0.8517, 0.8520]$ against a truth of 0.85 from initialisations spanning $[-0.4, 0.6]$ — direct evidence that $\alpha$ is estimated, not supplied. Right: RMSE of the recovered $\alpha$ and innovation variance against the number of training sequences, with an $n^{-1/2}$ reference line. Source: `outputs/exp_18/` and `outputs/exp_06_em_parameter_recovery/`.

— Gaussian closure, grid truncation, or an under-converged fit — does to the posterior mean is amplified by $e^{-t}/\Delta_t \sim 1/2t$ in the score, so small diffusion times remain the regime where any deficiency is most visible.

## 8.5 What convergence actually costs

The last piece of machinery this chapter reports is not a theorem but a measurement, and it is the one that most directly qualifies the results of Chapter 9: how many EM iterations the algorithm of Section 8.3 actually needs.

Two arms of the same chains were fit under identical initialisation and budget, one seeing clean data and the other once through the OU channel:

| iteration | clean | | through the channel | |
|---|---|---|---|---|
| | $\alpha$ | kurtosis | $\alpha$ | kurtosis |
| 1 | 0.8473 | 1.470 | 0.4276 | 0.112 |
| 30 | 0.8467 | 2.858 | 0.8366 | 1.527 |
| 120 | 0.8473 | 3.014 | 0.8464 | 3.164 |
| 600 | 0.8475 | 3.132 | 0.8466 | 3.096 |

(truth: $\alpha = 0.85$, innovation kurtosis 3.0). Both arms reach the truth; what differs is the *rate*. On clean data $\alpha$ is converged after a single M-step, since nothing about it is hidden at the chain level. Through the channel it takes on the order of 30–60 iterations — the missing-information mechanism of Dempster et al. (1977) appearing directly in the convergence rate, where the theory places it. The innovation shape is slower still, and on *clean* data already: tens of iterations against one for $\alpha$, because the mixture label is itself a latent variable even without an observation channel. The channel then compounds this. A dedicated sweep measuring exactly how many iterations the slowest configuration needs found roughly **2000** — about 50× a budget of 40 iterations, which several of the experiments reported in Chapter 9 used as a fixed default before this measurement existed.

*Remark* 8.5 (Fixed budgets compare convergence rates, not estimators)*.* This is the single methodological lesson of the chapter, and it recurs throughout the next one. Whatever coordinate of a fit converges slower looks worse under a fixed iteration count, and a slower-converging configuration under-iterated by the same budget produces what looks like a genuine deficit rather than symmetric noise — which is precisely the signature that makes it easy to mistake for a real effect. An eight-seed replication at the same fixed budget of 200 iterations found no channel penalty on converged shape recovery once compared correctly (paired clean − noised difference $-0.052 \pm 0.217$, consistent with zero) — retiring an earlier claim that had rested on a single non-converged draw. Where Chapter 9 reports a result obtained at a fixed, possibly insufficient budget, it says so explicitly and states which quantities are and are not affected.

## 8.6 Summary

The estimator combines two facts that hold for any innovation law — Proposition 8.2 and Proposition 8.3 — with a generalised M-step and Fisher's identity to turn exact BP conditional expectations into a marginal-likelihood gradient. Everything about the *posterior inference* step is exact up to the grid discretisation of Section 8.4, which is controlled to far below any effect measured. What is not free is optimisation time: the innovation shape is an order of magnitude slower to converge than the autoregressive coefficient, and by a further order of magnitude once observed only through noise. Chapter 9 reports what this estimator buys — in data efficiency and, separately, in the fidelity of what it generates — and where, along axes of model capacity and model misspecification, it stops buying anything.

# Chapter 9

# What Structure Buys, and Where It Stops Paying

Chapter 8 built an estimator that is exact in its inference step and approximate only in its numerical representation and its optimisation budget. This chapter asks what that estimator is worth. Four questions organise it, in the order the evidence answers them most cleanly.

> How much does the structured estimator reduce denoising error at a given amount of data; does that pointwise advantage survive when the estimated score is used to *generate*, by integrating the reverse process; how much mixture capacity does the model need; and how far can the Markov assumption itself be wrong before the structure stops helping?

The first and last are answered. The middle two are not, and the sections that address them say why the experiments as run cannot settle them; the summary at the end of the chapter states each of the four at the strength its evidence actually supports.

Every comparison below fits the mixture kernel of (8.2) by the EM procedure of Chapter 8 and compares it against neural denoisers trained by ordinary denoising score matching on the same noised data, with no access to the Markov factorisation, the linear kernel form, or homogeneity across sites — three structural facts the estimator is given and the networks are not. Two network architectures recur throughout: a fully connected MLP with no locality bias, and a local convolutional network matched in receptive field to the estimator's own locality. The comparison measures the value of the structure supplied, not a general claim about neural denoisers.

## 9.1 Pointwise data efficiency

Under the structural assumptions of Section 8.1, at the data budgets tested, the structured estimator attains lower relative denoising error than the global MLP by a factor of 7.0–17.6 across the 8 training-set sizes (Table 9.1, Figure 9.1). The factor is *not* monotone in the budget, and it is the largest of several defensible aggregations of the same cells — Appendix I.1 gives the other four and the paired intervals. These figures are generated by `tools/make_tab_efficiency.py` from the frozen outputs rather than transcribed, so the table, the figure and this sentence cannot drift apart.

That comparison is against a network with no locality prior, and most of the margin is attributable to that rather than to the estimator. A locality-respecting weight-shared baseline narrows it to a small multiple; how much survives against an architecture that can propagate across the whole chain, rather than one confined to a fixed window, is discussed with the scoping it requires in Section 9.2. The direction is consistent with the locality results of Chapter 7 and the receptive-field argument of Chapter 5.

Both selections behind this comparison are made on a disjoint validation split rather than on the test bundle: the network parameterisation (predicting the noise or the clean signal) and, in a separate sweep, the receptive-field radius. Validation selection agrees with the test-set choice in 33 of 35 cells for the parameterisation and 54 of 60 for the two-axis receptive-field sweep, and the quantities it moves are small — the mean advantage $11.65\times \to 11.66\times$, the receptive-field ratio $3.846 \to 3.862$, both under $1.005\times$. The protocol is the defensible one independently of the size of the correction it produces.

**Table 9.1:** Relative denoising error against the deterministic grid-BP reference under the true kernel, averaged over the noise schedule; 16 seeds ± one standard error, aggregated per seed. Both arms are tuned on a disjoint validation bundle — the network selects its parameterisation *and* training length, EM–BP its iteration count — agreeing with the test-set optimum in 81.1% and 77.6% of cells. EM–BP is more accurate in 1,518 of 1,520. The $N = 4096$ row uses a raised budget, the caps having been set for $N \leq 2048$; rerunning $N = 2048$ at that budget on the same seeds moves its ratio by $-0.16 \pm 0.18$. 16 of 1536 cells are excluded because the narrowest fitted mixture component is under two grid cells wide; including them changes no ratio by more than 0.08. $N = 8192$ was in earlier drafts and is withdrawn (Appendix C).

| sequences $N$ | network | EM–BP (24 free) | ratio |
|---:|---:|---:|---:|
| 32 | $0.4031 \pm 0.0059$ | $\mathbf{0.0589 \pm 0.0026}$ | 7.0 |
| 64 | $0.3285 \pm 0.0026$ | $\mathbf{0.0501 \pm 0.0041}$ | 7.8 |
| 128 | $0.2759 \pm 0.0024$ | $\mathbf{0.0343 \pm 0.0017}$ | 8.8 |
| 256 | $0.2268 \pm 0.0010$ | $\mathbf{0.0258 \pm 0.0014}$ | 9.7 |
| 512 | $0.1878 \pm 0.0006$ | $\mathbf{0.0219 \pm 0.0011}$ | 9.0 |
| 1024 | $0.1569 \pm 0.0006$ | $\mathbf{0.0162 \pm 0.0012}$ | 12.5 |
| 2048 | $0.1326 \pm 0.0005$ | $\mathbf{0.0132 \pm 0.0008}$ | 15.5 |
| 4096 | $0.1171 \pm 0.0004$ | $\mathbf{0.0094 \pm 0.0007}$ | 17.6 |

## 9.2 How much of the margin is the architecture?

A ratio against a fully connected network cannot separate knowing the Markov structure from sharing weights across sites, and the second is the cheaper explanation. Measured against a weight-shared window head instead — radius and parameterisation selected on validation, the estimator refitted on the same chains — *most of the headline margin is architectural*, and what is left is a small multiple rather than an order of magnitude.

That residual is reported here as a bound on what remains to be explained, not as a measurement of what the structural assumption is worth, and the reason is specific. A window head trained to a fixed number of gradient steps sees each chain progressively fewer times as the training set grows: at batch 64 and 8000 steps the expected presentations per chain fall from 8000 at $N = 32$ to 250 at $N = 2048$. A residual that widens with $N$ therefore cannot be separated, under that protocol, from the optimisation budget contracting. The window head's own error falls throughout, from 0.116 to 0.052. A widening residual is therefore not evidence that the convolution *fails to acquire* the structure as

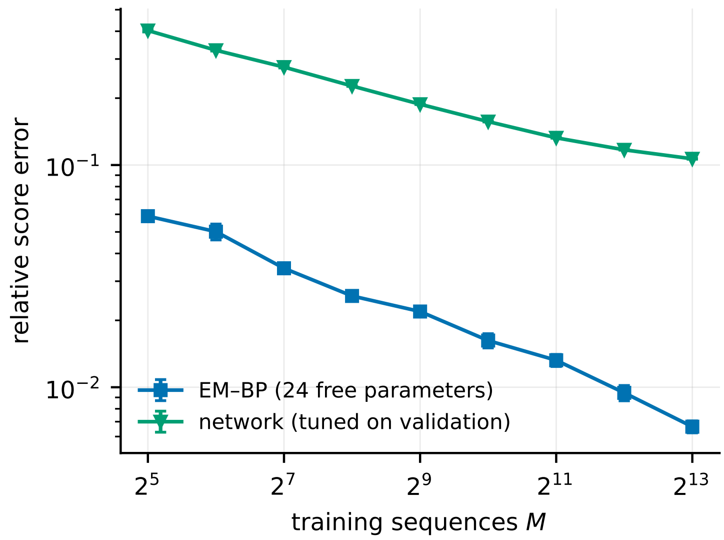


**Figure 9.1:** Relative score error against training-set size, mean over 16 replicates with standard errors, for the EM/BP estimator and the validation-tuned MLP. The window-head baseline of Section 9.2 ran at a different protocol and is not plotted here. Drawn from the same certified frozen outputs, over the same size grid and with the same seed-first aggregation, as Table 9.1.

data grows: this design cannot distinguish that mechanism from its own budget schedule, and no reading of the trend is asserted here.

Two further limits belong with the number. The comparison scored a single coordinate — the centre site of the chain — rather than the whole sequence, and it ran at a shorter chain, five noise levels instead of twelve, and a four-component mixture, so its ratio is not directly comparable with Table 9.1. A corrected experiment at the headline protocol, scoring all sites and a predeclared interior region, selecting both arms' budgets on validation, and adding baselines that can propagate across the whole chain rather than a fixed window, is specified in `experiments/exp_31_structured_baseline.py`. Its results are not in this thesis. What can be said without it is the direction and its order: against an unstructured network the estimator's advantage is roughly an order of magnitude, and against a structure-aware one it is considerably smaller.

## 9.3 How much mixture capacity is enough

This section reports two things that must be kept apart. The *evidence-based* plateau below stands, in the weak form stated there — a gain this design cannot resolve, not a demonstrated equivalence. The *shape-based* capacity curves that follow it do not: they are confounded with convergence rate, as Remark 9.1 sets out. The distinction is metric, not rhetoric — held-out evidence settles early in the optimisation and the fitted shape does not, so the same fits support one statement and not the other.

Enlarging $C$ in (8.2) should help until the mixture can represent the true innovation law and stop helping after. Held-out log-evidence per edge, fit and scored on disjoint data, rises with $C$ and then *stops rising detectably*: paired against $C = 1$ on identical training and evaluation data within each cell (six paired seeds, $N = 128$),

| $C$ | paired $\Delta$(evidence/edge) | s.e. | beyond 2 s.e.? | effective $C$ |
|---|---:|---:|---:|---:|
| 2 | +0.000013 | 0.000090 | no | 2.00 |
| 4 | +0.001286 | 0.000107 | yes | 3.82 |
| 8 | +0.001735 | 0.000136 | yes | 7.70 |
| 16 | +0.001740 | 0.000152 | yes | 15.50 |

$C = 2$ buys nothing over a single Gaussian innovation and capacity begins to pay at $C = 4$. Beyond that the reading has to be narrow. $C = 8$ and $C = 16$ differ by $5 \times 10^{-6}$, inside one standard error — but the relevant contrast is the *paired* quantity $\Delta_{16,8}$, not two separate contrasts against $C = 1$, and a difference this design cannot resolve is not a demonstration that there is none. Six seeds at one sample size, with no predeclared equivalence region, cannot establish equivalence; failing to reject is not evidence of no effect. So: **at $N = 128$ this experiment does not resolve a held-out-evidence improvement from $C = 8$ to $C = 16$, and most of the gain relative to $C = 1$ is already present by $C = 8$.** That is a failure to resolve a further gain, not saturation, and it is in any case a statement about held-out evidence rather than about the converged innovation shape, which must not be restated as the latter. The same pattern holds on the primary pointwise metric —

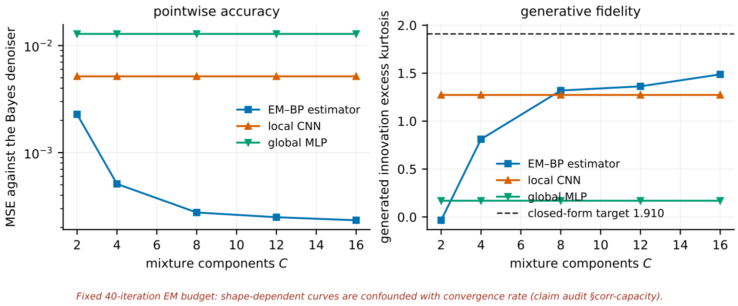


**Figure 9.2:** Held-out denoising MSE and generated-sample fidelity against mixture capacity $C$, three seeds, at a fixed EM budget of 40 iterations. Read alongside the caveat below: the shape-dependent curves in this figure are confounded with EM convergence rate; the evidence-based plateau reported in the text, from a separate paired design, is not. Source: `outputs/exp_16/`.

held-out denoising MSE against the Bayes denoiser, paired the same way — where $C = 8$ is the paired optimum at both $N = 128$ and $N = 512$, $C = 16$ is not distinguishable from it, and at $N = 512$ the smallest capacity $C = 2$ is measurably *worse* than $C = 1$: extra parameters cost more than they buy below the capacity the data supports. This is not a resolution artefact — the narrowest fitted component stays at least 8.8 grid spacings wide in every one of these cells.

*Remark* 9.1 (This capacity picture is qualified, and the qualification matters). Every fit contributing to Figure 9.2 and to any curve in $C$ that depends on the *fitted or generated shape* of the innovation (as opposed to held-out evidence or runtime) used the fixed EM budget of 40 iterations. Section 8.5 measured that the innovation shape can need on the order of 2000 iterations to settle — a factor of roughly 50 — and, applied to this specific sweep, found that across 72 seed-configurations **93% had not settled in the shape coordinate by iteration 40**. Because a larger $C$ has more shape parameters to move within the same fixed budget, under-iteration alone produces a curve that looks exactly like a capacity effect whether or not capacity is doing anything. A direct check attributes roughly **96% of the apparent capacity effect** in this configuration to convergence rate at the fixed budget, not to capacity. The held-out-evidence plateau reported in the main text above survives this concern, because it is a paired comparison scored by an average over the whole density that settles early in the optimisation; the runtime measurement below is unaffected for the same reason. Anything resting on fitted or generated *shape* at $C > 1$ in the rest of this section, and in Section 9.4 where those same 40-iteration fits reappear, should be read as evidence for a real phenomenon at an under-converged operating point, not as a claim about the converged estimator. The direct fix is to re-run to a convergence criterion rather than a fixed count; that is specified in `experiments/exp_32_capacity_equivalence.py` and its results are not in this thesis.

Runtime is close to flat across the whole capacity sweep, 13.5 $\rightarrow$ 14.3 seconds per iteration from $C = 2$ to $C = 16$ while the number of free parameters grows 6 $\rightarrow$ 48, confirming that the E-step's $O(nM^2)$ cost dominates and only the M-step scales with $C$ — this measurement, like the evidence plateau above, is unaffected by Remark 9.1.

## 9.4 Pointwise accuracy and generative fidelity

Whether the pointwise advantage transfers to generation is not settled by the experiments available here, and the reason is the budget rather than the measurement. Every fit in this comparison used a fixed budget of 40 EM iterations, while Section 8.5 measures the fitted innovation shape — which is precisely what the generative metric scores — as needing on the order of 2000 iterations to settle. The comparison therefore scores converged estimators against unconverged ones on the one coordinate that had not converged, and a dissociation between pointwise and generative quality cannot be separated from that asymmetry.

The dissociation may well be real, and the design cannot distinguish it from its own iteration budget. No claim in either direction is made here. The measured values, the within-arm form of the comparison and the convergence diagnostic that settles it are recorded in the companion compendium's claim audit, for whoever runs the corrected version.

## 9.5 Where the Markov assumption stops paying

Every result so far assumes the clean chain is exactly first-order Markov. Two controlled violations were measured against an exact reference (linear Gaussian algebra for the contaminated model, so the estimator being wrong is not confused with the reference being wrong): a rank-one global latent shared across all sites at strength $\beta$, and a long-range coupling at strength $\gamma$ added on top of the chain.

| violation | strength | CNN/EM-BP | Chow–Liu | reading |
|---|---|---|---|---|
| global latent | $\beta = 0$ | $11.4 \to 5.4$ | $-0.0009$ | Markov control |
| global latent | $\beta = 0.5$ | $4.1 \to 3.2$ | $+0.0091$ | degraded, still winning |
| global latent | $\beta = 1.0$ | $2.1 \to 3.1$ | $+0.0128$ | **never inverts** |
| long-range | $\gamma = 0$ | $25.4 \to 10.5$ | $+0.0014$ | Markov control |
| long-range | $\gamma = 0.10$ | $0.77 \to 1.32$ | $+0.0574$ | **inverts at low $t$** |
| long-range | $\gamma = 0.40$ | $0.47 \to 1.03$ | $+0.1010$ | **inverts across most of it** |

The two violations behave qualitatively differently, and the difference is principled rather than incidental. A global latent shared across every site makes the score residual *exactly* rank one, by a Woodbury identity; a chain-structured estimator can absorb a rank-one correction into its fitted coefficients and does — the fitted $\alpha$ rises and the fitted innovation kurtosis flattens as $\beta$ grows, exactly tracking the contamination it is misattributing to the chain. Long-range coupling is only *approximately* low rank, and a chain family cannot represent it at all: the advantage is gone by $\gamma \approx 0.05$–$0.10$ and the structured estimator's own relative error reaches 57% at high noise and the strongest tested coupling. The minimum CNN/EM-BP ratio over the full noise schedule never falls below 2.08 for the rank-one violation at any strength up to $\beta = 1.0$ — where half the marginal variance is a shared global constant — and falls to 0.47 for the long-range one. The Chow–Liu deviation column corroborates the mechanism independently: the fitted chain stays close to the best possible Markov approximation of the truth under rank-one contamination and departs from it under long-range coupling, exactly where a chain can no longer represent what generated the data. The same qualitative split holds when the innovation law is

Laplace rather than Gaussian: the rank-one violation still never inverts the advantage, with the minimum ratio at $\beta = 1.0$ equal to 1.45 against an exact non-Markov reference.

*Remark* 9.2. This extends the lesson of Chapter 6: there, what broke under a harder innovation law was the *representability of messages* in a finite-dimensional family, while the tree, the two-sweep schedule, and the score–posterior identity survived unconditionally. Here, the further question is what breaks when the graph is not a tree to begin with. The answer is not "misspecification degrades the advantage" uniformly — it is that a chain-structured estimator tolerates the one violation it can absorb into its own parameters and fails on the one it structurally cannot represent, and the two are not interchangeable amounts of "how wrong the model is": they are different in kind.

## 9.6 Shape is intrinsically harder to identify than correlation

Section 8.5 found that the innovation shape converges far slower than the autoregressive coefficient under EM. A direct information-theoretic measurement gives a structural reason: on a generalised-Gaussian innovation family where a shape parameter $\beta$ moves the fourth moment at *exactly fixed* variance, the efficient Fisher information for $\beta$ — after projecting out the nuisance coordinates $(\alpha, q)$, since raw diagonal entries alone are misleading when parameters are correlated — decays two to three orders of magnitude faster with diffusion noise than the information for $\alpha$ itself:

| true $\beta$ | info loss, $\alpha$ | info loss, $q$ | info loss, $\beta$ (shape) |
|---|---|---|---|
| 1.0 | 78× | 235× | **8,738**× |
| 1.5 | 83× | 459× | **64,227**× |
| 3.0 | 56× | 522× | **68,460**× |

(information loss measured between the smallest and largest diffusion time tested). This is not merely a restatement of the convergence finding: it is a property of the noised likelihood itself, independent of any optimiser, and it explains *why* the shape coordinate is slow — there is structurally less information about it in noised data at any fixed sample size, at every diffusion time past the smallest ones. It is also the reason the capacity sweep of Section 9.3 is exactly the kind of measurement where an insufficient iteration budget is dangerous: the coordinate that is hardest to move is also the one with the least signal per observation.

## 9.7 Scope not covered here

The estimator of Chapter 8 was also extended, within the same research package, to video sequences and to natural-image patches, and to a wavelet-domain hidden-Markov-tree generalisation of the chain structure used throughout this thesis. Those extensions are reported in the package's own notebooks rather than reproduced here. The image comparison in particular fits the two kernel families with different and not directly comparable factories, so it does not support a ratio between them, and none is quoted in this thesis.

## 9.8 Summary

Two results are established, and two of the four questions this chapter opened with are answered more narrowly than they were asked.

At equal training-set size the structured estimator attains relative denoising error lower than the unstructured MLP by a factor of 7.0–17.6, across 8 sizes and 16 replicates, and this survives the validation-split correction to the oracle that originally selected the comparison. This is a ratio of errors at fixed data, not a ratio of the data required to reach a fixed error: the two curves do not cross in the range measured, so no data-equivalence factor is quoted. Most of that margin is attributable to the network's lack of any locality prior rather than to the estimator; against a weight-shared window head the advantage is a small multiple rather than an order of magnitude, and Section 9.2 gives the reasons that residual is a bound on what remains to be explained rather than a measurement of what the Markov assumption is worth.

The Markov structure the estimator relies on is not a single quantity of "how much structure”. It survives a violation it can absorb into its own parameters — a rank-one global latent, where the advantage never inverts even when half the marginal variance is a shared constant — and fails, well before generic intuition about misspecification would suggest, on a long-range coupling it structurally cannot represent.

The remaining two questions are open rather than answered. Whether the pointwise advantage transfers to generation is not resolved here: the experiment that appeared to answer it scored converged estimators against unconverged ones on the single coordinate that had not converged, and Section 9.4 withdraws the reading rather than reversing it. On capacity, held-out evidence and paired pointwise MSE both show most of the gain relative to $C = 1$ already present by $C = 8$, with no resolvable improvement from $C = 8$ to $C = 16$ at $N = 128$; six seeds at one sample size with no predeclared equivalence region cannot convert that into a claim of saturation, and any capacity statement resting on fitted or generated *shape* at the fixed 40-iteration budget is confounded with convergence rate (Remark 9.1).

# Chapter 10

# Discussion and Conclusions

## 10.1 Findings

**Derived exactly.** For the stationary AR(1) chain under coordinatewise OU corruption, the joint score is $S(x,t) = -Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$, equal by the score–posterior identity to $(e^{-t}\mathbb{E}[a|x] - x)/\Delta_t$ (RQ1). Its coupling structure has a closed-form lifecycle: exactly tridiagonal at $t = 0$, filling one diagonal per order in $t$ by the band-fill law (5.13), and returning to the identity at rate $e^{-2t}$. In the bulk, the single quantity $q(\alpha, t) \in (0, |\alpha|]$ governs both the posterior correlation length and the geometric decay of evidence influence, with limits $q \to 0$ as $t \to 0$ and $q \to |\alpha|$ as $t \to \infty$. The posterior of the noisy chain lives on a tree-shaped factor graph, on which one forward and one backward BP sweep compute all smoothing marginals; on the Gaussian chain the messages close algebraically in the Gaussian family, and the recursion is the Kalman filter followed by the RTS smoother (RQ2). The influence of evidence at distance $d$ on the posterior mean decays exactly as $q^d$; the radius-$r$ local score estimator's error is measured to decay at the same rate over the tested range, which is stated as a measurement rather than derived (RQ4). On the same model, the per-node AMP/TAP closure returns the exact score (the mean fixed point is closure-independent) but a different variance, with existence boundary, breakdown time $t_c(\alpha)$, and critical coupling $\alpha_c = \sqrt{2} - 1$ in closed form (Appendix B); all of these formulas were re-verified independently.

**Measured numerically.** The two Gaussian score routes agree to $10^{-15}$–$10^{-14}$ across the tested $(K, \alpha, t)$ grid, with an independently written implementation reproducing the agreement. On Laplace chains, Gaussian-projected BP measured against validated grid quadrature has median relative score error 0.24 at $t = 0.05$, falling below $10^{-2}$ by $t \approx 1$ and below $10^{-3}$ by $t \approx 2.1$; no trial exceeds 10% relative error beyond $t \approx 0.98$. The error is amplified at small $t$ by the exact factor $e^{-t}/\Delta_t$, is worst for bimodal innovation laws, and decreases with stronger chain correlation, while the score direction remains accurate (median cosine $\geq 0.976$ throughout, worst single trial 0.879) (RQ3). At equal locality budget, truncating the inference is more accurate than truncating the score matrix at small $t$ (12.3% against 21.0% at $t = 0.05$).

**The Gaussian-to-Laplace comparison.** The structural lesson is where exactness is lost: the tree, the two-sweep schedule, and the score–posterior identity survive any innovation law; what breaks is the representability of messages in a finite-dimensional family. The practical lesson is when it matters: the Gaussian closure is essentially free at moderate and large diffusion times, and expensive precisely in the small-$t$ regime where the score magnitude is amplified, though even there the score direction is preserved much better than its magnitude.

**Learning the innovation law.** Chapters 8 and 9 relax the assumption, held throughout the rest of this thesis, that the innovation law is known. An EM procedure built directly on the exact chain inference of Chapter 5 estimates a mixture innovation from noised data alone, with a structurally exact E-step for any innovation law and a generalised M-step.

Answering RQ6, the resulting estimator attains between 7.0 and 17.6 times lower pointwise denoising error than networks trained by denoising score matching on identical data, at equal training-set size, across the 8 sizes tested. This is a ratio of errors at fixed data and not a ratio of the data each method needs to reach a fixed error: the two curves do not cross in the measured range, so no data-equivalence factor follows from it. The ratio is *not* monotone in the training-set size, and the reported summary is the largest of several defensible aggregations of the same cells, which Appendix I.1 sets out. At the larger sizes each side often selects the largest budget on offer; that is not the same as being limited by it, since an arm selects its last checkpoint whenever validation error is still falling at any rate at all. Rerunning the 2048-sequence case with both budgets tripled moves the ratio by $-0.16 \pm 0.18$, so the caps are selected without binding. Both arms choose their budget on a disjoint validation bundle — the network its parameterisation and training length, the estimator its iteration count — because the training length is a bias–variance knob for both and selecting one while pinning the other is not a neutral protocol. The numbers are generated from the frozen outputs by `tools/make_tab_efficiency.py` rather than transcribed.

What that comparison does *not* establish bears repeating here, because the figure is quotable and the caveat is not. The two arms are asymmetric in supplied information by construction: the estimator receives the Markov factorisation, the autoregressive form and homogeneity, and the networks receive none of them. The measurement is of what a correct structural assumption is worth on a family where it holds exactly. It is not a ranking of learning algorithms.

In particular the comparison is against an *unstructured* network, and most of the margin turns out to be attributable to that choice rather than to estimation. A weight-shared window baseline — the same object as a one-layer convolution — closes the greater part of the gap, which means the headline figure should be read as an upper bound on the value of the structural assumption and not as its size. What survives against a baseline that can actually represent chain inference, rather than one restricted to a fixed window, is the quantity of real interest, and this thesis does not measure it: the comparison available here scores one site, at a shorter chain and a coarser noise schedule, under a budget that contracts as the training set grows (Section 9.2). The corrected experiment is specified in `experiments/exp_31_structured_baseline.py` and its results are not in this thesis. What can be said without it is the direction and its order: roughly an order of magnitude against an unstructured network, and considerably less against a structure-aware one.

The chain-Markov assumption itself tolerates a violation it can absorb into its own fitted parameters (a shared global latent, up to the point where it accounts for half the marginal variance) and fails, well before generic intuition about misspecification would suggest, on one it structurally cannot represent (long-range coupling, by a coupling strength of roughly one tenth).

**Two questions this thesis leaves open rather than answers.** Whether mixture capacity saturates near eight components, and whether pointwise accuracy and generative fidelity can move in opposite directions within one estimator, are both unresolved here, and for the same reason: the experiments that address them share a fixed iteration budget under which most shape coordinates had not settled, and the shape is what both questions turn on. On capacity there is a second, independent limitation, worth separating because it would apply even to a fully converged rerun — failing to resolve a difference between $C = 8$ and $C = 16$ at six seeds is not evidence that none exists, and no equivalence region was predeclared against which "none" could be judged. A budget-limited comparison answers "how do these behave after this much optimisation", which is not the question either was asked; that distinction is itself one of the findings under RQ5, and the convergence measurement that establishes it (Section 8.5) is what the corrected reruns are designed around.

## 10.2 What the model suggests but does not establish

Within the Gaussian benchmark, local score computations are near-optimal away from intermediate diffusion times, the required receptive field scales as $\xi \log(1/\varepsilon)$ with $\xi = 1/\log(1/q)$, and a local algorithm is preferable to a banded linear readout. These are properties of exact inference in one controlled model. Whether trained score networks on real sequence data inherit them is untested here and would require learning experiments.

## 10.3 Limitations

(i) The exact solutions concern one-dimensional state per frame and linear AR(1) dynamics, extended to a two-dimensional nonlinear state only in Chapter 7. (ii) The non-Gaussian results are numerical: grid BP is a reference only within its validated resolution regime ($\sqrt{2t} \gtrsim 3\,\mathrm{d}x$), and the Laplace closure error is a measured curve, not a formula. (iii) The locality laws are proved for the Gaussian bulk; their non-Gaussian counterparts are only structurally delimited; in particular no low-rank structure is asserted for the Laplace $K = 2$ posterior Hessian, which holds only in a narrow regime. (v) The EM-learning results of Chapters 8 and 9 use a fixed, generous but ultimately insufficient iteration budget for several of the reported sweeps; any conclusion in those chapters resting on fitted or generated *shape* at that budget is flagged explicitly as confounded with convergence rate rather than resolved. (vi) The misspecification results of Section 9.5 probe exactly two violations of the Markov assumption; they establish that the two are not interchangeable in effect, not a general theory of how much non-Markov structure a chain estimator tolerates. (vii) The image and video extensions of the EM/BP estimator, explored in the underlying research package, are outside the scope of this thesis and are not reported here.

## 10.4 Future work

Mixture messages, in the form of a fully estimated $C$-component innovation kernel rather than a fixed two-component closure, are carried out in Chapters 8 and 9. Reverse dynamics remains open: Section 9.4 integrates the reverse SDE under the estimated EM/BP score and under two trained neural scores, but on fits the convergence analysis shows to be unsettled, so the comparison needs redoing with estimators selected on a convergence criterion before it can answer the question it was built for. Discrete chains remain untouched by this thesis: with a finite alphabet the BP messages are finite vectors and inference is exact with no closure at all, which would give a second solvable benchmark alongside the Gaussian one.

What the new chapters measured opens further directions of their own. Re-running the capacity sweep of Section 9.3 to a convergence criterion rather than a fixed iteration count, now that Section 8.5 has measured what that criterion should be, would settle the one result in this thesis that is explicitly flagged as confounded. The misspecification study of Section 9.5 probes two violations at a time; a paired design across violation strengths, and further violation types beyond a global latent and long-range coupling, would generalise the finding that the two mechanisms are not interchangeable. And the reverse-dynamics comparison itself used a fixed, small chain length and one innovation family; whether pointwise accuracy and generative fidelity dissociate at all once both arms are converged, and if so whether that behaviour scales with sequence length or survives other non-Gaussian innovation laws, is the question Section 9.4 leaves open.

# Appendix A

# Gaussian Identities Used Throughout

This appendix collects, with proofs, the handful of Gaussian identities on which Chapter 5 relies. Notation: a Gaussian density in *moment* parameters is $\mathcal{N}(a; m, v)$; in *natural* parameters it is $\exp(-\frac{\lambda}{2}a^2 + \theta a + \text{const})$ with $\lambda = 1/v$, $\theta = m/v$.

## A.1 Quadratic forms and Gaussian normalisation

For $J \succ 0$ (symmetric positive definite) and any $h$,

$$\int_{\mathbb{R}^K} \exp\Big(-\tfrac{1}{2}a^\mathsf{T} J a + h^\mathsf{T} a\Big)\,\mathrm{d}a = \frac{(2\pi)^{K/2}}{\sqrt{\det J}} \exp\Big(\tfrac{1}{2}h^\mathsf{T} J^{-1} h\Big), \tag{A.1}$$

by completing the square, $-\frac{1}{2}a^\mathsf{T} J a + h^\mathsf{T} a = -\frac{1}{2}(a - J^{-1}h)^\mathsf{T} J (a - J^{-1}h) + \frac{1}{2}h^\mathsf{T} J^{-1} h$, and translating. Consequently a density $p(a) \propto \exp(-\frac{1}{2}a^\mathsf{T} J a + h^\mathsf{T} a)$ is the Gaussian with mean $m = J^{-1}h$ and covariance $J^{-1}$: *reading off* $(J, h)$ *from a log-density is the fastest route to posterior parameters*, used in Proposition 5.5.

## A.2 Products of Gaussians in the same variable

$$\mathcal{N}(a; m_1, v_1)\,\mathcal{N}(a; m_2, v_2) \;\propto\; \mathcal{N}\Big(a;\, m, v\Big), \qquad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \qquad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}. \quad \text{(A.2)}$$

*Proof.* Add the exponents; the sum of two quadratics in $a$ is a quadratic in $a$ whose $a^2$-coefficient is $-\frac{1}{2}(1/v_1 + 1/v_2)$ and whose linear coefficient is $m_1/v_1 + m_2/v_2$; apply (A.1) in one dimension. In natural parameters the rule is addition: $(\lambda, \theta) = (\lambda_1 + \lambda_2, \theta_1 + \theta_2)$, which is why the BP updates of Chapter 5 are two additions per message. □

## A.3 Affine propagation (marginalisation)

If $a' = \alpha a + \eta$ with $\eta \sim \mathcal{N}(0, \sigma^2)$ independent of $a \sim \mathcal{N}(m, v)$, then

$$\int \mathcal{N}(a'; \alpha a, \sigma^2)\,\mathcal{N}(a; m, v)\,\mathrm{d}a = \mathcal{N}\Big(a';\, \alpha m,\, \alpha^2 v + \sigma^2\Big). \quad \text{(A.3)}$$

*Proof.* $a'$ is a sum of independent Gaussians, hence Gaussian; its mean is $\alpha m$ and its variance $\alpha^2 v + \sigma^2$ by linearity of expectation and independence. Read in the reverse direction (solving for $a$ given a Gaussian summary of $a'$), the same computation gives $\mathcal{N}\Big(a;\, m'/\alpha,\, (v' + \sigma^2)/\alpha^2\Big)$, the backward half-step of the BP sweep. □

## A.4 Gaussian conditioning

For a jointly Gaussian pair with precision blocks $\left(\begin{smallmatrix} J_{11} & J_{12} \\ J_{12}^{\top} & J_{22} \end{smallmatrix}\right)$, the conditional of block 1 given block 2 is Gaussian with precision $J_{11}$ (conditioning *reads off* the precision block) and mean shifted by $-J_{11}^{-1} J_{12}\, a_2$; marginalisation, dually, is trivial in covariance parameters. This precision/covariance duality (conditioning is local in $J$, marginalising is local in $\Sigma$) is the algebraic engine behind the cavity recursion (B.1): integrating a neighbour out of a tridiagonal precision produces the rank-one correction $-\beta^2/\lambda$ on the adjacent diagonal entry, by the Schur complement

$$\begin{pmatrix} \lambda & \beta \\ \beta & J_d \end{pmatrix} \longrightarrow J_d - \frac{\beta^2}{\lambda}. \tag{A.4}$$

## A.5 Woodbury and the two routes to the score

The Woodbury identity $(A+UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1}+VA^{-1}U)^{-1}VA^{-1}$, specialised to $\Sigma_t = \Delta_t I + e^{-2t}\Sigma_0$, connects the noisy precision $Q_t$ and the posterior precision $J = (e^{-2t}/\Delta_t)I + Q_0$:

$$Q_t = \frac{1}{\Delta_t}\left(I - \frac{e^{-2t}}{\Delta_t}\, J^{-1}\right), \tag{A.5}$$

which is exactly the operator form of the score–posterior identity: substituting the posterior mean $m = J^{-1}h = J^{-1}(e^{-t}/\Delta_t)x$ into $S = (e^{-t}m - x)/\Delta_t$ gives $S = -Q_t x$ identically. The machine-precision agreement of the two score routes (Section 5.3) is the numerical shadow of (A.5).

# Appendix B

# AMP, TAP, and the Bulk Fixed Point

This appendix develops the per-node Gaussian closures of statistical physics, AMP and the TAP equations, on the Gaussian chain, and proves the exact breakdown announced in Remark 5.13. The results are complete, audited, and independently re-verified; they are collected here rather than in Chapter 5 because they branch off the thesis's main line. Summary: on the chain, AMP still returns the exact score, but its variance closure fails past an exact breakdown time $t_c(\alpha)$, with critical coupling $\alpha_c = \sqrt{2} - 1$.

## B.1 The bulk cavity fixed point of BP

Chapter 5 solved the bulk covariance by matrix algebra; the same formulas re-emerge from the message recursion, which is worth exhibiting because it localises where BP and AMP differ. In the bulk, parametrise the forward message (evidence absorbed) by its precision $\lambda_k$; the closure lemmas of Section 5.6 give the scalar recursion

$$\lambda_{k+1} \;=\; J_d - \frac{\beta^2}{\lambda_k}, \tag{B.1}$$

with $J_d, \beta$ the bulk parameters (5.20): absorbing one neighbour's summary costs the Gaussian-integration correction $-\beta^2/\lambda$. The fixed points are $\lambda^{\pm} = \left(J_d \pm \sqrt{J_d^2 - 4\beta^2}\right)/2$, and since $J_d - 2|\beta| = e^{-2t}/\Delta_t + (1-|\alpha|)^2/\sigma_\eta^2 > 0$ (Chapter 5), the attractive upper fixed point

$$\boxed{\lambda^* = \frac{J_d + \sqrt{J_d^2 - 4\beta^2}}{2}} \tag{B.2}$$

exists for every $\alpha \in (-1, 1)$ and $t > 0$, with $|f'(\lambda^*)| = \beta^2/\lambda^{*2} < 1$: the sweep converges geometrically from any initialisation. Recombining the two directions and the local term, and using the fixed-point relation $\beta^2/\lambda^* = J_d - \lambda^*$,

$$\Lambda_{\text{bulk}} = J_d - \frac{2\beta^2}{\lambda^*} = 2\lambda^* - J_d = \sqrt{J_d^2 - 4\beta^2} \quad \Longrightarrow \quad V_{\text{exact}} = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \tag{B.3}$$

in exact agreement with Theorem 5.14. Belief propagation on the Gaussian chain has no phase transition: the cavity fixed point exists always, and the discriminant $J_d^2 - 4\beta^2$ is the quantity to watch as the closure is modified.

## B.2 AMP/TAP: same score, different variance

AMP, the algorithmic form of the TAP equations (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016), is the message passing one uses when the factor graph is dense: each message is then a sum of many weak contributions, approximately Gaussian by a central-limit argument, and one tracks per-node means and variances instead of per-edge messages, with the Onsager term correcting the self-influence. Rigorous derivations of AMP from BP exist in that dense limit (Genovese and Piana, 2026). The chain is the opposite regime in every respect (diagonal measurement, coupling in the prior, degree two), so it isolates what the per-node closure costs. Viewing the posterior as a Gaussian Markov field with precision $J$ and field $h$, the three schemes solve, at each node,

$$\text{mean field: } V_i = \frac{1}{J_{ii}}; \qquad \text{AMP/TAP: } V_i = \frac{1}{J_{ii} - \sum_{k \neq i} J_{ik}^2 V_k}; \qquad \text{BP: } V_i = (J^{-1})_{ii}. \tag{B.4}$$

The AMP closure is the BP cavity with the exclusion of the receiving neighbour dropped ($V_{k\to i} \approx V_k$): an $O(1/N)$ change on a dense graph, but a change of fixed, order-one size on a chain with two neighbours.

**Theorem B.1** (The score is closure-independent)**.** *For a Gaussian posterior, the fixed-point condition for the means is the same linear system $Jm = h$ under mean field, BP, and AMP. All three schemes hence return the exact posterior mean and, via the score–posterior identity, the same exact joint score, for every $(K, \alpha, t)$, regardless of whether their variance closures are accurate or even solvable.*

The proof is one line: in each scheme the mean update at node $i$ reads $m_i \leftarrow (h_i - \sum_{k\neq i} J_{ik} m_k)/J_{ii}$ at the fixed point; the closures modify the uncertainty bookkeeping, not the linear relation among means, and the unique solution of $Jm = h$ is the exact posterior mean. Numerically, AMP means agree with exact means to $4.3 \times 10^{-12}$ and AMP scores with BP scores to $1.7 \times 10^{-12}$ across all tested configurations.

For the variances the story splits. In the bulk the AMP closure (B.4) reads $V = 1/(J_d - 2\beta^2 V)$; the BP cavity is $V_{\text{cav}} = 1/(J_d - \beta^2 V_{\text{cav}})$. The entire difference between BP and AMP on the chain is a factor of two: both neighbours included where BP excludes one.

**Theorem B.2** (AMP bulk variance, breakdown time, critical coupling)**.** *The AMP bulk closure is the quadratic $2\beta^2 V^2 - J_d V + 1 = 0$, with physical root*

$$V_{\text{AMP}} = \frac{J_d - \sqrt{J_d^2 - 8\beta^2}}{4\beta^2}, \tag{B.5}$$

*which exists iff $J_d \geq 2\sqrt{2}\,|\beta|$, strictly stronger than BP's always-true $J_d > 2|\beta|$. Since $J_d(t)$ decreases monotonically in $t$, the existence condition fails, if ever, past a single time; solving it as an equality gives*

$$\boxed{t_c(\alpha) = -\tfrac{1}{2}\log\frac{g}{1+g}, \qquad g(\alpha) = \frac{2\sqrt{2}\,|\alpha| - 1 - \alpha^2}{1-\alpha^2},} \tag{B.6}$$

*finite exactly when $g > 0$, i.e.*

$$\boxed{|\alpha| \;>\; \alpha_c \;=\; \sqrt{2}-1 \;\approx\; 0.4142.} \tag{B.7}$$

*For $|\alpha| \le \alpha_c$ the AMP variance has a fixed point at every diffusion time; for $|\alpha| > \alpha_c$ it ceases to exist for all $t > t_c(\alpha)$. In the weak-coupling regime the two closures agree to second order, with*

$$V_{\mathrm{AMP}} - V_{\mathrm{exact}} = \frac{2\beta^4}{J_d^5} + O\Big(\frac{\beta^6}{J_d^7}\Big) \;>\; 0: \tag{B.8}$$

*AMP always overestimates the bulk variance, at fourth order in the coupling.*

**Toolbox B.1 — Roots, branch, and the phase boundary**

*Branch selection.* The quadratic has roots $V_\pm = (J_d \pm \sqrt{J_d^2 - 8\beta^2})/(4\beta^2)$; as $\beta \to 0$, $V_- \to 1/J_d$ (the decoupled answer) while $V_+$ diverges, so $V_-$ is the physical branch, and the attractive fixed point of the damped iteration used in practice. *Critical coupling.* $g > 0 \iff \alpha^2 - 2\sqrt{2}|\alpha| + 1 < 0 \iff |\alpha| \in (\sqrt{2}-1, \sqrt{2}+1)$; intersecting with $|\alpha| < 1$ leaves $|\alpha| > \sqrt{2}-1$. Consistency check at $t = \infty$: there the posterior is the bare prior, and the existence condition applied to $Q_0$ itself reads $(1+\alpha^2) \ge 2\sqrt{2}|\alpha|$, i.e. AMP applied to a bare AR(1) chain has a variance fixed point iff $|\alpha| \le \sqrt{2}-1$, which is (B.7) read backwards. *Weak coupling.* With $\epsilon := \beta^2/J_d^2$: $(1-4\epsilon)^{-1/2} = 1 + 2\epsilon + 6\epsilon^2 + O(\epsilon^3)$ for the exact variance; $1 - \sqrt{1-8\epsilon} = 4\epsilon + 8\epsilon^2 + O(\epsilon^3)$, divided by $4\epsilon J_d$, gives $V_{\mathrm{AMP}} = (1 + 2\epsilon + 8\epsilon^2 + O(\epsilon^3))/J_d$; subtract. $\square$

Numerically, on a $K = 12$ chain at $\alpha = 0.8$: the AMP score error stays at $10^{-13}$–$10^{-12}$ for every $t$, while the variance error is $1.3 \times 10^{-4}$ at $t = 0.05$, $3.5 \times 10^{-2}$ at $t = 0.2$, and past $t \approx 0.23$ the variance update has no positive fixed point at all. The audit further verifies the closed form (B.5) against the converged iteration to $2.5 \times 10^{-12}$; the existence condition at all 180 points of an $(\alpha, t)$ scan; $t_c$ against the bisected flip point of the iteration to $6.6 \times 10^{-6}$ relative; convergence below $\alpha_c$ even at $t = 50$; and the weak-coupling ratio $(V_{\mathrm{AMP}} - V_{\mathrm{exact}})/(2\beta^4/J_d^5) \to 1$ within 0.4%. The interpretation is not that AMP is wrong; on a dense benchmark (random $N = 600$ symmetric positive-definite

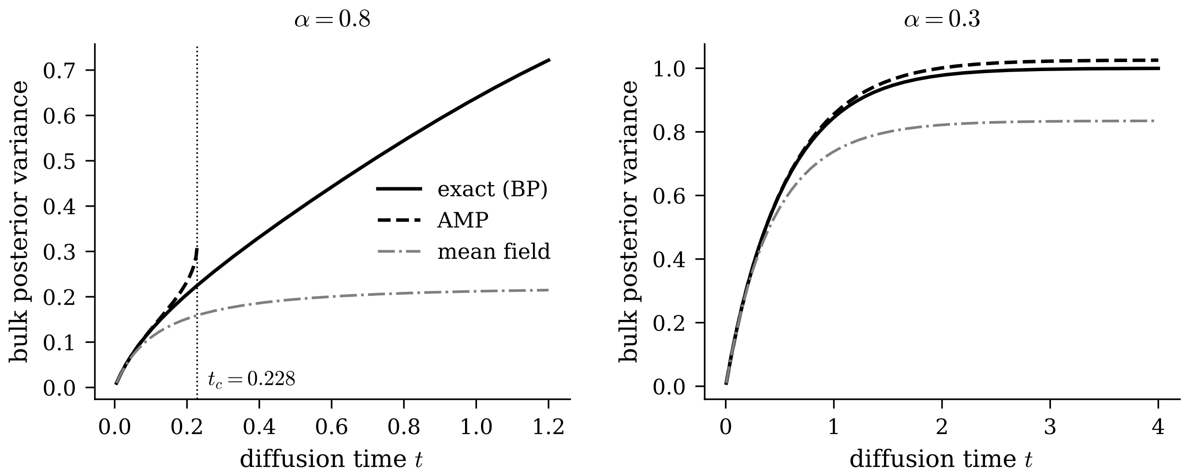


**Figure B.1:** The three closures in closed form. Left ($\alpha = 0.8 > \alpha_c$): exact = BP, AMP (B.5), and mean field; the AMP branch terminates with a square-root singularity at the breakdown time $t_c(0.8) = 0.228$ from (B.6) (dotted line); beyond it AMP has no variance fixed point. Right ($\alpha = 0.3 < \alpha_c$): AMP exists at every $t$ and slightly overestimates the variance, as predicted by (B.8).

precision) the same closure gets variances to $1.7 \times 10^{-3}$ where mean field errs at $4 \times 10^{-2}$. It is built for a different graph: the central-limit justification of the per-node closure needs many incoming messages, and a chain provides two.

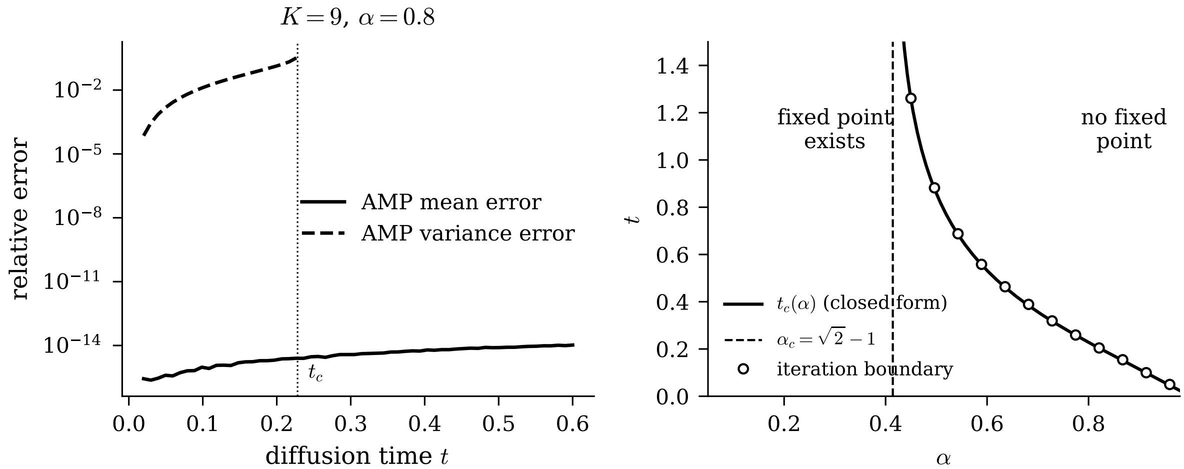


**Figure B.2:** BP against AMP on a finite chain ($K = 9$, $\alpha = 0.8$). Left: the AMP mean error (about $10^{-12}$, flat in $t$: Theorem B.1) against the AMP variance error, which grows with $t$ up to the breakdown time $t_c$ (dotted line). Right: the phase boundary of the AMP variance fixed point in the $(\alpha, t)$ plane: the solid curve is the closed form $t_c(\alpha)$ of (B.6), the dashed line is $\alpha_c = \sqrt{2} - 1$, and the open circles mark the breakdown as located independently by damped fixed-point iteration; the audit suite performs the same comparison over 180 points of this plane.

# Appendix C

# Reproducibility

## C.1 Repository layout

All source material lives in a single repository; the thesis directory is self-contained (LaTeX source, figures, and the figure script are committed together).

| Path | Content |
|---|---|
| `thesis/` | this document: LaTeX source, figures, figure script |
| `research/gaussian-bp/` | Gaussian chain note, code, 72-check audit |
| `research/bp-from-scratch/` | independent BP derivation and implementation |
| `research/unified-note/` | consolidated working note, code, 55-check audit |
| `research/nongaussian-bp/` | grid BP, EM, and the estimator comparisons |
| `src/`, `experiments/` | implementation and one driver per experiment |
| `hpc/`, `tools/` | deployment/provenance, and the table and figure generators |
| `outputs/frozen/` | the certified run directories the generators read |
| `research/gaussian-ar1-bp/` | Gaussian AR(1) package (BP = matrix score) |
| `research/initial-experiments/` | early experiments and toy-model material |
| `research/experiment1-rotating-ring/` | rotating-ring package: code, data, validation |

## C.2 Audit protocol

Every closed-form claim is verified by a deterministic check against an independently computed reference (brute-force linear algebra, dense grid quadrature, or finite differences), collected in audit drivers that exit non-zero on any failure. The audit driver of `research/gaussian-bp` runs 72 checks: assembly and inversion of $\Sigma_t$; the score–posterior route against the matrix score; the $K = 2$ closed forms; Convention-A BP against the matrix posterior (means, variances, scores); bulk variance, covariance decay $q^d$, and the recombination identity; AMP mean and score closure-independence; the AMP variance formula, existence boundary, breakdown time, and weak-coupling law; the locality slopes; the band-fill coefficients (with the factorial variant explicitly rejected); spectral preservation and the large-$t$ return. The audit driver of `research/unified-note` runs 55 checks on the consolidated note, including the Laplace $K = 1$ quadrature checks and the one-step boundary-message identity. `research/nongaussian-bp/` contains experiment drivers with logged master-error-identity residuals ($10^{-15}$–$10^{-12}$ throughout), grid self-convergence gates, and per-trial CSV outputs; the grid calibration of Section 6.4 is experiment 01, the closure measurement of Section 6.5 experiment 02. `research/experiment1-rotating-rin` carries the package behind Chapter 7: the exact surrogate score and Jacobian, the polar-grid recursion, the response diagnostics in a single module, and a validation driver that writes `validation.json` and exits non-zero on any failure. Its five checks are the ones reported in Table 7.1, and every number quoted in Chapter 7 is regenerated from the curated CSVs of that package rather than entered by hand. All checks were re-executed on the build machine at the time of writing: 72/72, 55/55 and 5/5 pass, and all experiment gates are green.

## C.3 Figure-to-code map

The figures of Chapters 5 to 6 and of the appendices are regenerated by one script, `figures/make_figures.py`, which recomputes each panel from the closed forms or quadrature of the corresponding chapter; the only one of those that reads stored experimental data is Figure 6.2, built from the per-trial CSV of `nongaussian-bp` experiment 02. The figures of Chapter 7 are regenerated by `figures/make_ring_figures.py`, in the same style; their closed-form panels are recomputed at build time and their measured panels read the curated CSVs under `research/experiment1-rotating-ring/data/`. All figures are vector PDF with white background; randomness, where used, is seeded.

| Figure | File | Source of the numbers |
|---|---|---|
| 5.1 | `fig_band_fill.pdf` | closed forms, Ch. 5 |
| 5.2 | `fig_tridiag_loss.pdf` | closed forms, Ch. 5 |
| 5.3 | `fig_local_vs_full.pdf` | closed forms, Ch. 5 |
| 6.1 | `fig_laplace_k1.pdf` | quadrature, Ch. 6 |
| 6.2 | `fig_laplace_closure.pdf` | `nongaussian-bp` exp. 02 CSV |
| B.1 | `fig_bulk_variance.pdf` | closed forms, App. B |
| B.2 | `fig_bp_vs_amp.pdf` | closed forms + iteration, App. B |
| 7.1 | `fig_ring_model.pdf` | closed forms + simulation, Ch. 7 |
| 7.2 | `fig_ring_gauge.pdf` | closed forms + simulation, Ch. 7 |
| 7.3 | `fig_ring_surrogate_response.pdf` | exact surrogate score, Ch. 7 |

## C.4 The learning experiments of Chapters 8 and 9

The closed-form and quadrature studies above are deterministic and run on a laptop. The estimator comparisons are neither, and they are gated differently.

**Provenance.** Every reported run is deployed by `hpc/deploy_clean.sh`, which refuses to build from a dirty tree, ships a `git archive` of the commit rather than the working tree, records the SHA-256 of that tarball, and extracts on the cluster under the commit id so that a run cannot overwrite or edit its own source. The motivating failure is preserved in that script's header: an earlier two-step stamp-then-rsync path produced `outputs/exp_12_scaled/`, whose recorded commit cannot have run the recorded command, and whose numbers are therefore not recoverable and are not cited in this thesis.

`tools/provenance_gate.py` intersects a run's recorded dirty-file list with the import closure of its own experiment module, so that uncommitted work elsewhere in the tree is distinguished from uncommitted work on the code path that produced the numbers; the table generators call it and mark any figure or table it cannot certify.

**Generated numbers.** No efficiency, structured-baseline or aggregation figure in this thesis is typed. Each is defined once in `overleaf/shared/sections/` by a generator (`tools/make_tab_efficiency.py`, `tools/make_tab_structured.py`, `tools/make_aggregation_robu` `tools/make_figures.py`) reading the frozen outputs, and the prose cites the macro. The figures and the tables therefore share a size grid and an aggregation rule by construction rather than by inspection.

**Withdrawn inputs.** `nseq` = 8192 is excluded from both Table 9.1 and Figure 9.1: the intersection of its dirty list with `exp_07`'s import closure contains the experiment, the frozen configuration and `src/em.py` themselves. The `nseq` = 4096 rows are retained, with the reason given in the table caption — their intersection is empty, so no edit in that tree could have moved those numbers.

## C.5 Environment

Python $\geq$ 3.10 with NumPy, SciPy, pandas and Matplotlib. The neural baselines of Chapter 9 are implemented in pure NumPy (`src/nnet.py` for the fully connected denoiser, `src/seq_nets.py` and `src/local_head.py` for the locality-respecting architectures) with an explicit `numpy.random.Generator` driving initialisation and minibatch order, so training is stochastic but reproducible from the recorded seed. Two optional dependencies are used and neither is on the exactness path: CuPy, selected by the `BP_DEVICE` environment variable, runs the same single BP implementation on a GPU — `tests/test_backend_parity.py` asserts the two devices agree, so a device change fails a test rather than moving a number quietly — and PyTorch, imported lazily in `src/fid.py` alone, for the generative metric. Arithmetic is float64 throughout.

The sweeps of Chapters 8 and 9 run under SLURM from the batch scripts in `hpc/`, one per experiment. The document compiles with a single run of `tectonic main.tex`.

# Appendix D

# Derivations from Statistical Mechanics

These are the derivations underlying Chapter 2. They are placed here so that the body carries the argument and the appendix carries the work; nothing in them is omitted.

## D.1 Hamiltonian mechanics and phase space

Consider a mechanical system with $n$ degrees of freedom, described by generalised coordinates $q = (q_1, \ldots, q_n)$ and conjugate momenta $p = (p_1, \ldots, p_n)$. Its dynamics is generated by a single scalar function, the *Hamiltonian* $H(q, p)$, which for the systems considered here has the mechanical form

$$H(q,p) \;=\; \underbrace{\sum_{i=1}^{n} \frac{p_i^2}{2m_i}}_{\text{kinetic}} \;+\; \underbrace{U(q_1, \ldots, q_n)}_{\text{potential}}, \tag{D.1}$$

and equals the total energy of the system. The state of the system at any instant is the pair $(q, p)$, a point in the $2n$-dimensional *phase space* $\Gamma \subseteq \mathbb{R}^{2n}$. The evolution is given by

*Hamilton's equations*,

$$\dot{q}_i \;=\; \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i \;=\; -\frac{\partial H}{\partial q_i}, \qquad i = 1, \ldots, n. \tag{D.2}$$

Two structural properties of (D.2) are the pillars on which statistical mechanics rests.

**Energy conservation.** Along any trajectory,

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \sum_{i=1}^{n} \left( \frac{\partial H}{\partial q_i}\,\dot{q}_i + \frac{\partial H}{\partial p_i}\,\dot{p}_i \right) = \sum_{i=1}^{n} \left( \frac{\partial H}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i}\frac{\partial H}{\partial q_i} \right) = 0. \tag{D.3}$$

The motion is therefore confined to the *energy shell* $\Gamma_E = \{(q,p) : H(q,p) = E\}$, a $(2n-1)$-dimensional hypersurface.

**Incompressibility of the phase flow.** The phase-space velocity field

$$v(q,p) \;=\; (\dot{q}, \dot{p}) \;=\; \left( \frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q} \right) \tag{D.4}$$

has identically vanishing divergence:

$$\nabla \cdot v \;=\; \sum_{i=1}^{n} \left( \frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) \;=\; \sum_{i=1}^{n} \left( \frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) \;=\; 0. \tag{D.5}$$

Hamiltonian flow behaves like an incompressible fluid in phase space: it can stretch and fold a region, but never change its volume. This identity is the seed of the probabilistic construction that follows.

## D.2 Liouville's theorem

A single trajectory is useless for macroscopic physics: for a gas, $n \sim 10^{23}$ and neither the initial condition nor the solution of (D.2) is accessible. The move that founds statistical mechanics is to describe our *ignorance* of the microstate by a probability density $\rho(q,p,t)$ on phase space: $\rho(q,p,t)\,\mathrm{d}q\,\mathrm{d}p$ is the probability that the system is in the infinitesimal cell around $(q, p)$ at time $t$. Liouville's theorem states how this density evolves.

**Theorem D.1** (Liouville)**.** *Under the Hamiltonian flow* (D.2)*, the phase-space density obeys*

$$\frac{\partial \rho}{\partial t} \;=\; -\{\rho, H\} \;=\; \sum_{i=1}^{n} \left( \frac{\partial H}{\partial q_i} \frac{\partial \rho}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial \rho}{\partial q_i} \right), \tag{D.6}$$

*where* $\{\cdot, \cdot\}$ *is the Poisson bracket. Equivalently, the density is constant along trajectories:* $\mathrm{d}\rho/\mathrm{d}t = 0$*.*

**Toolbox D.1 — Proof of Liouville's theorem, step by step**

The density is locally conserved (probability mass is neither created nor destroyed, only transported by the flow), so it satisfies the continuity equation familiar from fluid dynamics:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho\, v) = 0, \qquad v = (\dot{q}, \dot{p}).$$

Expand the divergence of the product using $\nabla \cdot (\rho v) = v \cdot \nabla \rho + \rho\, (\nabla \cdot v)$:

$$\frac{\partial \rho}{\partial t} + \underbrace{\sum_{i=1}^{n} \left( \dot{q}_i \frac{\partial \rho}{\partial q_i} + \dot{p}_i \frac{\partial \rho}{\partial p_i} \right)}_{v \cdot \nabla \rho} + \rho \underbrace{\sum_{i=1}^{n} \left( \frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right)}_{\nabla \cdot v} = 0.$$

By the incompressibility identity (D.5) the last term vanishes. Substituting Hamilton's equations $\dot{q}_i = \partial H / \partial p_i$, $\dot{p}_i = -\partial H / \partial q_i$ into the middle term:

$$\frac{\partial \rho}{\partial t} + \sum_{i=1}^{n} \left( \frac{\partial H}{\partial p_i} \frac{\partial \rho}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial \rho}{\partial p_i} \right) = 0 \quad \Longleftrightarrow \quad \frac{\partial \rho}{\partial t} = -\{\rho, H\}.$$

Finally, the total (convective) derivative along a trajectory is

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\partial \rho}{\partial t} + v \cdot \nabla \rho = \frac{\partial \rho}{\partial t} + \{\rho, H\} = 0,$$

which is the second form of the statement: an observer riding a phase-space trajectory sees a constant local density. □

Liouville's theorem has an immediate consequence: *any density that is a function of* *conserved quantities only is stationary.* If $\rho(q,p) = f\big(H(q,p)\big)$ for some function $f$, then $\{\rho, H\} = f'(H)\,\{H, H\} = 0$, so $\partial\rho/\partial t = 0$. Equilibrium statistical mechanics is the study of which $f$ nature chooses.

## D.3 From dynamics to ensembles

An *ensemble* is a probability distribution over microstates that is stationary under the dynamics and compatible with the macroscopic constraints we impose (fixed energy, fixed temperature, and so on). Two ensembles matter for this thesis.

### D.3.1 The microcanonical ensemble

For a perfectly isolated system, the energy is fixed at $E$ and Liouville's theorem suggests the simplest choice compatible with it: a uniform density on the energy shell,

$$\rho_{\mathrm{mc}}(q,p) \;=\; \frac{1}{\Omega(E)}\,\delta\Big(H(q,p) - E\Big), \qquad \Omega(E) = \int_\Gamma \delta\Big(H(q,p) - E\Big)\,\mathrm{d}q\,\mathrm{d}p, \tag{D.7}$$

where $\Omega(E)$ counts the microstates at energy $E$. The postulate that all accessible microstates are equally likely (the *equal a priori probability* postulate) is motivated, not proved, by ergodicity: for ergodic dynamics, time averages along a single trajectory converge to averages over the shell, so the uniform measure is the one a long experiment effectively samples. Boltzmann's entropy attaches a number to this count,

$$S(E) \;=\; k_B \log \Omega(E), \tag{D.8}$$

with $k_B$ the Boltzmann constant. Entropy is the logarithm of the number of microscopic ways a macroscopic state can be realised.

### D.3.2 The canonical ensemble and the Boltzmann distribution

Isolated systems are an idealisation; the physically ubiquitous situation is a system exchanging energy with a much larger environment (a *heat bath*) at temperature $T$. The stationary distribution of the system alone is then no longer uniform: it is the *Boltzmann–Gibbs distribution*,

$$\boxed{\rho_\beta(x) \;=\; \frac{1}{Z(\beta)}\, e^{-\beta H(x)}, \qquad Z(\beta) = \int e^{-\beta H(x)}\, \mathrm{d}x, \qquad \beta = \frac{1}{k_B T},} \tag{D.9}$$

where $x = (q, p)$ collects the microstate and $Z(\beta)$, the *partition function*, normalises the density. Equation (D.9) can be derived in two independent ways, and both derivations are instructive enough to be carried out in full: the first (physical) traces the exponential to the entropy of the bath; the second (information-theoretic) characterises it as the least-committal distribution compatible with a fixed mean energy, a perspective due to Jaynes (1957) that returns, almost verbatim, in modern machine learning.

**Toolbox D.2 — The Boltzmann distribution from the heat bath**

Let the total system (system $S$ together with bath $B$) be isolated with total energy $E_{\text{tot}}$, and let the coupling be weak, so $H_{\text{tot}} = H_S + H_B$. Apply the microcanonical postulate to the total system: the probability that $S$ is in a *specific* microstate $x$ with energy $H_S(x) = \epsilon$ is proportional to the number of bath microstates carrying the remaining energy,

$$\mathbb{P}(x) \;\propto\; \Omega_B\Big(E_{\text{tot}} - \epsilon\Big) \;=\; \exp\Big[\tfrac{1}{k_B} S_B\Big(E_{\text{tot}} - \epsilon\Big)\Big],$$

using the definition (D.8) of the bath entropy. Since the bath is huge, $\epsilon \ll E_{\text{tot}}$, expand $S_B$ to first order:

$$S_B(E_{\text{tot}} - \epsilon) = S_B(E_{\text{tot}}) - \epsilon\, \frac{\partial S_B}{\partial E}\Big|_{E_{\text{tot}}} + O\Big(\epsilon^2/E_{\text{tot}}\Big).$$

The derivative of the bath entropy with respect to its energy *defines* the bath temperature, $\partial S_B/\partial E = 1/T$. Substituting,

$$\mathbb{P}(x) \;\propto\; \underbrace{e^{S_B(E_{\text{tot}})/k_B}}_{\text{constant in } x} \cdot e^{-\epsilon/(k_B T)} \;\propto\; e^{-\beta H_S(x)}.$$

The quadratic remainder is suppressed by the bath's heat capacity and vanishes in the thermodynamic limit. Normalising yields (D.9). Note what happened: the exponential form is nothing but the first-order Taylor expansion of the bath's entropy. The Boltzmann factor is the price, in bath microstates, of lending energy $\epsilon$ to the system. □

**Toolbox D.3 — The Boltzmann distribution from maximum entropy**

Forget physics; ask, with Jaynes (1957): among all densities $\rho$ with a given mean energy $\mathbb{E}_\rho[H] = U$, which one assumes the least beyond the constraint? "Least" is measured by the Gibbs–Shannon entropy $S[\rho] = -k_B \int \rho \log \rho$. Maximise $S[\rho]$ subject to $\int \rho = 1$ and $\int \rho H = U$ with Lagrange multipliers $\lambda_0, \lambda_1$:

$$\mathcal{L}[\rho] = -k_B \int \rho \log \rho \,\mathrm{d}x \;-\; \lambda_0\Big(\int \rho \,\mathrm{d}x - 1\Big) \;-\; \lambda_1\Big(\int \rho H \,\mathrm{d}x - U\Big).$$

Take the functional derivative and set it to zero pointwise:

$$\frac{\delta \mathcal{L}}{\delta \rho(x)} = -k_B\Big(\log \rho(x) + 1\Big) - \lambda_0 - \lambda_1 H(x) = 0 \quad\Longrightarrow\quad \rho(x) = \exp\Big[-1 - \tfrac{\lambda_0}{k_B} - \tfrac{\lambda_1}{k_B} H(x)\Big].$$

The density is an exponential of the constrained quantity. This is completely general: constraining $\mathbb{E}[\phi_j(x)]$ for features $\phi_j$ produces $\rho \propto \exp[-\sum_j \beta_j \phi_j(x)]$, the exponential family. The multiplier $\lambda_0$ is fixed by normalisation and produces $Z$; renaming $\lambda_1/k_B = \beta$ and identifying $\beta = 1/(k_B T)$ through thermodynamics recovers (D.9) exactly. Second-order check: $\delta^2 \mathcal{L}/\delta \rho^2 = -k_B/\rho < 0$, so the stationary point is the unique maximum (the entropy is strictly concave). □

The second derivation deserves a remark. The Boltzmann distribution is not a statement about molecules but about *inference under constraints*. Any time we posit a model of the form $p(x) \propto e^{-E(x)}$, as we will for Markov random fields (Chapter 3) and energy-based models (Appendix F), we are doing statistical mechanics, whether the variables are spins, pixels, or the frames of a video.

## D.4 The partition function and free energy

The partition function $Z(\beta)$ in (D.9) looks like a mere normaliser; it is in fact the central computational object of the theory, and the reason approximate inference exists as a field. Its logarithm defines the *Helmholtz free energy*,

$$F(\beta) \; = \; -\frac{1}{\beta} \log Z(\beta). \tag{D.10}$$

**Toolbox D.4 — Everything is a derivative of $\log Z$**

Differentiate $\log Z$ with respect to $\beta$, carrying the derivative through the integral:

$$-\frac{\partial \log Z}{\partial \beta} = -\frac{1}{Z}\frac{\partial}{\partial \beta}\int e^{-\beta H} = \frac{\int H\, e^{-\beta H}}{\int e^{-\beta H}} = \mathbb{E}_{\rho_\beta}[H] \; = \; U,$$

the mean energy. Differentiate once more:

$$\begin{aligned}\frac{\partial^2 \log Z}{\partial \beta^2} &= \frac{\partial}{\partial \beta}\left(-\frac{\int H e^{-\beta H}}{Z}\right) = \frac{\int H^2 e^{-\beta H}}{Z} - \left(\frac{\int H e^{-\beta H}}{Z}\right)^2 \\ &= \mathbb{E}[H^2] - \mathbb{E}[H]^2 \; = \; \mathrm{Var}(H) \; \geq \; 0.\end{aligned}$$

So $\log Z$ is convex in $\beta$, its first derivative gives the mean energy, its second the energy fluctuations; the latter equals $k_B T^2 C_V$ with $C_V$ the heat capacity. This is a *fluctuation–dissipation* relation: how a system responds to heating is determined by its spontaneous equilibrium fluctuations.

The same identity holds for any observable coupled linearly to a field. If the Gibbs weight carries a factor $e^{h \cdot a}$, then $Z(h) = \int \nu(a) e^{h \cdot a}\, da$ is a cumulant generating function and

$$\frac{\partial\langle a_k\rangle}{\partial h_j} = \frac{\partial^2 \log Z}{\partial h_j \partial h_k} = \mathrm{Cov}(a_k, a_j), \tag{D.11}$$

so a susceptibility is a covariance. Susceptibilities defined this way are the standard probe of phase transitions, and they are the instrument used by Bachtis et al. (2024) to locate the transitions discussed in Section 2.3. Chapter 7 shows that the response of a diffusion score to its own observations is exactly of the form (D.11), with the observation playing the role of the field. Equation (D.11) is a static identity within one fixed measure, and is not the two-time dynamical fluctuation–dissipation theorem of non-equilibrium statistical mechanics. Finally, rewriting the Gibbs–Shannon entropy of $\rho_\beta$:

$$S[\rho_\beta] = -k_B \mathbb{E}\Big[\log \rho_\beta\Big] = -k_B \mathbb{E}\Big[-\beta H - \log Z\Big] = \frac{U}{T} + k_B \log Z,$$

and multiplying by $-T$: $-TS = -U + F$, i.e.

$$F = U - TS.$$

The free energy is the exact trade-off between energy and entropy: at low $T$ minimising $F$ means minimising energy (ordered states win), at high $T$ it means maximising entropy (disordered states win). Phase transitions are the non-analyticities of $F$ where the winner changes. $\square$

There is one more characterisation of $F$ worth recording, because it is the statistical-mechanics form of the variational principle that underlies both the training objective of diffusion models (Chapter 2) and the Bethe free energies of message passing. For *any* trial density $\mu$,

$$\underbrace{\mathbb{E}_\mu[H] - \tfrac{1}{\beta} S[\mu]/k_B}_{\text{variational free energy } F[\mu]} \;=\; F \;+\; \tfrac{1}{\beta}\, D_{\mathrm{KL}}(\mu \,\|\, \rho_\beta) \;\;\geq\;\; F, \tag{D.12}$$

with equality iff $\mu = \rho_\beta$: the Boltzmann distribution is the unique minimiser of the variational free energy, and the gap is exactly a Kullback–Leibler divergence. Every variational method in machine learning is a restriction of (D.12) to a tractable family of trial densities.

## D.5 The Hopfield dynamics descends the energy

Moved here from Chapter 2, where the statement is used but the two lines of algebra behind it are not the point.

Let $h_i = \sum_{j \neq i} J_{ij}\sigma_j$ be the local field on neuron $i$ and suppose the update (2.5) flips it, so $\sigma_i' = -\sigma_i$. Only the terms of $H$ containing $\sigma_i$ change, hence

$$\Delta H \;=\; H(\sigma') - H(\sigma) \;=\; -\,(\sigma_i' - \sigma_i)\,h_i \;=\; -\,2\,\sigma_i'\,h_i.$$

The update sets $\sigma_i' = \mathrm{sign}(h_i)$, so $\sigma_i' h_i = |h_i| \geq 0$ and $\Delta H \leq 0$, strictly whenever a flip actually occurs. $H$ is bounded below on a finite configuration space, so the dynamics terminates in a local minimum and *every* initial condition falls into an attractor.

The single-pattern case $P = 1$, the Mattis model (Mattis, 1976), is gauge-equivalent to the uniform ferromagnet: the relabelling $\sigma_i \to \xi_i\sigma_i$ maps the stored pattern onto the all-up state. It returns in Section 2.13 as the exactly solvable teacher in a modern analysis of how energy-based models learn.

## D.6 First- and second-order transitions

The order of a transition is the order of the derivative of the free energy that first misbehaves.

At a *first-order* transition the first derivative jumps: the order parameter is discontinuous, latent heat is absorbed, the two phases coexist at the transition, and correlation lengths stay finite. At a *second-order* transition the first derivative is continuous and the second diverges: the order parameter grows continuously from zero, the susceptibility (D.11) and the correlation length diverge, and observables acquire power laws.

The distinction is what makes the learning cascade of Section 2.13 detectable through susceptibilities rather than through jumps, and it is also why the finite models of this thesis cannot exhibit either in the strict sense: both notions require the thermodynamic limit, and a correlation length that grows until it reaches the system size and then stops is a finite-size ceiling.

## D.7 The learning cascade in detail

Supporting Section 2.13. The chapter states that RBM training is a cascade of second-order transitions and what that means for this thesis; the mapping that establishes it, and the empirical evidence, are here.

gradient (2.6). The central idea is to track training through the singular value decomposition of the weight matrix,

$$W \;=\; \sum_{\gamma=1}^{r} w_\gamma \, u_\gamma \bar{u}_\gamma^{\mathsf{T}}, \tag{D.13}$$

whose modes $(u_\gamma, \bar{u}_\gamma)$ act as candidate order parameters: the magnetisation of the visible layer along mode $\gamma$, $m_\gamma = u_\gamma \cdot v/\sqrt{N_v}$, measures whether the learned Gibbs measure has developed structure in that direction. At initialisation the weights are small and the model is a paramagnet: no direction in configuration space is preferred. The claim of Bachtis et al. (2024) is that training is an *effective cooling*: as the singular values grow, the model crosses a sequence of genuine second-order phase transitions, one for each feature it learns, each signalled by a diverging susceptibility and falling in the mean-field (Curie–Weiss) universality class. The mechanism is visible in a two-line calculation.

**Toolbox D.5 — The retrieval transition of a one-mode RBM**

Following Bachtis et al. (2024), take a single Gaussian hidden unit of variance $1/N_v$ coupled to $N_v$ binary visibles through a weight vector $w$: the energy is $E(v,h) = -h\, w \cdot v + \frac{N_v}{2} h^2$. Integrating out the Gaussian $h$ gives the visible marginal

$$p(v) \;\propto\; \exp\!\left[\tfrac{1}{2N_v}\,(w \cdot v)^2\right] :$$

a Mattis/Hopfield measure (Section 2.2) whose single pattern is $w$ itself, made extensive by the $1/N_v$ variance scaling. Write $w = \bar{w}\,\xi$ with $\xi \in \{-1,+1\}^{N_v}$ and let $m = \frac{1}{N_v}\sum_i \xi_i v_i$ be the overlap; then $p(v) \propto \exp[\frac{N_v \bar{w}^2}{2} m^2]$. Counting configurations at fixed overlap ($\asymp \exp[N_v H(\frac{1+m}{2})]$ with $H$ the binary entropy) gives the free-entropy density

$$\phi(m) \;=\; \frac{\bar{w}^2}{2}\,m^2 + H\Big(\frac{1+m}{2}\Big) \;=\; \log 2 + \frac{\bar{w}^2 - 1}{2}\,m^2 - \frac{m^4}{12} + O(m^6).$$

This is a Landau expansion: the quadratic coefficient changes sign at $\bar{w}^2 = \|w\|^2/N_v = 1$. Below the threshold the maximiser is $m = 0$ (paramagnet); above it two symmetric maximisers $\pm m^*$ appear continuously (retrieval). The transition is second order with mean-field exponents, and the susceptibility along $w$ diverges as $\|w\|^2/N_v \uparrow 1$. Growing a singular value during training is therefore literally a quench through a Curie–Weiss critical point. $\square$

**Why a cascade.** With data generated by a Hopfield teacher whose two patterns are correlated combinations $\xi^{1,2} = \eta_1 \pm \eta_2$ of orthogonal directions, the data covariance is rank two with unequal strengths, and the early-time gradient flow of the student's SVD amplitudes along $\eta_1$ and $\eta_2$ is exponential with *two different rates.* The stronger direction (the centre of mass of the clusters) reaches its critical magnitude first: the learned measure splits into two lumps along $\pm\eta_1$. The weaker direction keeps growing at its own rate and crosses criticality later, splitting each lump again: four modes, the full teacher, retrieved in hierarchical order. Each crossing is a distinct second-order transition with its own critical time, its own diverging susceptibility, and hysteresis below it. Learning proceeds coarse-to-fine not by accident of optimisation but because the critical times are ordered by the spectrum of the data.

**Evidence on real data.** On MNIST, CelebA, and human-genome data, Bachtis et al. (2024) track the susceptibilities of individual SVD modes during training of binary–binary RBMs and observe the same signatures: successive susceptibility peaks that sharpen with system size, finite-size-scaling collapses consistent with mean-field exponents $(\gamma, \nu) = (1, \frac{1}{2})$ at upper critical dimension four, and hysteresis, the operational fingerprints of genuine transitions rather than smooth crossovers. This confirms and sharpens the earlier picture of RBM learning as progressive alignment with the data's principal components (Decelle et al., 2017), of feature emergence as condensation (Tubiana and Monasson, 2017), and of sequential mode learning in linear networks (Saxe et al., 2014); a broad review of the EBM–physics interface is Carbone (2025).

# Appendix E

# Stochastic Calculus and the Diffusion Channel

The Itô calculus, the Fokker–Planck equation, sampling by dynamics, and the AR(1) chain in full, supporting Chapter 2.

## E.1 The AR(1) chain: the running example of the thesis

We now introduce the specific Markov chain that the research chapters solve completely: the first-order autoregressive (AR(1)) process, the canonical Gaussian Markov chain (Brockwell and Davis, 1991). Let $\alpha \in (-1, 1)$ and let $(\eta_k)$ be i.i.d. $\mathcal{N}(0, \sigma_\eta^2)$ innovations; define

$$a_{k+1} \;=\; \alpha\, a_k + \eta_k, \qquad k = 0, 1, \ldots, K-2. \tag{E.1}$$

The transition kernel is Gaussian, $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$, and the joint law of the trajectory follows from (2.8). Throughout the thesis we use the stationary normalisation $\sigma_\eta^2 = 1 - \alpha^2$ and start the chain in its stationary law, $a_0 \sim \mathcal{N}(0, 1)$, so that every frame has unit marginal variance. A trajectory $a = (a_0, \ldots, a_{K-1})$ is *one* data point, the prototype of a video or a time series; its frame index $k$ is an internal time, to be kept distinct from the diffusion time $t$ introduced in Section 2.7. Everything about this chain is computable in closed form; the next toolbox derives the three facts used throughout.

**Toolbox E.1 — Stationary distribution and autocovariance of AR(1)**

*(i) Stationary variance.* Suppose $a_k \sim \mathcal{N}(0, \sigma^2)$ and impose that $a_{k+1}$ has the same variance. Since $\eta_k$ is independent of $a_k$,

$$\mathrm{Var}(a_{k+1}) = \alpha^2 \,\mathrm{Var}(a_k) + \sigma_\eta^2 \overset{!}{=} \mathrm{Var}(a_k) \quad \Longrightarrow \quad \sigma_\infty^2 = \frac{\sigma_\eta^2}{1-\alpha^2}.$$

The fixed point exists precisely because $|\alpha| < 1$; the map $\sigma^2 \mapsto \alpha^2\sigma^2 + \sigma_\eta^2$ is a contraction, so *any* initial variance converges to $\sigma_\infty^2$ geometrically. Since Gaussianity is preserved by the affine recursion, the stationary law is $\pi = \mathcal{N}(0, \sigma_\infty^2)$, which equals $\mathcal{N}(0,1)$ under the stationary normalisation.

*(ii) Autocovariance.* In the stationary regime, for $d \geq 1$, unroll the recursion $d$ times: $a_{k+d} = \alpha^d a_k + \sum_{j=0}^{d-1} \alpha^j \eta_{k+d-1-j}$. All innovation terms are independent of $a_k$, hence

$$\mathrm{Cov}(a_k, a_{k+d}) = \alpha^d \,\mathrm{Var}(a_k) = \alpha^d \,\sigma_\infty^2,$$

and by symmetry $\mathrm{Cov}(a_i, a_j) = \alpha^{|i-j|}\sigma_\infty^2$ for all $i, j$: correlations decay *geometrically* with the lag, with rate $\alpha$. The covariance matrix of $K$ consecutive frames,

$$(\Sigma_0)_{ij} = \sigma_\infty^2 \,\alpha^{|i-j|},$$

is symmetric Toeplitz, as stationarity requires (Section 2.5).

*(iii) Detailed balance.* With $\pi = \mathcal{N}(0, \sigma_\infty^2)$ and $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$, both sides of (2.9) are proportional to

$$\exp\!\left[-\frac{a^2}{2\sigma_\infty^2} - \frac{(a'-\alpha a)^2}{2\sigma_\eta^2}\right] = \exp\!\left[-\frac{a^2 + (a')^2 - 2\alpha\, aa'}{2\sigma_\eta^2}\right],$$

where the exponent on the right is *symmetric* in $(a, a')$. Hence the stationary AR(1) chain is reversible: statistically, the movie of the chain looks the same played in either direction. □

The AR(1) chain is thus a working miniature of everything this chapter discusses: a Markov kernel with an explicit stationary law, geometric convergence, reversibility, and, decisive for us, exact Gaussian algebra at every step.

## E.2 Sampling by dynamics: MCMC and Langevin

Sampling inverts the logic of the previous sections. There, a kernel $M$ was given and we asked for its stationary law $\pi$. In sampling problems we are given $\pi$, typically a Boltzmann distribution $\pi(x) \propto e^{-\beta H(x)}$ known only up to its partition function, and we must *design* a kernel whose stationary law is $\pi$; the ergodic theorem (2.10) then converts one long trajectory into expectation values. This is how the negative phase of Boltzmann-machine learning (2.6) is estimated in practice, and the idea originates, fittingly, in computational statistical mechanics (Metropolis et al., 1953).

The *Metropolis–Hastings* chain (Metropolis et al., 1953; Hastings, 1970) moves from $x$ by proposing $x' \sim g(\cdot|x)$ from any tractable proposal kernel and accepting the move with probability

$$A(x \to x') \;=\; \min\left\{1,\; \frac{\pi(x')\, g(x \mid x')}{\pi(x)\, g(x' \mid x)}\right\}, \tag{E.2}$$

staying at $x$ otherwise. The target enters only through the *ratio* $\pi(x')/\pi(x)$, so the intractable partition function cancels: Metropolis–Hastings samples from $e^{-\beta H}/Z$ while only ever evaluating energy differences. The acceptance rule is engineered so that the resulting kernel satisfies detailed balance (2.9) with respect to $\pi$; the verification is a short computation (Robert and Casella, 2004, Chapter 7). When $x$ is high-dimensional but each conditional $\pi(x_i|x_{\setminus i})$ is tractable, the *Gibbs sampler* (Geman and Geman, 1984) cycles through the coordinates resampling each from its conditional, a Metropolis–Hastings move whose acceptance probability is identically one. Gibbs sampling thrives exactly when the model has sparse conditional structure: for a Markov chain, $\pi(x_k|x_{\setminus k})$ depends only on the two neighbours, and for the bipartite RBM of Section 2.2 each layer updates in one parallel step. Inference cost follows graph structure, a principle that Chapter 3 makes systematic.

For continuous $x$ and differentiable energy, a different design principle uses gradients. Consider the discrete-time update

$$x_{k+1} \;=\; x_k + \epsilon\,\nabla \log \pi(x_k) \;+\; \sqrt{2\epsilon}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0, I), \tag{E.3}$$

the (unadjusted) *Langevin algorithm*. The drift pushes towards high probability; the noise maintains exploration. Two observations prepare what follows. First, (E.3) requires only the *score* $\nabla \log \pi$; the normalising constant is never needed. This is precisely why the score is the natural object for generative modelling (Chapter 2), and why so much of this thesis is devoted to computing scores exactly. Second, as $\epsilon \to 0$ the recursion (E.3) becomes a stochastic differential equation, the Langevin diffusion, whose stationary law is exactly $\pi$. Making that limit precise requires the Itô calculus and Fokker–Planck theory that occupy the rest of the chapter, where the claim is verified by direct computation.

## E.3 From ODEs to SDEs: the Itô calculus

An ordinary differential equation $\dot{x} = f(x, t)$ moves a point along a deterministic velocity field. Physical systems, however, feel rapid, unresolved forces (collisions, thermal agitation) whose cumulative effect is well modelled by adding noise:

$$\mathrm{d}X_t \;=\; \underbrace{f(X_t, t)\,\mathrm{d}t}_{\text{drift}} \;+\; \underbrace{g(t)\,\mathrm{d}W_t}_{\text{diffusion}}. \tag{E.4}$$

Here $W_t$ is a *Wiener process* (standard Brownian motion): the unique process with $W_0 = 0$, independent increments, $W_t - W_s \sim \mathcal{N}(0, t-s)$, and continuous paths. Equation (E.4) is shorthand for the integral equation $X_t = X_0 + \int_0^t f(X_s, s)\,\mathrm{d}s + \int_0^t g(s)\,\mathrm{d}W_s$, and the whole subtlety lives in the last term: Brownian paths are nowhere differentiable, so $\int g\,\mathrm{d}W$ cannot be a Riemann–Stieltjes integral. Itô's construction (Itô, 1944) defines it as the limit of left-endpoint sums,

$$\int_0^t g(s)\,\mathrm{d}W_s \;=\; \lim_{n\to\infty}\sum_{j=0}^{n-1} g(s_j)\left(W_{s_{j+1}} - W_{s_j}\right), \tag{E.5}$$

where the evaluation at the *left* endpoint $s_j$ makes the integrand independent of the coming increment. Two consequences follow immediately: the Itô integral is a martingale (zero mean), and its variance is given by the *Itô isometry*, $\mathbb{E}\left[(\int_0^t g\,\mathrm{d}W)^2\right] = \int_0^t g(s)^2\,\mathrm{d}s$, both because cross terms $\mathbb{E}[g(s_j)(W_{s_{j+1}} - W_{s_j})]$ vanish by independence.

It is worth recording why no convention-free definition is available, since the same two sums reappear in Chapter 7. Fix $[0,t]$ and a partition of mesh tending to zero. The *quadratic variation* $Q_n = \sum_j (W_{s_{j+1}} - W_{s_j})^2$ has mean $\sum_j \Delta s_j = t$ and variance $2\sum_j(\Delta s_j)^2$, which vanishes in the limit, so $Q_n \to t$ in mean square. The *total variation* $V_n = \sum_j |W_{s_{j+1}} - W_{s_j}|$ instead diverges, because $Q_n \le (\max_j |W_{s_{j+1}} - W_{s_j}|)\,V_n$ and the maximum increment tends to zero by continuity: a bounded $V_n$ would force $Q_n \to 0$, contradicting $Q_n \to t > 0$. A Riemann–Stieltjes integral $\int f\,\mathrm{d}g$ requires $g$ to have bounded variation, precisely so that the limit does not depend on where inside each subinterval $f$ is evaluated. For Brownian motion that guarantee is absent, different evaluation points give genuinely different limits, and a convention must be fixed.

The price of the construction is a new chain rule. Its origin is the quadratic variation just computed: increments scale as $\mathrm{d}W_t \sim \sqrt{\mathrm{d}t}$, so second-order terms in $\mathrm{d}W$ are first-order in $\mathrm{d}t$ and cannot be discarded.

**Toolbox E.2 — Itô's lemma, and where the extra term comes from**

Let $Y_t = \phi(X_t, t)$ with $\phi$ twice differentiable and $X_t$ solving (E.4). Taylor-expand $\phi$ to second order:

$$\mathrm{d}Y = \frac{\partial\phi}{\partial t}\,\mathrm{d}t + \frac{\partial\phi}{\partial x}\,\mathrm{d}X + \frac{1}{2}\frac{\partial^2\phi}{\partial x^2}\,(\mathrm{d}X)^2 + \dots$$

Substitute $\mathrm{d}X = f\,\mathrm{d}t + g\,\mathrm{d}W$ and expand the square:

$$(\mathrm{d}X)^2 = f^2(\mathrm{d}t)^2 + 2fg\,\mathrm{d}t\,\mathrm{d}W + g^2(\mathrm{d}W)^2.$$

Now use the multiplication table of Itô calculus, which is nothing but the scaling $\mathrm{d}W \sim \sqrt{\mathrm{d}t}$ made systematic:

$$(\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\,\mathrm{d}W = 0, \qquad (\mathrm{d}W)^2 = \mathrm{d}t.$$

The last identity is the rigorous statement that the quadratic variation of $W$ over $[0,t]$ equals $t$: for a partition of mesh $\to 0$, $\sum_j (W_{s_{j+1}} - W_{s_j})^2 \to t$ in mean square, since the sum has mean $\sum_j (s_{j+1} - s_j) = t$ and variance $2\sum_j (s_{j+1} - s_j)^2 \to 0$. Keeping the surviving terms:

$$\mathrm{d}Y = \left(\frac{\partial\phi}{\partial t} + f\,\frac{\partial\phi}{\partial x} + \frac{g^2}{2}\,\frac{\partial^2\phi}{\partial x^2}\right)\mathrm{d}t \;+\; g\,\frac{\partial\phi}{\partial x}\,\mathrm{d}W.$$

The half-Laplacian term, absent from the classical chain rule, is the engine of everything that follows: it is why densities spread, why the Fokker–Planck equation has a second-order term, and why exponential transformations of SDEs pick up variance corrections. □

## E.4 The Fokker–Planck equation: transporting densities

The SDE (E.4) moves *samples*; the corresponding motion of the *density* $p_t(x)$ is governed by the Fokker–Planck (or forward Kolmogorov) equation. This change of viewpoint, from trajectories to densities, is the same one Liouville's theorem performed for Hamiltonian flow, and the derivation is parallel, with the Itô correction supplying a new, diffusive term.

**Toolbox E.3 — Derivation of the Fokker–Planck equation**

Let $\phi$ be an arbitrary smooth test function of compact support, and compute $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\phi(X_t)]$ in two ways.

*Way 1: through the samples.* By Itô's lemma,

$$\mathrm{d}\phi(X_t) = \Big(f\,\phi' + \tfrac{g^2}{2}\,\phi''\Big)\mathrm{d}t + g\,\phi'\,\mathrm{d}W_t,$$

and taking expectations kills the martingale term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathbb{E}[\phi(X_t)] = \mathbb{E}\Big[f(X_t,t)\,\phi'(X_t)\Big] + \frac{g(t)^2}{2}\,\mathbb{E}\Big[\phi''(X_t)\Big].$$

*Way 2: through the density.* $\mathbb{E}[\phi(X_t)] = \int \phi(x)\,p_t(x)\,\mathrm{d}x$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathbb{E}[\phi(X_t)] = \int \phi(x)\,\frac{\partial p_t(x)}{\partial t}\,\mathrm{d}x.$$

Equate the two, and integrate Way 1 by parts to move all derivatives off $\phi$ (boundary terms vanish by compact support): once for the drift term, twice for the diffusion term,

$$\int \phi\,\frac{\partial p_t}{\partial t} = \int \phi\left[-\frac{\partial}{\partial x}\Big(f\,p_t\Big) + \frac{g^2}{2}\,\frac{\partial^2 p_t}{\partial x^2}\right]\mathrm{d}x.$$

Since $\phi$ is arbitrary, the integrands must agree pointwise:

$$\boxed{\frac{\partial p_t}{\partial t} = -\frac{\partial}{\partial x}\Big(f(x,t)\,p_t(x)\Big) + \frac{g(t)^2}{2}\,\frac{\partial^2 p_t(x)}{\partial x^2}.}$$

The first term transports probability along the drift (as in Liouville); the second spreads it diffusively. The diffusive term is the macroscopic footprint of the Itô correction. □

In vector form, and for later reference, the Fokker–Planck equation of (E.4) reads

$$\partial_t p_t = -\nabla\cdot\Big(f\,p_t\Big) + \tfrac{1}{2}g^2\,\nabla^2 p_t. \tag{E.6}$$

### E.4.1 Stationary law of Langevin dynamics: closing the loop

Apply the Fokker–Planck equation to the *Langevin SDE* associated with a potential $U$ (the continuous-time limit of the algorithm (E.3)):

$$\mathrm{d}X_t = -\nabla U(X_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t. \tag{E.7}$$

A stationary density solves $0 = \nabla \cdot (\nabla U\, p + \beta^{-1}\nabla p)$, i.e. the probability current $J = -(\nabla U)\,p - \beta^{-1}\nabla p$ is divergence-free; imposing the natural (decaying) boundary conditions forces $J \equiv 0$, hence $\nabla p/p = -\beta\nabla U$, hence

$$p_\infty(x) \;\propto\; e^{-\beta U(x)} : \tag{E.8}$$

*Langevin dynamics equilibrates to the Boltzmann distribution* of Chapter 2. This is the two-line proof of the claim left pending in Section E.2, and it explains the deep kinship of sampling and physics: a Langevin sampler *is* an overdamped physical system at temperature $1/\beta$. Note also that the drift of (E.7) is (up to $\beta$) the *score* of the target, $-\nabla U = \nabla \log p_\infty$: dynamics that sample a distribution are driven by its score.

For the OU process, $U(x) = x^2/2$ and $\beta = 1$: the stationary law is $\mathcal{N}(0,1)$, confirming the $t \to \infty$ limit of Section 2.7.

# Appendix F

# Derivations for Diffusion Models

Energy-based models, the DDPM variational bound, score matching in its two equivalent forms, and the probability-flow ODE, supporting Chapter 2.

## F.1 Energy-based models

Modern generative modelling asks a single question: given samples of an unknown distribution $p_0$, build a machine that produces new ones. The last decade produced several answers: variational autoencoders (Kingma and Welling, 2014), adversarial networks (Goodfellow et al., 2014), normalizing flows (Rezende and Mohamed, 2015), diffusion models. It is easy to present them as unrelated tricks. The energy-based view of LeCun et al. (2006) is the antidote: it is the direct continuation of the Boltzmann-machine programme of Section 2.2, and, as this chapter unfolds, diffusion models will appear as the member of the energy-based family that finally solved its oldest problem.

An *energy-based model* (EBM) over $x \in \mathcal{X}$ assigns to each configuration a scalar energy $E_\theta(x)$ measuring how badly $x$ fits the model's idea of the data: low energy for plausible configurations, high for implausible ones. In LeCun et al.'s formulation the energy is the primitive object: inference means finding low-energy configurations, and learning means shaping the landscape, pushing the energy of observed data down and the energy of everything else up. A probabilistic model is recovered whenever one normalises with the

Gibbs measure of Chapter 2,

$$p_\theta(x) \;=\; \frac{e^{-E_\theta(x)}}{Z(\theta)}, \qquad Z(\theta) = \int_{\mathcal{X}} e^{-E_\theta(x)}\,\mathrm{d}x, \tag{F.1}$$

the Boltzmann distribution (D.9) with $\beta$ absorbed into the energy and the physics replaced by a parametric family. The appeal of (F.1) is that $E_\theta$ is unconstrained: any function defines a valid unnormalised density. The price is the partition function, and every basic operation hits it. *Evaluating* $p_\theta(x)$ requires $Z(\theta)$, intractable in general (Section 2.2). *Sampling* requires the MCMC machinery of Section E.2, whose mixing time can be exponential in the presence of metastability. *Learning* by maximum likelihood requires the gradient

$$-\nabla_\theta \log p_\theta(x) = \nabla_\theta E_\theta(x) - \mathbb{E}_{p_\theta}\Big[\nabla_\theta E_\theta\Big], \tag{F.2}$$

the "push down on data, pull up on the model's own samples" rule, the continuous-variable form of the Boltzmann-machine rule (2.6), whose second term again demands sampling from $p_\theta$. LeCun et al. (2006) frame the alternatives systematically: either fight the negative phase with *contrastive* losses that pull up only on well-chosen offending configurations (contrastive divergence (Hinton, 2002) being the MCMC-flavoured member of the family), or design architectures and regularisers that bound the volume of low-energy space so that no pull-up is needed. The lesson of the tutorial is that the partition function is not an accounting detail; it is *the* design constraint of the field.

There are two classical escape routes, and this thesis walks both. The first is the *score*: since $\nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x)$, the gradient of the log-density in $x$ is free of $Z$, because differentiating in $x$ kills the constant. Sections F.3–2.11 build the modern edifice standing on this observation. The second is *structure*: when the energy decomposes into local terms, marginals and even $Z$ itself become computable by dynamic programming; that route (Markov random fields, factor graphs, message passing) is developed in Chapter 3.

## F.2 Denoising diffusion probabilistic models

The discrete-time formulation (Sohl-Dickstein et al., 2015; Ho et al., 2020) discretises (2.15) into a Markov chain $q(x_s|x_{s-1}) = \mathcal{N}(x_s; \sqrt{1-\beta_s}\, x_{s-1}, \beta_s I)$ with a variance schedule $\beta_1, \ldots, \beta_S$, and learns a reverse chain $p_\theta(x_{s-1}|x_s)$. Training maximises a variational lower bound on the data log-likelihood; the physicist will recognise the Gibbs variational principle (D.12) in modern dress.

**Toolbox F.1 — The evidence lower bound of a diffusion model**

For any latent-variable model with joint $p_\theta(x_{0:S})$ and any inference distribution $q(x_{1:S}|x_0)$, Jensen's inequality gives

$$\begin{aligned}
\log p_\theta(x_0) &= \log \int p_\theta(x_{0:S})\, \mathrm{d}x_{1:S} = \log\, \mathbb{E}_q\left[\frac{p_\theta(x_{0:S})}{q(x_{1:S}|x_0)}\right] \\
&\geq\ \mathbb{E}_q\left[\log \frac{p_\theta(x_{0:S})}{q(x_{1:S}|x_0)}\right]\ =:\ \text{ELBO}.
\end{aligned}$$

Insert the two Markov factorisations $p_\theta(x_{0:S}) = p(x_S)\prod_s p_\theta(x_{s-1}|x_s)$ and $q(x_{1:S}|x_0) = \prod_s q(x_s|x_{s-1})$, then regroup telescopically using Bayes' rule $q(x_s|x_{s-1}) = q(x_{s-1}|x_s, x_0)\, q(x_s|x_0)/q(x_{s-1}|x_0)$:

$$\begin{aligned}
-\text{ELBO}\ &= D_{\text{KL}}(q(x_S|x_0)\,\|\,p(x_S))\ -\ \mathbb{E}_q \log p_\theta(x_0|x_1) \\
&\quad + \sum_{s=2}^{S} \mathbb{E}_q\, D_{\text{KL}}(q(x_{s-1}|x_s, x_0)\,\|\,p_\theta(x_{s-1}|x_s))\,,
\end{aligned}$$

one denoising step per summand. The decisive point: because the forward chain is Gaussian and conditioned on *both* endpoints, each target $q(x_{s-1}|x_s, x_0)$ is an explicit Gaussian. Matching it with a Gaussian $p_\theta$ reduces each KL term to a weighted regression of the posterior mean; concretely, to predicting the noise $\varepsilon$ that was added (Ho et al., 2020). The thermodynamic reading of the ELBO gap as an entropy-production bound goes back to Sohl-Dickstein et al. (2015). □

The practical upshot of Ho et al. (2020) is the loss

$$\mathcal{L}(\theta) = \mathbb{E}_{s,\, a\sim p_0,\, \varepsilon}\Big\|\varepsilon - \varepsilon_\theta\Big(\underbrace{a\, e^{-t_s} + \sqrt{\Delta_{t_s}}\,\varepsilon}_{x_{t_s}},\ s\Big)\Big\|^2, \tag{F.3}$$

a denoising regression, and the empirical discovery that this simplified objective, with each noise level weighted equally and all the variational bookkeeping dropped, trains photorealistic image models. Its connection with the score is not incidental, as the next section shows: predicting the noise *is* estimating $\nabla \log p_t$ up to scale.

## F.3 Score matching

Suppose we model the score directly by a function $s_\theta(x)$ and wish it to match $\nabla \log p(x)$ for data from $p$. The naive objective $\mathbb{E}_p\|s_\theta - \nabla \log p\|^2$ seems to require knowing the very score we lack. Two classical results remove the obstruction.

**Toolbox F.2 — Implicit score matching (Hyvärinen)**

Expand the square and keep only $\theta$-dependent terms:

$$\mathbb{E}_p \left\|s_\theta - \nabla \log p\right\|^2 = \mathbb{E}_p \left\|s_\theta\right\|^2 - 2\,\mathbb{E}_p \left\langle s_\theta, \nabla \log p\right\rangle + \text{const.}$$

The cross term integrates by parts. Componentwise, using $p\,\partial_i \log p = \partial_i p$ and assuming $p\, s_{\theta,i} \to 0$ at infinity:

$$\int p\ s_{\theta,i}\ \partial_i \log p\ \mathrm{d}x = \int s_{\theta,i}\ \partial_i p\ \mathrm{d}x = -\int p\ \partial_i s_{\theta,i}\ \mathrm{d}x.$$

Therefore

$$\mathbb{E}_p \left\|s_\theta - \nabla \log p\right\|^2 = \mathbb{E}_p\Big[\ \left\|s_\theta(x)\right\|^2 + 2\,\nabla\cdot s_\theta(x)\Big] + \text{const}:$$

an objective involving only the model and data samples (Hyvärinen, 2005). Elegant, but the divergence term is expensive for large networks, which is why diffusion models use the denoising variant instead. □

**Toolbox F.3 — Denoising score matching (Vincent)**

Fix $t$ and consider the noised marginal (2.16), written as $p_t(x) = \int p_0(a)\, q_t(x|a)\, \mathrm{d}a$ with Gaussian kernel $q_t(x|a) = \mathcal{N}(x; ae^{-t}, \Delta_t I)$. Then

$$\nabla_x \log p_t(x) = \frac{\int p_0(a)\, \nabla_x q_t(x|a)\, \mathrm{d}a}{p_t(x)} = \int \underbrace{\frac{p_0(a)\, q_t(x|a)}{p_t(x)}}_{=\, p(a|x)} \nabla_x \log q_t(x|a)\, \mathrm{d}a,$$

so that

$$\nabla_x \log p_t(x) \;=\; \mathbb{E}\Big[\nabla_x \log q_t(x|a) \,\Big|\, x\Big] :$$

the score of the *marginal* is the posterior average of the score of the *kernel*, and the latter is trivial, $\nabla_x \log q_t(x|a) = -(x - ae^{-t})/\Delta_t$. A standard $L^2$-projection argument turns this into a regression: for any $s_\theta$,

$$\mathbb{E}_{x\sim p_t} \left\| s_\theta(x) - \nabla \log p_t(x) \right\|^2 = \mathbb{E}_{a,x} \left\| s_\theta(x) - \nabla_x \log q_t(x|a) \right\|^2 + \text{const},$$

because conditional expectation is exactly the $L^2$ projection onto functions of $x$ (Vincent, 2011). Substituting the Gaussian kernel score and $x = ae^{-t} + \sqrt{\Delta_t}\varepsilon$ gives $\nabla_x \log q_t = -\varepsilon/\sqrt{\Delta_t}$: *learning the score of noisy data is learning to predict the noise*, and the DDPM loss (F.3) is denoising score matching in disguise. $\square$

The continuous-time synthesis (Song and Ermon, 2019; Song et al., 2021) trains one network $s_\theta(x, t)$ against denoising score matching at all noise levels and samples with the reverse SDE (2.14); an accessible exposition is Song (2021), and Karras et al. (2022) systematise the design space. This is the resolution, promised in Section F.1, of the energy-based model's oldest problem: Song and Ermon replace the energy by its gradient, the equilibrium sampler by a finite-time reverse SDE, and the intractable likelihood by a regression whose targets are exact.

## F.4 Flows: the deterministic side of the same coin

Anderson's theorem has a deterministic twin. As noted in Section 2.8, the marginals $p_t$ of the forward SDE are also transported by the noise-free *probability-flow ODE* with velocity $f - \frac{g^2}{2}\nabla \log p_t$ (Song et al., 2021): the same score, consumed by an ODE instead of an SDE, turns generation into a smooth, invertible map from Gaussian noise to data. This places diffusion models inside an older family. *Normalizing flows* (Rezende and Mohamed, 2015; Papamakarios et al., 2021) construct an invertible map $T_\theta$ and evaluate likelihoods through the change-of-variables formula; their continuous-time version parametrises the velocity field of an ODE directly (Chen et al., 2018). What diffusion brought to this family is a *simulation-free training signal*: *flow matching* (Lipman et al., 2023) and stochastic interpolants (Albergo and Vanden-Eijnden, 2023) show that one can regress the velocity field of any prescribed interpolation between noise and data (the OU path (2.15) being one choice among many) with the same conditional trick as denoising score matching, never integrating the ODE during training; a systematic introduction is Holderrieth and Erives (2025). The unifying statement is worth recording: DDPMs, score SDEs, and flow-matching models all learn a vector field whose ground truth is a *conditional expectation under the noising channel*; they differ in the path of marginals and in whether sampling is stochastic or deterministic. Every exact result of this thesis therefore speaks to the whole family at once: the object we compute in closed form, the conditional expectation behind the field, is the training target of all of them.

# Appendix G

# Gaussian Vectors and Matrix Tools

The multivariate Gaussian in both parametrisations, the spectral theorem, and the matrix-inversion identities used throughout the thesis.

## G.1 The multivariate Gaussian, in both parametrisations

**Definition G.1** (Multivariate Gaussian)**.** A random vector $a \in \mathbb{R}^K$ is Gaussian with mean $m$ and (positive definite) covariance $\Sigma$, written $a \sim \mathcal{N}(m, \Sigma)$, if its density is

$$\mathcal{N}(a;\, m, \Sigma) \;=\; \frac{1}{(2\pi)^{K/2}\sqrt{\det \Sigma}} \exp\Big[ -\tfrac{1}{2}(a-m)^{\mathsf{T}}\Sigma^{-1}(a-m) \Big]. \tag{G.1}$$

Expanding the quadratic form shows that the same density can be written in *natural* (information) parameters,

$$\mathcal{N}(a;\, m, \Sigma) \;\propto\; \exp\Big[ -\tfrac{1}{2}\, a^{\mathsf{T}} J\, a + h^{\mathsf{T}} a \Big], \qquad J = \Sigma^{-1}, \qquad h = \Sigma^{-1} m, \tag{G.2}$$

with $J$ the *precision matrix* and $h$ the precision-weighted mean. The two parametrisations are dual, and the duality organises everything that follows (Bishop, 2006; Koller and Friedman, 2009):

- *Linear maps.* If $a \sim \mathcal{N}(m, \Sigma)$ then $Aa + b \sim \mathcal{N}(Am + b,\, A\Sigma A^{\mathsf{T}})$: Gaussianity is preserved by affine transformations, with moments transforming linearly. In particular a sum of independent Gaussians is Gaussian with means and covariances adding.

- *Marginalisation is easy in $\Sigma$.* The marginal of a sub-block $a_1$ of $a = (a_1, a_2)$ is Gaussian with mean $m_1$ and covariance $\Sigma_{11}$: simply read off the block.

- *Conditioning is easy in $J$.* Partitioning $J = \left(\begin{smallmatrix} J_{11} & J_{12} \\ J_{12}^{\mathsf{T}} & J_{22} \end{smallmatrix}\right)$, the conditional law of $a_1$ given $a_2$ is Gaussian with precision $J_{11}$ (read off the block) and mean shifted by $-J_{11}^{-1} J_{12}\, a_2$. Eliminating a variable instead produces the *Schur complement* correction on its neighbours. Proofs of both statements are in Appendix A.

- *Products are easy in $J$.* Multiplying two Gaussian densities in the same variable adds their natural parameters, $(J, h) \mapsto (J_1 + J_2,\, h_1 + h_2)$. Bayesian updating with Gaussian likelihoods is addition.

The dual reading of zeros is used constantly in this thesis. A zero in the *covariance* means marginal independence of two coordinates. A zero in the *precision* means *conditional* independence given all the others: from the conditioning rule, if $J_{ij} = 0$ then $a_i$ and $a_j$ do not interact once the rest is fixed. A Gaussian vector can be, and in this thesis always is, globally correlated (dense $\Sigma$) while conditionally local (sparse $J$). Which matrix one should look at depends on the question, and confusing the two is a common source of error in reasoning about Gaussian dependence.

## G.2 Matrix tools: the spectral theorem and matrix inversion

Since Gaussian computation is matrix computation, we record the two linear-algebra facts the research chapters use constantly.

**Theorem G.2** (Spectral theorem for symmetric matrices)**.** *Every symmetric $A \in \mathbb{R}^{K\times K}$ has an orthonormal basis of eigenvectors: $A = U\,\mathrm{diag}(\omega_1, \ldots, \omega_K)U^\intercal$ with $U^\intercal U = I$ and real eigenvalues $\omega_i$.*

Three consequences matter here. First, analytic operations act on eigenvalues alone: if no eigenvalue vanishes,

$$A^{-1} = U\,\mathrm{diag}(1/\omega_i)\,U^\intercal, \qquad \text{and more generally} \qquad f(A) = U\,\mathrm{diag}\left(f(\omega_i)\right)U^\intercal.$$

Second, affine images share the eigenbasis: $(cA + dI)\,u_i = (c\,\omega_i + d)\,u_i$, so a whole family of matrices of the form $c(t)A + d(t)I$ is diagonalised once and for all by the $U$ of $A$. Chapter 5 uses exactly this fact to diagonalise the diffusion problem at every noise level simultaneously. Third, for the symmetric Toeplitz covariances produced by stationary processes (Section 2.5), the eigenvectors are asymptotically the Fourier modes of the sequence (Brockwell and Davis, 1991): stationary Gaussian problems decouple, in the large-$K$ limit, into independent scalar problems per frequency.

Inversion, when the inverse is not wanted eigenvalue by eigenvalue, has two workhorse expansions. The *Neumann series*

$$(I - B)^{-1} \;=\; \sum_{m\geq 0} B^m, \qquad \text{convergent for } \;\|B\| < 1, \tag{G.3}$$

expresses the inverse of a near-identity matrix as a sum of powers; it is the tool behind the small-$t$ expansions of Chapter 5, where powers of a banded matrix have growing but controlled bandwidth. The *Woodbury identity* relates the inverse of a matrix to the inverse of a low-rank or structured perturbation of it; its statement, proof, and the specific consequence used by the research chapters are collected in Appendix A. Finally, for the tridiagonal Toeplitz matrices that Markov chains produce, the inverse is available in closed form through a transfer-matrix (geometric-ansatz) computation, carried out where it is needed (Chapter 5).

## G.3 Gaussian belief propagation and the Kalman filter

The Gaussian family is the canonical closed message family, and its closure is pure algebra.

**Toolbox G.1 — The two closure lemmas of Gaussian message passing**

*Lemma 1 (products).* For Gaussians in the same variable,

$$\mathcal{N}(a; m_1, v_1) \times \mathcal{N}(a; m_2, v_2) \ \propto\ \mathcal{N}(a; m, v), \qquad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \qquad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}:$$

the sum of two quadratics in $a$ is a quadratic in $a$; precisions and precision-weighted means add, as the natural-parameter rule of Section G.1 requires.

*Lemma 2 (affine marginalisation).* Pushing a Gaussian through a linear-Gaussian transition,

$$\int \mathcal{N}(a'; \alpha a, \sigma^2)\, \mathcal{N}(a; m, v)\, \mathrm{d}a \ =\ \mathcal{N}(a';\, \alpha m,\, \alpha^2 v + \sigma^2),$$

because $a' = \alpha a + \eta$ is a sum of independent Gaussians. Read in the reverse direction the same computation gives $\mathcal{N}\Big(a;\, m'/\alpha,\, (v' + \sigma^2)/\alpha^2\Big)$.

A model whose factors are Gaussian in each variable performs, in (3.3), only these two operations. For such models every message is Gaussian, carried exactly by two numbers, and each update is a few arithmetic operations. □

The oldest and most famous instance predates factor graphs by decades. For linear-Gaussian state-space models (a Gaussian Markov chain observed through linear Gaussian noise), the forward sum–product sweep is the *Kalman filter* (Kalman, 1960), alternating a *predict* step (Lemma 2, push the belief through the dynamics) and an *update* step (Lemma 1, absorb the new observation, with the correction weighted by the Kalman gain); the backward sweep combined with the forward one yields the fixed-interval smoother of Rauch et al. (1965), which conditions each state on *all* observations. That the classical filter and smoother are exactly the Gaussian specialisation of sum–product on the chain's factor graph was made explicit by Kschischang et al. (2001); textbook treatments are

Bishop (2006); Koller and Friedman (2009). Chapter 5 derives the filter from the BP sweep, line by line, for our diffusion posterior, where the observation factors take a specific form (a pseudo-observation whose strength depends on the diffusion time), and verifies the equivalence numerically to double-precision rounding.

One remark, recorded here because it resolves a doubt raised repeatedly in supervision: the Gaussian closure of this section is *algebra*, valid at any node degree, and invokes no limit theorem. It must not be confused with the central-limit justification of the *approximate* Gaussian closures of the next section, which does require many incoming messages and which a chain, with two neighbours per node, does not provide.

## G.4 From BP to TAP and AMP

Statistical physics developed its own message-passing calculus for *dense* graphs, where every node interacts weakly with many others. On such graphs the incoming messages to a node are numerous and nearly independent, so their product is governed by a central-limit argument: the cavity field is approximately Gaussian, and each message can be summarised by a mean and a variance even when the underlying variables are not Gaussian. Historically the resulting self-consistency equations appeared first as the Thouless–Anderson–Palmer (TAP) equations of the Sherrington–Kirkpatrick spin glass (Thouless et al., 1977), whose distinctive ingredient is the Onsager reaction term, the correction that removes a node's own influence from the field it feels. Three decades later the same structure re-emerged in compressed sensing as *approximate message passing* (AMP) (Donoho et al., 2009): a reduction of BP on a dense bipartite graph to iterations on means and variances, with the Onsager term reappearing as a memory correction. The derivation of AMP from BP by moment expansion is carried out for the $\ell_2$ minimisation problem by Genovese and Piana (2026); the general technology, its state-evolution analysis, and its phase diagrams are reviewed by Zdeborová and Krzakala (2016) and, in lecture-note form, by Krzakala and Zdeborová (2024). Mézard (2017) derived the TAP equations of the Hopfield model, connecting this calculus back to the networks of Chapter 2.

The logic of the CLT justification should be kept explicit, because this thesis probes exactly its boundary. On a dense graph with $N$ weak interactions per node, each message removes one interaction out of $N$, message and marginal differ at order $1/N$, and sums of many weak, nearly independent terms are Gaussian: the per-node Gaussian closure is controlled. On a chain every node has two neighbours: no aggregation takes place, each message carries the entire influence of half the system, and nothing justifies a Gaussian form for it beyond the special cases where it is exact. What happens when one imposes the closure anyway is a quantitative question. This thesis answers it on the chain in two complementary ways: exactly, for the per-node (AMP/TAP) closure on the Gaussian chain, where the breakdown admits a closed-form phase boundary (Appendix B); and numerically, for the Gaussian-projected messages on genuinely non-Gaussian chains (Chapter 6).

# Appendix H

# Derivations for the Rotating Ring

The mechanics behind Chapter 7. The chapter states the gauge reduction, the exact surrogate score and their consequences, which is what the argument needs; the algebra establishing them is here, where it can be checked without interrupting that argument.

## H.1 Rotations preserve the standard isotropic Gaussian

Proof of Lemma 7.2.

*Proof.* A linear map of a Gaussian is Gaussian, so $R\eta$ is fixed by its first two moments: $\mathbb{E}[R\eta] = R\,\mathbb{E}[\eta] = 0$ and $\mathrm{Cov}(R\eta) = R\,I_2\,R^\top = RR^\top = I_2$. Equivalently, in terms of densities, $(2\pi)^{-1}e^{-\|z\|^2/2}$ depends on $z$ only through $\|z\|$, and the change of variables $z \mapsto Rz$ changes neither $\|R^{-1}z\| = \|z\|$ nor $|\det R^{-1}| = 1$: the level sets are circles centred at the origin and a rotation maps each circle to itself. □

## H.2 The gauge reduction

Proof of Lemma 7.3.

*Proof.* Substituting (7.10) and using the composition rule of (7.12),

$$y_{k+1} = R_{-(k+1)\psi}\Big(R_\psi z_k + \sigma\eta_{k+1}\Big) = \underbrace{R_{-(k+1)\psi}R_\psi}_{=R_{-k\psi}} z_k + \sigma R_{-(k+1)\psi}\eta_{k+1} = y_k + \sigma\widetilde{\eta}_{k+1},$$

and by Lemma 7.2 each $\widetilde{\eta}_{k+1} \sim \mathcal{N}(0, I_2)$; they stay independent because each is a fixed linear map of a different independent $\eta$. Next, $U_\psi$ is block diagonal with orthogonal blocks, so $U_\psi^\top U_\psi = \mathrm{diag}(I_2, \ldots, I_2) = I_{2K}$ and $\det U_\psi = \prod_k \det R_{-k\psi} = 1$. Applying $U_\psi$ to (7.8) gives $U_\psi x = e^{-t}U_\psi a + \sqrt{\Delta_t}\, U_\psi\xi$ with $U_\psi\xi \sim \mathcal{N}(0, I_{2K})$, so $U_\psi x$ has exactly the law of the noised non-rotating trajectory: the de-rotation commutes with the channel. For the densities, $z = U_\psi^\top y$ and $P_t(z \mid \psi) = P_t^{(0)}(U_\psi z)\,|\det U_\psi|$, the Jacobian factor being one. Differentiating that identity componentwise, with $\partial y_b/\partial z_c = (U_\psi)_{bc}$,

$$\frac{\partial}{\partial z_c}\log P_t^{(0)}(U_\psi z) = \sum_b (U_\psi)_{bc}\left(\nabla \log P_t^{(0)}\right)_b(U_\psi z) = \Big[U_\psi^\top S^{(0)}(U_\psi z, t)\Big]_c.$$

The transpose is not cosmetic: a score is a gradient, so it transforms with $U_\psi^\top$ while a point transforms with $U_\psi$. □

## H.3 The clean law after the gauge

Iterating (7.13) and letting the intermediate terms cancel,

$$y_k - y_0 = \sum_{r=1}^{k}(y_r - y_{r-1}) = \sigma\sum_{r=1}^{k}\widetilde{\eta}_r, \qquad \text{so} \qquad y_k = A + \sigma\sum_{r=1}^{k}\widetilde{\eta}_r, \tag{H.1}$$

with $A := y_0 \sim p_\lambda$ and the empty sum equal to zero. Every frame contains the *same* random anchor, drawn once on the ring, plus a partial sum of independent innovations; the anchor is the only non-Gaussian ingredient of the model.

The clean law factorises over the ring prior and the one-step transitions, so $\log P_0(y) = -(\|y_0\|-1)^2/2\lambda - (2\sigma^2)^{-1}\sum_{k=0}^{K-2}\|y_{k+1} - y_k\|^2 + \text{const}$. Differentiating is a counting exercise: an interior $y_k$ appears in exactly two summands, the edge arriving at it and the edge leaving it, whose derivatives are $-(y_k - y_{k-1})/\sigma^2$ and $+(y_{k+1} - y_k)/\sigma^2$. Adding them, and using $\nabla_y \|y\| = y/\|y\|$ for the ring term,

$$\begin{aligned} S_0(y,0) &= \frac{1-\|y_0\|}{\lambda}\frac{y_0}{\|y_0\|} + \frac{y_1 - y_0}{\sigma^2}, \\ S_k(y,0) &= \frac{y_{k-1} - 2y_k + y_{k+1}}{\sigma^2}, \qquad 1 \le k \le K-2, \\ S_{K-1}(y,0) &= \frac{y_{K-2} - y_{K-1}}{\sigma^2}. \end{aligned} \tag{H.2}$$

The interior clean score is a discrete second derivative along internal time, that is a discrete Laplacian: each block measures how far frame $k$ departs from the average of its neighbours. It is strictly local, and it vanishes identically on the model's own noiseless solution, because a trajectory rotating at radius exactly one maps under the gauge to a constant $y_k$, for which every second difference vanishes and, with $\|y_0\| = 1$, so does the ring term. That vanishing is verified to machine zero (Table 7.1).

## H.4 The exact surrogate score

Proof of Proposition 7.4.

*Proof.* Expanding the exponent of (7.16) in powers of $\alpha$, with the correlated covariance $C_t \otimes I_2$, gives (7.17). Writing $\log P_t(x) = \text{const} - \frac{1}{2}x^\mathsf{T}(C_t^{-1} \otimes I_2)x + \log Z(h(x))$, the gradient of the first term is $-(C_t^{-1}x)_k$ in block $k$, while $\partial h/\partial x_k = e^{-t}q_k I_2$ and $\nabla_h \log Z = \mathbb{E}[A \mid x]$, the first cumulant identity for the exponential family in $h$. Differentiating once more, so that the second derivative of $\log Z$ returns $\mathrm{Cov}(A \mid x)$, gives the response. $\square$

# Appendix I

# How the headline ratio was aggregated

The ratio quoted in Chapter 9 and in the conclusions is one of several summaries the same cells support, and it is systematically the largest of them. That is a fact about the estimand rather than about the estimator, so it belongs with the number wherever the number appears. The section below is generated by `tools/make_aggregation_robustness.py` from the same frozen outputs that produce the table itself, and is shared verbatim with the paper — a second, hand-kept copy is precisely how two documents come to disagree.

## I.1 Does the headline depend on how the ratio was averaged?

A ratio is a nonlinear function of two means, so its value depends on the order of averaging. Table 9.1 reports the mean of per-cell ratios, aggregated per seed. That is the right inferential unit — the twelve noise levels share a training set and a fitted model — but being the right unit does not make it the only defensible summary, so here is the same data under five.

**Table I.1:** The headline under five estimands. (1) is what Table 9.1 reports. The interval is a paired bootstrap over seeds, 20,000 resamples; seeds are resampled rather than cells, because cells within a seed share a fitted model.

| $N$ | (1) per-cell | (2) ratio of means | (3) geometric | (4) median | (5) paired 95% CI |
|---|---|---|---|---|---|
| 32 | 7.0 | 6.8 | 6.1 | 6.8 | [6.1, 8.1] |
| 64 | 7.8 | 6.6 | 6.7 | 6.6 | [6.3, 9.5] |
| 128 | 8.8 | 8.0 | 7.8 | 8.6 | [7.9, 9.6] |
| 256 | 9.7 | 8.8 | 8.6 | 9.2 | [8.5, 10.9] |
| 512 | 9.0 | 8.6 | 7.9 | 8.2 | [8.1, 9.9] |
| 1024 | 12.5 | 9.7 | 10.8 | 11.1 | [10.5, 14.7] |
| 2048 | 15.5 | 10.0 | 13.2 | 12.9 | [13.0, 18.9] |
| 4096 | 17.6 | 12.4 | 15.6 | 16.0 | [15.4, 19.9] |

Two things follow, and the second is the one worth stating plainly.

The conclusion does not depend on the choice: every estimand at every size exceeds 6.1, every bootstrap interval sits well clear of 1, and no aggregation produces a reversal anywhere.

But **the estimand we report is systematically the largest of the four**, and by a margin that grows with $N$ — up to 5.5 at $N = 2048$. The ratio of per-seed averaged errors, which is the summary a reader is most likely to have in mind, runs 6.6 to 12.4 rather than 7.0 to 17.6.

That ordering is a measurement, not a theorem, and it is worth being precise about why no inequality delivers it. Jensen gives $\mathbb{E}[1/Y] \geq 1/\mathbb{E}[Y]$, but $\mathbb{E}[X/Y] \geq \mathbb{E}[X]/\mathbb{E}[Y]$ does *not* follow for arbitrary positive correlated $X$ and $Y$ — the dependence between numerator and denominator can push the comparison either way. What holds here is empirical: denominator heterogeneity across noise levels interacts with that dependence, and in this dataset the mean of cellwise ratios comes out largest. It is a property of these cells, which is exactly why the estimand has to be named rather than assumed.

Resolving the average over noise levels shows where the aggregation hides structure: the ratio is not flat in $t$ but peaks near $t \approx 0.5$ and falls at both ends. Two different "weakest" quantities have to be kept apart. The weakest seed-averaged $(N, t)$ aggregate in the grid is 3.6; the weakest *individual* cell is 0.92, and EM–BP is beaten outright in 2 of the $1,520$ cells summarised here. The estimator's advantage is largest at moderate noise, where the posterior is dominated by neither the likelihood nor the prior — which is where knowing the transition structure should help most.

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